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Interference-Based Forgetting in Spatial Navigation: Latent Behavioral Signatures and the Impact of Synaptic Plasticity Disruption.

"Oblit basat en la interferència en la navegació espacial dirigida a recompensas: empremtes latents de comportament i l'impacte de la disrupció de la plasticitat sinàptica"

Doctoral thesis presented by:

Paula Peixoto Moledo

This work was developed at the Pathogenesis of Autoimmune Neuronal Disorders laboratory in the Institut d'Investigacions Biomèdiques August Pi i Sunyer (IDIBAPS) and the Neurobiología Funcional y de Sistemas laboratory in the Instituto Cajal- Consejo Superior de Investigaciones Científicas(CSIC).

Thesis Directors:
Pablo E. Jercog Josep Dalmau i Obrador

Academic Tutor:
Josep Dalmau i Obrador

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“Science and everyday life cannot
and should not be separated.”
Rosalind Franklin (1920–1958)

“Na cidade cren que fora dela non hai máis
ca paisaxe, patacas e leite;
ignoran que tamén existe unha cultura
noble, antiga e insubornable”
Castelao (1886 -1950)

Á miña familia, que na terriña
esperan sempre a que a súa nena volva.

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List of abbreviations

- **AMPA / AMPAR** – α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor
- **AP** – Anterior–Posterior
- **CEEA** – Comitè Ètic d'Experimentació Animal
- **CI** – Confidence Interval
- **CO_2** – Carbon Dioxide
- **DV** – Dorso–Ventral
- **DPPX** – Dipeptidyl-peptidase-like protein 6
- **EM** – Expectation–Maximization
- **GLM** – Generalized Linear Model
- **GLM–HMM** – Generalized Linear Model–Hidden Markov Model
- **HBSS** – Hank's Balanced Salt Solution
- **HMM** – Hidden Markov Model
- **IgG** – Immunoglobulin G
- **ITI** – Inter-Trial Interval
- **LTP** – Long-Term Potentiation
- **LGI1** – Leucine-rich Glioma-Inactivated 1
- **MI** – Memory Index
- **ML** – Medio–Lateral

- **mGluR5** – Metabotropic Glutamate Receptor 5
- **NMDAR** – N-Methyl-D-Aspartate Receptor
- **NR** – Non-Rewarded
- **OCT** – Optimal Cutting Temperature compound
- **PBS** – Phosphate-Buffered Saline
- **PFA** – Paraformaldehyde
- **POMDP** – Partially Observable Markov Decision Process
- **rhuMab** – Recombinant Human Monoclonal Antibody
- **SEM** – Standard Error of the Mean

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Abstract

Memory is a core adaptive function that enables animals to anticipate outcomes and guide behavior beyond immediate sensory cues. In this thesis, we use the 8-port maze (Morales et al., 2020) as a paradigm to disentangle learning from memory, to track the coexistence of multiple temporal traces, and to dissect the mechanisms of memory interference and strategy dynamics in recall.

In Chapter 1, we demonstrate that learning performance during training does not predict recall strength. Using the Memory Index (MI), we establish a robust and training-independent measure of memory, showing that animals recall memories from both 2h and 24h delays within the same session, thus providing behavioral evidence that multiple temporal traces coexist rather than overwrite one another.

In Chapter 2, we uncover a spatial distance-dependent pattern of interference: memories from nearby rewarded locations reinforce one another, whereas distant ones interfere and weaken recall. A generative model partially reproduced this effect, suggesting that spatial relationships between memories contribute to, but do not fully account for, their stability.

In Chapter 3, we apply a GLM-HMM to identify the latent strategies underlying recall. Four distinct strategies emerge—goal-directed recall (2h), outdated recall (24h), memory-avoidance, and lapses/disengagement. These strategies are expressed dynamically within and across sessions, explaining variability in behavior beyond average performance. Importantly, the spatial patterns of latent strategies show that interference is not a single, uniform effect, but becomes evident through the use of particular states, especially those guided by memory.

In Chapter 4, we show that anti-NMDAR antibody infusion produces a reorganization of strategy dynamics: recall behavior becomes more dominated by 2h strategies at the expense of flexibility, and the spatial interference pattern of memory-based strategies is inverted.

In Chapter 5, we find that anti-LGI1 antibodies also disrupt recall dynamics but in a qualitatively different manner: increases in 2h recall are more modest, while 24h recall and avoidance strategies persist in altered proportions, and interference spatial modulation shows non-monotonic patterns. These results reveal distinct disruptions of memory organization across autoimmune encephalitis models.

Taken together, this thesis supports—based on behavioral, computational, and experimental evidence—that memory recall is not a fixed process, but can emerge from the flexible combination of multiple latent strategies. The 8-port maze, combined with latent-state modeling, offers a useful framework to bridge synaptic dysfunction and behavioral memory organization, while also opening the door to correlating behavior with neuronal imaging data.

Keywords: Neuroscience, Computational Neuroscience, Memory, Machine Learning, Autoantibodies ,memory interference,latent strategies, GLM-HMM.

Resum

La memòria és una funció adaptativa fonamental que permet als animals anticipar resultats i guiar el comportament més enllà dels senyals sensorials immediats. En aquesta tesi utilitzem el 8-port maze (Morales et al., 2020) com a paradigma per dissociar aprenentatge i memòria, seguir la coexistència de múltiples traços temporals i disseccionar els mecanismes d'interferència i la dinàmica d'estratègies durant el record.

Al Capítol 1, demostrem que el rendiment durant l'entrenament no prediu la força del record. Mitjançant l'Índex de Memòria (MI) establim una mesura robusta i independent de l'entrenament, mostrant que els animals recorden tant després de 2 hores com de 24 hores dins de la mateixa sessió. Això aporta evidència conductual que múltiples traços temporals poden coexistir en lloc de sobreescriure's mútuament.

Al Capítol 2, revelem un patró d'interferència dependent de la distància espacial: els records de localitzacions recompensades properes es reforcen mútuament, mentre que els llunyans interfereixen i debiliten el record. Un model generatiu reprodueix parcialment aquest efecte, suggerint que les relacions espacials entre records hi contribueixen, però no en donen una explicació completa.

Al Capítol 3, apliquem un GLM-HMM per identificar les estratègies latents que subjauen al record. Emergeixen quatre estratègies distintes: record dirigit (2h), record obsolet (24h), evitació de memòria i lapses/desconnexió. Aquestes estratègies s'expressen dinàmicament dins i entre sessions, explicant la variabilitat del comportament més enllà del rendiment mitjà. És important destacar que els patrons espacials de les estratègies latents mostren que la interferència no és un efecte únic i uniforme, sinó que es manifesta a través de l'ús d'estats concrets, especialment aquells guiats per la memòria.

Al Capítol 4, mostrem que la infusió d'anticossos anti-NMDAR produeix una reorganització de les dinàmiques d'estratègies: el comportament de record queda més dominat per les estratègies de 2h, a costa de la flexibilitat, i el patró d'interferència espacial de les estratègies basades en la memòria s'inverteix.

Al Capítol 5, trobem que els anticossos anti-LGI1 també alteren les dinàmiques de record, però d'una manera qualitativament diferent: l'augment en l'ús del record de 2h és més moderat, mentre que el record de 24h i les estratègies d'evitació persisteixen en proporcions alterades, i la modulació espacial de la interferència mostra patrons no monòtons. Aquests resultats revelen disrupcions distintes en l'organització de la memòria segons el model d'encefalitis autoimmune.

En conjunt, aquesta tesi dona suport —a partir d'evidències conductuals, computacionals i experimentals— a l'idea que el record no és un procés fix, sinó que pot sorgir de la

combinació flexible de múltiples estratègies latents. El 8-port task, combinat amb el modelatge d'estats latents, ofereix un marc útil per connectar la disfunció sinàptica amb l'organització de la memòria conductual, alhora que obre la porta a correlacionar el comportament amb les dades d'imatge neuronal.

Paraules clau: Neurociències, Neurociència computacional, Memòria, Aprenentatge automàtic , Autoanticossos,interferència de la memòria, estratègies latents, GLM-HMM, hipocamp.

BACKGROUND

From Adaptive Memory to Experimental Models

Memory plays a fundamental role in survival by enabling organisms to use past experiences to guide future behavior. By recalling where food was found or where a predator was last encountered, animals can anticipate outcomes and act in advance, reducing reliance on immediate sensory cues or slow trial-and-error learning. In this sense, memory provides an adaptive advantage: it supports anticipation, planning, and flexible adjustment to changing environments.

From a computational perspective, memory does more than store past events: it provides a compressed representation of experience that the brain can reuse to make decisions more efficiently. Some computational models use internal or hidden states to link sensory inputs with actions, offering a way to describe behavior. Instead of recalculating action values from raw input each time, these internal models draw on prior knowledge to simulate likely outcomes — a process that reflects model-based rather than model-free inference (Gershman & Daw, 2017). By reducing the cost of evaluating options, memory enables faster, more accurate, and more adaptive responses in dynamic and uncertain contexts.

From a biological perspective, memory is therefore not just a record of the past, but a tool to bridge time with behavior. It enables goal-directed actions over seconds, hours, or even days, and thus has been a central focus of neuroscience — from synaptic mechanisms to systems-level computations.

How can such memory based computations be studied in the laboratory? First of all, researchers need species whose natural behavior strongly depends on memory, but that can also be observed and manipulated under controlled conditions. Rodents provide this balance exceptionally well. In the wild, they must remember food locations, predator territories, and safe shelters. In the lab, these same skills can be probed in tasks that mimic natural demands while allowing precise experimental control (National Research Council, 2011). For this reason, rodents have become a powerful model for linking behavior to the neural processes that underlie memory.

Within the model-based inference framework, episodic memory provides a particularly rich source of information for constructing and updating these internal models. Episodic memory refers to the ability to encode and retrieve detailed representations of specific past experiences, including their spatial and temporal context (Tulving, 1972). By reactivating stored episodes, the brain can simulate possible future scenarios and anticipate their consequences without relying exclusively on ongoing sensory input. This process relies heavily on the hippocampal–entorhinal circuit, where episodic traces are integrated with current context to generate predictions that guide behavior (Hassabis & Maguire, 2007; Stachenfeld, Botvinick & Gershman, 2017). In doing so, episodic memory supports efficient

predictive coding by providing a structured data source that the brain can exploit to generalize across situations, reduce computational demands, and improve the accuracy of decision-making.

To understand how these computations are implemented biologically, we turn to the hippocampus. The hippocampus stands out as a central hub for spatial memory. Its role was first revealed through human lesion studies. The most striking case is that of patient H.M., who lost the ability to form new episodic memories after bilateral removal of his medial temporal lobes to treat epilepsy (Scoville & Milner, 1957; Corkin, 2002). While his general cognition and older memories remained intact, he could no longer remember daily events — demonstrating that the hippocampus is crucial for forming long-term declarative memories (Squire & Zola-Morgan, 1991).

Yet clinical cases alone could not explain how the hippocampus encodes information. This required direct recordings of neural activity, which became possible in animal models. A landmark discovery in this context was the identification of hippocampal neurons that signal the animal's position in space. In their pioneering work with freely moving rats, John O'Keefe and Jonathan Dostrovsky (1971) recorded cells that fired reliably when the animal occupied a specific location in its environment. These location-tuned neurons became known as *place cells*, and the area in which each neuron was most active is referred to as its *place field*. The stability of this spatial tuning over time gives rise to the idea of place cells as the instrumental elements to generate memory . Together, populations of place cells have been proposed to be forming an internal "map" of the environment, a representation that supports navigation and memory for places (O'Keefe & Nadel, 1978). Importantly, place fields are not fixed: they remap with changes in environment, task demands, or reward locations (Muller & Kubie, 1987; Lever et al., 2002; Lee et al., 2006). Thus, hippocampal codes are flexible, embedding spatial information within a broader mnemonic framework that includes goals, experiences, and context. Notably, the long-term stability of place cells has been questioned by the observation of place cell turnover across days (Ziv et al., 2013), further underscoring the adaptive and dynamic nature of hippocampal representations.

To complete the picture it is important to add that during rest and sleep, the hippocampus often "replays" recent sequences in compressed time, reflecting both past trajectories and potential future ones (Foster & Wilson, 2006; Pfeiffer & Foster, 2013; Ólafsdóttir et al., 2018). This offline activity is thought to strengthen and reorganize memory traces, supporting consolidation and planning. Such persistence beyond immediate behavior raises a fundamental question: how can hippocampal representations be stabilized over days or weeks without fading or being overwritten?

Synaptic Plasticity and the Cellular Basis of Memory

If hippocampal maps can change with goals, context, and experience, how can such fleeting patterns of activity leave a trace that lasts for days or even weeks — i.e., a change that outlasts the immediate neural dynamics? A memory cannot depend only on the brief bursts of neural firing that occur during behavior. It must involve more stable changes in how neurons communicate with one another. This line of reasoning naturally points to the synapse as the site where experience leaves its imprint.

One mechanism that allows such persistence is synaptic plasticity — the ability of synaptic connections to become stronger or weaker depending on patterns of activity. In this way, groups of neurons that were active together during an event, sometimes referred to as engrams, can later be reactivated more easily, providing a cellular basis for recall. The best-studied example is long-term potentiation (LTP), first described in the hippocampus by Bliss and Lømo in 1973. LTP refers to a lasting increase in synaptic efficacy that arises when presynaptic input coincides with strong postsynaptic depolarization, and it has become a cornerstone model for how experience can leave a durable trace in the brain.

But what makes this process possible at the molecular level? In hippocampal circuits, a central role is played by N-Methyl-D-Aspartate (NMDA) receptors, which act as coincidence detectors. These receptors open only when presynaptic activity and postsynaptic depolarization converge, allowing calcium to enter the cell (Collingridge et al., 1983; Malenka & Nicoll, 1999). The calcium signal initiates intracellular cascades that strengthen the synapse, increasing the likelihood that the same neurons will fire together in the future. In line with this model, blocking NMDA receptors affects hippocampus-dependent learning and memory in rodents (Tsien et al., 1996; Whitlock et al., 2006).

What happens when this molecular machinery is disrupted? A striking example comes from anti-NMDAR encephalitis, a rare autoimmune disorder in which antibodies target and disable NMDA receptors (Dalmau et al., 2007; Dalmau et al., 2011). Among other cognitive symptoms, patients often exhibit profound impairments in forming new declarative memories. The severity of the outcome level of the patient's treatment correlates with antibody titers (Gresa-Arribas et al., 2014) and with functional and structural alterations of hippocampal connectivity (Finke et al., 2013). Because symptoms often improve as antibodies decline, the deficits are thought to arise from reversible receptor blockade rather than permanent damage, such as cell death. Consistently, rodent models in which NMDA receptor function is disrupted — either by passive transfer of patient antibodies (Planagumà et al., 2015), genetic manipulation (Tsien et al., 1996), or pharmacological blockade (Morris et al., 1986) — show comparable impairments in hippocampus-dependent tasks.

Yet NMDA receptors are only one part of a much broader system. LTP depends on a host of synaptic proteins that coordinate the initiation, propagation, and maintenance of

strengthened connections. Several of these have been identified as targets in other autoimmune encephalitides. The second most common, anti-LGI1 encephalitis, affects Leucine-rich Glioma-Inactivated 1 protein (LGI1), a synaptic organizer that stabilizes excitatory connections in hippocampal and temporal lobe circuits (Fukata et al., 2006; Lai et al., 2010). When antibodies disrupt LGI1, patients experience severe memory loss and temporal-lobe abnormalities, and animal models receiving patient-derived antibodies likewise show impaired hippocampal LTP and memory (Petit-Pedrol et al., 2018).

Although anti-NMDAR and anti-LGI1 encephalitis act on different molecular targets — one interfering with the initiation of plasticity, the other with its maintenance — both produce hippocampus-dependent memory deficits in humans and rodents. This convergence underscores a central principle: the stability of memory depends not only on the ability to form strong synaptic connections, but also on the capacity to preserve them over time, and translate into behavioral expression across different timescales. Later in this thesis, we will return to these models to examine how such molecular disruptions manifest in behavior, linking cellular mechanisms to measurable memory strength. But molecular mechanisms alone do not explain how memories are organized across longer timescales, or how they are retrieved under changing conditions. To understand these broader questions, we must move from the level of synapses to the level of systems neuroscience and computational models.

Theories of Memory Organization

The molecular and synaptic mechanisms described above form a much broader and complex biological machinery that allows experiences to be stored in the brain. However, on their own they do not explain how memories are organized across time, how they are updated with new information, or how they are accessed under varying conditions. To address these broader questions, researchers have developed systems-level theories that describe the architecture and dynamics of long-term memory at the level of brain networks and cognitive processes. Two particularly influential frameworks are Multiple Trace Theory (MTT) and Interference Theory (IT), each offering a distinct perspective on how memory traces are stored, maintained, and expressed.

To appreciate the contribution of MTT, it is useful to first recall the classical standard consolidation model (Squire & Alvarez, 1995). From this perspective, the hippocampus plays a role mainly in the early stages of episodic memory: it acts as a short-term registrar that grants access to recent experiences. As consolidation proceeds, memories are gradually relocated to the neocortex for long-term storage, so that later retrieval no longer requires hippocampal involvement.

But is this really the case? MTT (Nadel & Moscovitch, 1997; Winocur & Moscovitch, 2011) challenged this classical view by arguing that the hippocampus remains persistently involved in the storage and retrieval of episodic memories over the long term. Each time a memory is

retrieved — whether spontaneously during offline processing (for example during sleep) or actively during behavior — a new "trace" is created. This trace partially overlaps with existing ones but incorporates new contextual or temporal details. Over repeated retrievals, multiple distributed traces accumulate within the hippocampal–neocortical system. The redundancy afforded by these traces makes memories more resistant to degradation, while also allowing their gradual modification in light of new experiences. This framework accounts for findings showing that even remote episodic memories can depend on the hippocampus and that retrieval often strengthens or updates memories rather than merely replaying them (Moscovitch et al., 2005; Sekeres et al., 2018). MTT thus emphasizes the cumulative and dynamic nature of memory storage, with the hippocampus serving not as a transient staging area, but as a long-term component of the episodic memory system repository that actively participates in memory updating.

In contrast, IT focuses on the competitive interactions between memories, rather than their accumulation. Originating in early experimental psychology (Barnes and Underwood, 1959; Postman & Underwood, 1973) and later refined by cognitive and computational models (Norman & O'Reilly, 2003; Norman, Newman, & Detre, 2007), this theory posits that forgetting and retrieval failure often arise not from passive decay of memory traces, but from the presence of other memories that share overlapping features — whether in content, context, or retrieval cues. When multiple traces are sufficiently similar, they compete for retrieval, leading to partial suppression, distortion, or even complete failure.

But what forms can this interference take? It may be *proactive*, when older memories disrupt the retrieval of newer ones; *retroactive*, when new experiences overwrite or obscure older traces; or *retrieval-induced forgetting*, in which retrieving one memory actively suppresses others (Anderson et al., 1994). This framework highlights that memory retrieval is a selective and competitive process, constrained not only by the strength of individual traces but also by their similarity and overlap with others.

Recent computational models have begun to integrate elements of both perspectives. Within the hippocampus, two complementary processes — pattern separation and pattern completion — help manage interference. For example, competitive retrieval models and probabilistic sampling frameworks (Norman et al., 2007; Polyn et al., 2009; Bornstein & Norman, 2017) formalize the idea that the brain must balance the benefits of multiple stored traces (as in MTT) with the costs of interference when those traces overlap. Importantly, interference is not always detrimental: in some cases, overlap between memories can actually facilitate retrieval. Repeated or convergent experiences can reinforce a shared representation, making it more robust.

How, then, does the brain navigate this tension between competition and cooperation? One proposal is that hippocampal dynamics are regulated by the balance between pattern separation and pattern completion. Pattern separation refers to the process by which similar

inputs are orthogonalized into distinct neural representations, reducing overlap and minimizing interference; it is thought to be mediated primarily by the dentate gyrus (Leutgeb et al., 2007; Yassa & Stark, 2011). Pattern completion, in contrast, allows a partial or degraded cue to retrieve the full stored representation, a process attributed to the recurrent connectivity of CA3. Together, these operations enable the hippocampus to flexibly discriminate between similar experiences while also generalizing across them when appropriate.

When this balance is disrupted — whether by excessive overlap, insufficient balancing effect of inhibitory inputs, or degraded synaptic plasticity — the result can be increased interference and behavioral confusion. Disentangling how pattern separation and completion operate in behavior requires experimental paradigms that can (i) generate controlled levels of overlap between memory traces, (ii) separate memory encoding from retrieval, so that interference can be measured without contamination from new learning, and (iii) provide sufficient data to resolve these dynamics with statistical and computational tools.

Crucially, such paradigms must also preserve ecological validity. That is, they should mimic the complexity and natural conditions in which memory is used, engaging the navigation and decision-making processes critical for survival. Studying these processes in practice therefore requires tasks that can manipulate memory overlap while still preserving naturalistic navigation. These combined demands naturally point toward freely moving navigation tasks that offer both precise experimental control and the richness of unconstrained exploration. To bridge these theoretical considerations with empirical testing, we now turn to such behavioral paradigms. In our laboratory, this rationale previously motivated the design of a high-throughput navigation paradigm that systematically manipulates spatial overlap across days while separating new learning from memory retrieval.

Ecological Behavioral Paradigms for Studying Memory

Because of the requirements just listed, freely moving navigation tasks are especially well suited for studying memory in a way that captures the challenges animals face in natural environments. In such paradigms, animals can draw on the full repertoire of sensory, motor, and cognitive processes that shape memory formation and retrieval. They integrate visual landmarks, vestibular cues from head and body motion, proprioceptive feedback from limb position, and olfactory traces: cues that are crucial for hippocampal spatial representations and therefore for memory-guided navigation, which combined with internal representations, guide their choices. This sensory–motor integration engages key brain structures such as the hippocampus and entorhinal cortex, where spatial coding phenomena like place fields, grid fields, and theta phase precession naturally emerge (O'Keefe & Dostrovsky, 1971; Hafting et al., 2005; Buzsáki & Moser, 2013).

But what happens if these natural cues are restricted? Head-fixed paradigms, while offering exquisite experimental control and compatibility with high-resolution imaging techniques, impose severe limitations on movement and remove access to critical self-motion signals. These constraints can alter how hippocampal and cortical networks represent space and may suppress behaviors that, in the wild, are tightly linked to navigation and memory use (Aronov & Tank ,2014; Ravassard et al., 2013).

From the perspective of studying long-term memory, such differences are far from trivial. The ecological function of memory is to support flexible and adaptive behavior in changing environments—allowing animals to generalize across contexts when appropriate and discriminate between similar episodes when necessary—processes supported by hippocampal pattern completion and pattern separation. Experimental frameworks that impose strong restrictions on behavior can limit the engagement of the mechanisms most critical to memory's ecological function—navigating, exploring, and choosing under uncertainty (Allen & Fortin , 2013)

This point becomes especially important when probing the subtle cognitive effects of molecular or circuit-level perturbations. Differences between healthy and impaired function may be most visible in complex, self-guided behavior rather than in rigidly structured tasks. For this reason, the work presented in this thesis relies on the 8-port maze, a behavioral task developed in our lab. The maze was designed to combine the ecological richness of free navigation with the experimental control and high-throughput data collection needed to generate structured memory processes under reproducible conditions (Morales et al., 2020).

The 8-port Maze as a Framework for Memory Research

To study these questions in practice, we use the 8-port maze, a freely navigating task designed to capture the dynamics of memory recall under controlled conditions (Morales et al., 2020). In this task, eight nose-poke ports are arranged in a hexadecagonal arena, and on each day one port is designated as the rewarded location. Mice explore the maze in a trial-based manner, first learning the target port during training and later recalling it after a two-hour delay. A critical feature of the design is an initial "no-feedback" interval at the start of each session, during which pokes produce no reward. This prevents rapid re-learning to avoid contamination of pure recall, ensuring that pokes cannot provide information about the new goal location, thereby isolating recall from new learning . This naturally leads to the first question addressed in this thesis: **does training performance truly predict subsequent recall, or are encoding and retrieval partly independent processes? (*Objective 1*)**

The task also naturally creates conditions for memory interference. This leads to a second key question: **how do memories of different temporal ages coexist, and how does their spatial proximity shape retrieval? (*Objective 2*).** Because the rewarded location changes daily but is always drawn from the same set of eight ports, new memories can overlap with

those from the previous day — sometimes reinforcing one another, and sometimes competing, depending on their spatial distance, temporal distance, or similarity of contexts. This overlap mirrors the challenges described by interference theories and allows us to probe how the hippocampus resolves competition between past and present traces across different timescales (e.g. 2h vs. 24h), spatial distances, or experimental manipulations. Because the port changes daily in an unpredictable manner, mice cannot rely on habitual or model-free strategies and must draw on stored spatial memories.

Another strength of the 8-port maze is its high-throughput structure. Animals can self-initiate many short attempts to solve the task (also called trials) within a session, yielding large datasets without additional handling or stress. This wealth of data makes the task well suited for computational approaches that look beyond simple success rates, enabling fine-grained analyses of memory-guided choices and latent behavioral strategies.

To summarize recall performance, we use the Memory Index (MI), a simple measure of directional bias toward the rewarded port computed during the no-feedback interval to capture recall before new learning occurs (Morales et al., 2020). High MI values indicate accurate recall, values near zero reflect random exploration, and negative values suggest systematic avoidance. While the MI provides a clear first measure of memory strength, its simplicity also highlights the advantage of the 8-port maze: with many trials per session, we can go beyond a single summary score to ask how animals use memory on a choice-by-choice basis.

Our previous work (Morales et al 2020) has shown that mice often display measurable recall for both the port rewarded earlier the same day (2h delay) and the port rewarded on the previous day (24h delay).The coexistence of these memory traces aligns with theories proposing that multiple memories can compete or interact during retrieval. However, this observation also raises a key question: is recalling multiple memories an optimal solution to our task, or does it sometimes introduce competition that slows reward acquisition compared to random exploration? The analyses presented in this thesis aim to disentangle these possibilities.

Computational Models of Memory-Guided Behavior

The high-throughput nature of the 8-port maze makes it possible to move beyond simple summary measures and ask in more detail how animals use memory to guide their choices. But how can we test whether the biases we observe — toward the current day's memory or the previous day's — are truly adaptive, or simply artifacts of the task?

To address this, we first turned to a simple computational model. We formalized the maze as a partially observable Markov decision process (POMDP) and implemented a minimal reinforcement-learning agent with three basic strategies: pure exploration with no memory, recall of the current day's location, and recall of the previous day's location. A mixture of the

last two captured partial recall of both memories implemented as a stochastic mixture policy, selecting actions according to a weighted combination of the two memories and foraging. This deliberately simple setup was meant to answer a basic question: does the tendency to use both the current and previous day's memory truly influence behavior in a meaningful way, or is it merely a by-product of the task that provides no real benefit to the animals?

Simulations provided a clear prediction. Accurate recall of the current location dramatically reduced search time compared to random exploration (see Figure 1). In contrast, relying on the previous day's location impaired performance, sometimes even worse than chance. Interestingly, partial recall of both memories was still better than random search, though less efficient than perfect recall of the current target. These results illustrate how the structure of the 8-port maze can promote either optimal or suboptimal strategies depending on which memories are engaged, and why quantifying the cost of outdated recall is important. These results suggest that reliance on multiple memories could reflect, at least in part, a natural tendency for recently stored traces to be reinstated during retrieval.

Yet, animals are unlikely to be restricted to a handful of predefined strategies. Instead, their choices may reflect a richer and more flexible repertoire. This raises a third key question: **is recall governed by a single process, or does it emerge from multiple strategies that animals flexibly switch between? (Objective 3)**. To address this, we turned to latent-state modeling, specifically the generalized linear model–hidden Markov model (GLM-HMM) (Ashwood et al., 2022). In this framework, each poke is treated as arising from a hidden state corresponding to a distinct strategy or behavioral mode, such as accurate recall, reliance on yesterday's memory, or exploratory search. States evolve according to a Markov process, allowing us to detect whether mice persist in a strategy, switch rapidly, or transition in stereotyped ways. Crucially, this analysis operates at the level of individual pokes rather than session averages, revealing dynamics that remain invisible to aggregate measures like the Memory Index.

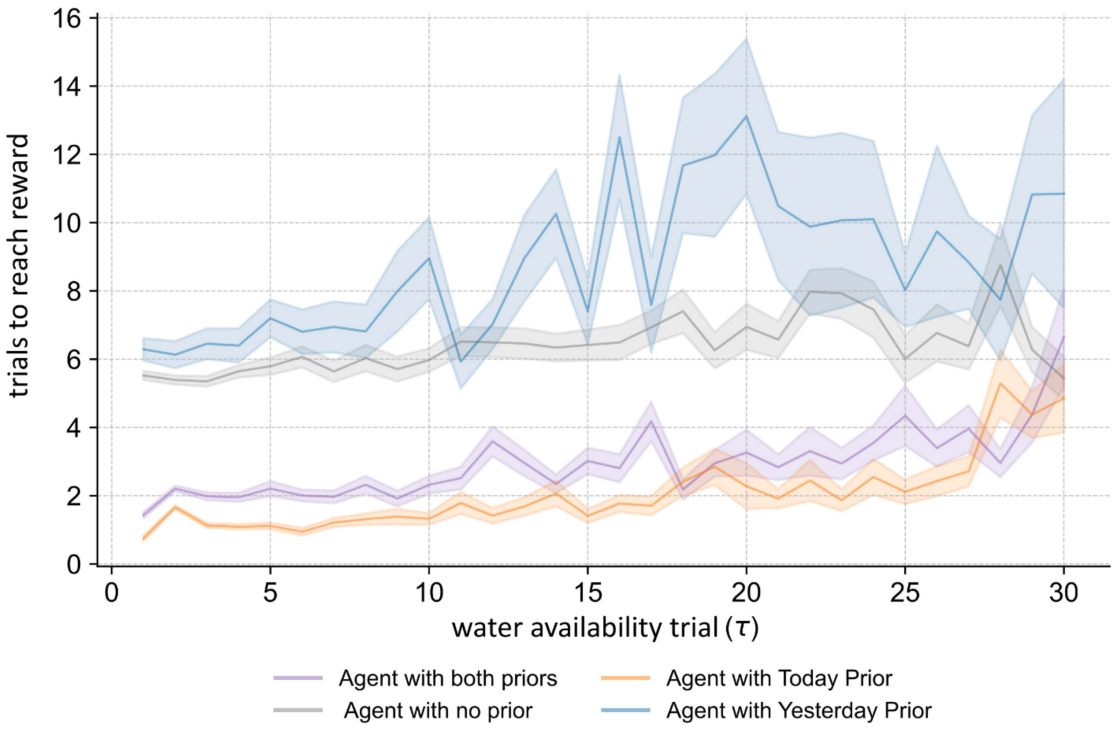

Figure 1. POMDP simulations reveal how prior knowledge shapes reward acquisition efficiency in the 8-port maze. Results from a Partially Observable Markov Decision Process (POMDP) model of agent behavior. The plot shows the mean number of trials required to obtain the reward as a function of the trial in which water became available (x-axis: water availability trial τ; y-axis: trials to reward). Shaded areas denote SEM across simulations (n=1500). The grey line shows performance of an agent without prior information, which relies on random exploration (~5 trials to reward). The orange line shows agents biased by the current day's reward location, which markedly improved performance (~2 trials). The blue line corresponds to agents biased by the previous day's reward location, which impaired performance relative to random exploration (~7 trials). The purple line shows agents combining both priors, yielding intermediate performance: more efficient than random exploration but less so than relying only on the current day's prior.

This approach also provides a powerful bridge to molecular mechanisms. It allows us to test whether disruptions of synaptic plasticity — such as those caused by antibodies against NMDA receptors or LGI1 — alter not only the strength of memory but also the flexibility of the strategies animals use. Since NMDA receptor–dependent plasticity and LGI1-mediated activity contribute to different aspects of hippocampal function — NMDA receptors enabling induction of plasticity, and LGI1 maintaining synaptic stability, they may leave distinct signatures in strategy dynamics. By combining the ecological richness of the 8-port maze with latent-state modeling, we can detect such changes with high temporal and strategic resolution.

In this thesis, we will use this framework to show that recall in the 8-port maze cannot be reduced to simple learning and recall performances. Instead, it reflects a dynamic

competition and cooperation between multiple memory traces, expressed through a repertoire of strategies that vary with memory age, spatial similarity, and molecular integrity. Finally, **by using antibody models of autoimmune encephalitis, this thesis asks how disruptions of two distinct synaptic mechanisms — NMDA receptor–dependent signaling and LGI1-mediated stabilization — alter the expression, flexibility, and spatial organization of memory-guided behavior (*Objectives 4 and 5*)**. The following chapters expand on these findings in detail, linking behavioral dynamics to synaptic mechanisms in health and disease.

OBJECTIVES

The main objective of this thesis is to define the rules that govern the acquisition and retrieval of spatial memories in the 8-port maze, with a focus on their temporal dynamics, their interactions during recall, and how these processes are modified under experimental perturbations of synaptic plasticity.

To accomplish the main objective, the specific subobjectives were as follows:

1. Understand how measures of task acquisition relate to subsequent recall strength in the 8-port maze.
2. To investigate how memories of different temporal ages coexist, and how spatial proximity between targets shapes their retrieval.
3. To characterize the diversity of recall strategies, their temporal dynamics, and how animals switch between them during memory-guided navigation.
4. To test how antibody-mediated disruption of NMDA receptor–dependent synaptic mechanisms affects the expression, flexibility, and spatial organization of recall-guided behavior.
5. To determine how disruption of LGI1-mediated synaptic stabilization alters the expression, flexibility, and spatial organization of memory-guided behavior.

METHODS

Author's contributions

The experiments were designed by Paula Peixoto Moledo, Pablo E. Jercog and Josep Dalmau. The behavioral experiments were conducted by Paula Peixoto Moledo. The surgeries for cannula implantation were performed by Pablo E. Jercog with the assistance of Paula Peixoto Moledo. Antibody infusion experiments and histology controls were conducted by Paula Peixoto Moledo. Figure 1 from the background section is an output of a reinforcement learning model designed by Gonzalo Hernandez and Yashar Ahmadian. Quantification of the behavioral data was done by Paula Peixoto Moledo. Model design was carried out by Yashar Ahmadian and Paula Peixoto Moledo,and was an adaptation of the GLM-HMM from the SSM Linderman's Lab library. Data analysis and figure preparation were carried out by Paula Peixoto Moledo, while the interpretation of the data was a collaborative work among Paula Peixoto Moledo, Pablo E. Jercog, and Yashar Ahmadian. Contributions for the refinement and corrections of this manuscript were performed by Pablo E. Jercog, Yashar Ahmadian, Myrna Rosenfield and Josep Dalmau.

Publications

The results from this thesis have not yet been published. Partial results from chapters 1 and 2 have been presented in FENS 2022 and IBRO 2023.

1. Experimental procedures and set-up.

1.1. Behavioral Task: The 8-port maze task

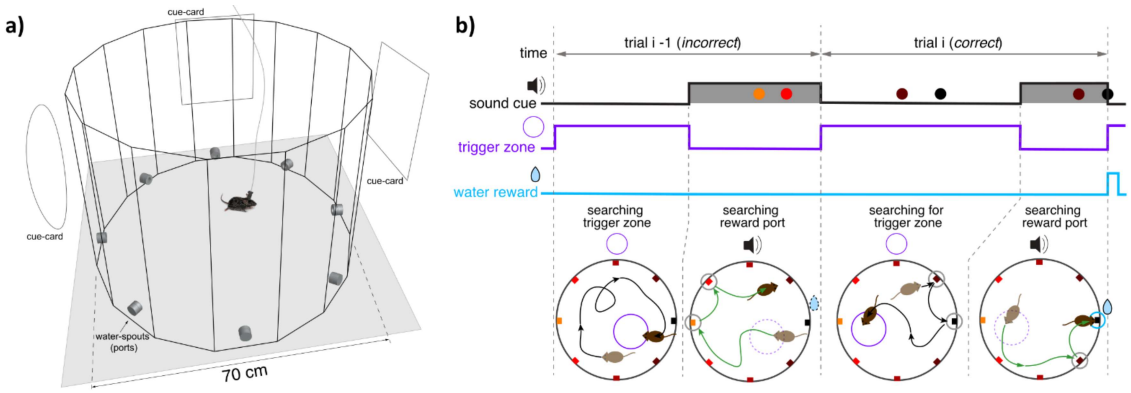

Figure 2. The 8-port maze behavioral task. (a) Schematic of the open-field arena (70 cm diameter) built from transparent acrylic walls and equipped with eight equidistant water-spout ports. Cue cards on the walls served as distal visual references, and the maze was housed within a sound-attenuating chamber. (b) Trial structure. Each trial began when the mouse entered a randomly located invisible trigger zone (violet), which initiated a sound cue and opened a 4-s reward window during which one designated port delivered water. If the mouse reached the correct port, the tone stopped and the trial was scored as correct; otherwise, the trial was incorrect. Example trajectories are shown for incorrect and correct trials. The rewarded port was fixed within each daily session but changed randomly across days. Adapted from Morales et al., 2020 (Front. Behav. Neurosci. 14:64).

1.1.1. Overview of the task.

The 8-port maze is a high-throughput, automated spatial memory task designed to assess long-term memory formation in rodents (Morales et al., 2020) (Figure 2, adapted from Morales et al., 2020). The behavioral box is an octagonal open-field arena equipped with a nose-poke port in each of its eight walls. The arena is transparent, allowing distal visual cues to be placed outside the structure to facilitate navigation. The automated design minimizes human intervention and enables the acquisition of large datasets suitable for robust statistical analyses.

Once a mouse enters the arena, it undergoes a brief acclimation period (approximately 40 sec) before testing begins. After this period, the animal self-initiates a trial by navigating until entering into an invisible trigger zone, as illustrated in the circular leftmost panel of Figure 2b. Entering this zone activates an auditory cue and opens a response window lasting a maximum of 4 seconds, during which the mouse can explore and interact with any of the eight ports. During this time, animals are free to poke multiple ports without penalty, as shown in the second circular panel of Figure 2b. Whether these pokes result in reward or not depends on the behavioral phase being tested (see Section 1.1.2).

When the auditory cue ends, the trial terminates, and the animal can initiate a new one by locating a newly repositioned trigger zone (third circular panel of Figure 2b). The trigger zone is relocated randomly by the computer control on each trial, ensuring that the mouse starts from different positions, promoting full arena exploration and preventing stereotyped trajectories. Each animal performs from ~30 and up to ~120 self-initiated trials per experimental day.

1.1.2. Rewarded and unrewarded periods

Each session includes two consecutive periods that differ in whether rewards are available or not. At the beginning of the session, animals experience an unrewarded period lasting 0–3 minutes, during which no water reward or feedback is provided. During this time, the auditory cue defining the 4-second response window, remains active for its full duration regardless of the animal's behavior or choices, since correct pokes do not trigger water delivery or tone cessation. The duration of the unrewarded phase is randomized across sessions to prevent temporal expectation and increase uncertainty. During this period, animals typically complete ~3–20 trials under unrewarded conditions. These trials serve as controlled probes of memory recall, allowing assessment of how strongly animals remember the rewarded port from previous learning, independent of new learning or reinforcement of previously learned rewarded locations.

Following the uncertainty period, animals enter the rewarded period, in which, unlike during the unrewarded phase, poking the correct port stops the auditory cue and immediately triggers water delivery. This event shortens the sound cue time from the maximum 4 seconds to the actual time taken by the animal to reach the correct port. Through repeated trials, animals learn to identify and revisit the rewarded port for that day, improving their performance as the session progresses, and presumably reinforcing the memory of the rewarded port.

1.1.3. Training and recall sessions.

Each experimental day consists of two sessions separated by a 2-hour delay: a training session (approximately 30 minutes) and a recall session (approximately 10 minutes). Both sessions include an initial unrewarded or uncertainty period followed by a rewarded period, following the structure described in Section 1.1.2.

During the *training session*, animals begin with an initial uncertainty period in which they often poke more frequently at ports that had been rewarded on previous days, providing a behavioral readout of residual memory from prior experiences. Once this unrewarded period ends and water becomes available, animals learn to identify which of the eight ports delivers water rewards for that day. The rewarded port remains constant throughout the session, allowing animals to gradually improve their performance through repeated experience. Each

training session lasts approximately 30 minutes, a duration chosen to maximize data acquisition while ensuring stable encoding of the correct port. Across training, mice typically complete 60 to 120 trials, depending on their motivation and exploration speed.

During the recall session, conducted 2 hours after training, animals are tested on their memory of the previously rewarded port during the training session. As in training, the session begins with an unrewarded followed by a rewarded period in which water is again available. The rewarded port is the same as in the preceding training session, allowing animals to use previously acquired information to guide their choices. Each recall session lasts approximately 10 minutes, during which mice typically complete 20 to 40 trials, depending on their motivation and activity level. This session provides a direct measure of recent memory, reflecting how well the animal remembers information from the same day.

Memory strength in both training and recall sessions is quantified using the Memory Index (MI) defined by Morales et al. (2020) and described in greater detail in Section 2.2 of these methods.

Across experimental days, the identity of the rewarded port changes pseudorandomly following a soft "no-repeat" rule, to reduce the likelihood of consecutive-day repetitions and prevents the formation of strong spatial biases.

1.1.4. 8-port maze strategy shaping.

Animals trained in the 8-port maze task undergo a structured pre-training process that shapes their behavior:

- *Habituation:*

During 3-4 days animals are exposed to the experimenter, the experimental room, and the behavioral box to reduce their stress levels.

- *Pre-training:*

The pre-training or shaping phase consists of several stages that teach and guide the animal through the associations and rules needed to solve the task correctly.

- **Stage 1. Basic port–reward association.** Once the animal entered the trigger zone, the correct port was illuminated (extra visual cue) and an extra reward (~5 µL of water) was delivered to signal availability. If the animal poked the illuminated port within 110 s, the light was turned off and an additional water drop (~5 µL) was delivered as reinforcement. This stage familiarized the animal with the port–reward contingency and encouraged exploratory poking.
- **Stage 1.2: Auditory guidance-learning:** In this stage, the duration of the auditory cue (and reward availability) was shortened from 110 s (Stage 1) to 10 s. As an additional aid, the rewarded port was illuminated only during the first

two trials of the session; in subsequent trials no extra visual cue was provided. Unlike Stage 1, no water drop was delivered upon entering the trigger zone—animals obtained reward only by poking the correct port within the cue window.

- **Stage 2.2. Precision shaping and uncertainty period.** No visual cues (port illumination) were provided. The reward window remained 10 s. The size of the trigger zone decreased from 30% to 26% of the arena's area, requiring more precise navigation. The unrewarded period (0–3 min randomized) was introduced at the start of training and recall sessions, during which correct pokes were not rewarded.

- **Stage 3. Final task phase.** The full experimental paradigm used for data collection. The only change was that the reward window was further reduced from 10 s to 4 s.

Each shaping stage typically lasted 2–4 days, although duration varied across animals depending on individual performance. Advancement from one stage to the next required animals to achieve >90% correct trials of the last 30 trials and with a minimum of 30 rewarded pokes across two consecutive training days. Across animals, full shaping required approximately 2–3 weeks before reaching Stage 3. All behavioral analyses presented in this thesis were performed during Stage 3, after animals had reached stable performance and had mastered the basic task rules.

1.2. Animals and ethical procedures.

All animals used for the studies described in this thesis were wild type Male C57BL6/J mice (*Janvier*), with an initial age of 8-10 weeks old (n=50, 25g-30g. Animals were kept in stable conditions of temperature (23 °C) and humidity (60%) with a constant light-dark cycle (12h/12h) and all the experiments were conducted in the early part of the light phase.

Animals were housed preferably in groups until surgery implantation, after which, animals were housed individually to avoid implant damage. The environment of the animals was enriched with one running wheel that served also the purpose of habituating them to a similar wheel that would be found in some phases of the training protocol. Animals were habituated to the animal room, the experimenter, and several aspects of the training one week before the beginning of the experiments.

Animals trained in the task that were water-rewarded had their availability to water restricted. For this reason, animal weight was recorded on a daily basis before starting the protocol to establish a baseline. This baseline was used as a reference for the experiment and animal weight was measured daily, avoiding drops of weight under 80% of the baseline. Any

animal that would go down such a limit was removed from training, examined for any health issue, and given water *ad libitum* until recovery.

All animals were guaranteed a minimum of 1 mL water per day. Most mice obtained this during their daily training session. On days when their intake during training was insufficient, supplemental water was provided immediately afterward, either manually by the experimenter or via an automated system that released two 0.5 mL aliquots at fixed time points. This ensured consistent hydration and animal welfare throughout the protocol.

All animal procedures were approved and executed following institutional guidelines (Generalitat de Catalunya: Autorització de projecte d'experimentació N◦ = 9997), and the Comitè Ètic d'experimentació Animal (CEEA) from University of Barcelona, Nº = 121/18.

1.3. Surgical Procedures: Bilateral cannula implantation in the brain ventricles.

Most of the animals included in the studies of this thesis had received either control or pharmacological treatment. For this reason, most of the animals had undergone a surgical procedure before starting the behavioral protocol. Before any of the surgeries, all animals were weighed to calculate the correct dosage for pain and antiseptic drug delivery.

First, animals were placed in an induction chamber with a continuous flow of 1.5% isoflurane and 0,8 L/min of oxygen concentration. Once animals were correctly sedated and showed a regular breathing pattern, they were placed in a stereotaxic frame with a maintaining flow of 0.5% isoflurane and 0.8L/min of oxygen concentration, stable anesthesia depth was ensured by continuous monitoring of respiration and reflexes. Animals were also injected intraperitoneally with a dose of 0.1mg/kg of Buprenorphine, an opioid analgesic. Following this, an ophthalmic antibiotic was placed on the animals' eyes to avoid infection. Body temperature was regulated and maintained during the surgery using a homeothermic blanket. Top-head fur was completely removed using depilatory cream to prevent infection. The scalp was cleaned and disinfected with iodine antiseptics. Afterward, the skin was cut and opened to access the skull. The skull was cleaned using a blade and H_20_2 to remove any fat and tissue and thus ensure better adherence to the acrylic cement. Bregma and lambda were set to the same *z-coordinate* using stereotactic equipment. Then, 4-6 anchoring screws (Component Supply Co., USA) were placed using a dental drill (Meisinger, Germany) to perform the holes in the skull.

A bilateral catheter cannula (#3260PDG-1.5/SPC, Gauge 26; length 3.5mm, Plastics One, Roanoke, VA) was implanted in the lateral ventricles after performing 2 additional and bigger holes in the skull using the stereotactic equipment with the following details: anterior-posterior (AP) = 0 mm; dorsoventral (DV) = −1.72mm; ML = +/-0.85mm coordinates relative to bregma (referenced to Paxinos & Franklin). The cannula and a stainless bar were

secured with dental cement. The stainless bar was needed to head-fix the animals during several steps of the behavior protocol. This bar allowed the intracranial infusion of antibodies or drugs during head fixation of the animal while running on a wheel to reduce stress. After acrylic or dental cement was solidified, any loose skin was closed and sutured using a self-dissolving suture. Post-operative care included subcutaneous meloxicam (1 mg/kg) and 1 mL saline immediately after surgery. Animals were monitored daily, provided with analgesia for at least 48 h post-surgery, and allowed a minimum recovery period of one week before resuming behavioral training.

For cannula placement verification, we used a functional angiotensin II assay. Human angiotensin II (Sigma-Aldrich, Steinheim, Germany; cat. no. A9525-1MG) was dissolved in HBSS to prepare a 1 µg/µL stock solution (stored at –20 °C) and diluted to a final concentration of 100 ng/µL on the day of use. A volume of 1 µL was infused intracerebroventricularly through the implanted cannula. Correct cannula placement was inferred when animals exhibited the expected water intake response, typically defined as consumption of at least 1 mL of water within 10 minutes of infusion, whereas control mice normally consume only about 0.04 to 0.05 mL in the same period. In addition, changes in body weight were used to corroborate water consumption. Each animal was weighed twice before the assay, with a 10 min interval to establish a stable baseline, and again 10 min after angiotensin II infusion. A post-infusion weight increase consistent with fluid intake confirmed that drinking had occurred. Only animals with verified cannula placement and confirmed water intake were included in subsequent behavioral experiments and analyses (Only 2 out of 50 animals were found to have incorrect cannula placement).

1.4. Autoimmune encephalitis animal models: anti-NMDAR and anti-LGI1 antibody infusion.

Pharmacological modulators were used to transiently alter neural activity and assess their effects on memory processes. These approaches were applied to investigate the functional role of targeted circuits in the 8-port memory task.

Following previously reported auto-synaptopathy animal models (Planagumá et al. 2015, Petit-Pedrol et al., 2018), memory was perturbed by the passive transfer of highly specific autoantibodies against NMDA receptors or LGI1 protein.

Due to the specific requirements of the 8-port maze task, we adapted the previously reported infusion protocol from Planagumá et al. 2015. Continuous delivery with osmotic pumps was unsuitable because our animals were maintained under controlled water restriction, which partially dehydrated them; osmotic pump implantation would have further increased fluid loss and compromised hydration. In addition, pump implantation would have required surgery mid-protocol, introducing variability between control and treatment

groups.Moreover, the use of daily acute infusions provided tighter control over the duration of antibody exposure, allowing us to precisely define the number of days the manipulation was maintained. To overcome these limitations, we opted for daily micro-infusions using a Hamilton pump coupled to dual 5 µL syringes, which allowed precise control of antibody delivery while preserving stable behavioral and hydration conditions. Instead of using osmotic pumps, an infusion pump (KDS 310, Stoelting) with two 5µL Hamilton syringes (Hamilton) was used. The dose of antibodies infused daily was equivalent to the one reported in *Planagumá et al. 2015*. To avoid infusing into the animals' ventricles an excessive amount of liquid, the antibody concentration was increased to 780 µg/mL, reducing the initial volume by a factor of seven.

Animals from the animal models underwent daily antibody infusions. Upon reaching the final experimental stage (stage 3), mice started the infusion protocol. Every afternoon, following the behavioral sessions, animals were gently head-restrained and allowed to run on a wheel, identical to the one used during habituation and in their home cages. To ensure acclimatization and reduce stress, mice were habituated over several days before the first antibody infusion to both, restrain and implant manipulation. A double cannula injector matching the dimensions of the guide cannula was used (Plastic Ones). Following the habituation protocol, a day of saline infusion was performed with an equivalent volume, 2 days previous to the infusion of antibodies. This helped further habituate animals and minimize possible first infusion side effects. Daily i.c.v. infusions were performed after the recall session to standardize exposure. A total amount of 0.873 µL was loaded of either an antibody solution (780 µg/ml) or a control solution containing the same concentration of nonspecific control human antibodies or saline depending on the experimental group. The infusion rate was established at 0.3 µL/min and took around 3 minutes to deliver the total amount of solution. The injector was left inside for another minute to allow for diffusion, and then slowly removed ensuring that no fluid was overflowing. Between animals, the injector was cleaned with ethanol and water before loading a new sample. The injection protocol was repeated daily over several days, depending on the experimental cohort: 24 consecutive days for animals infused with anti-NMDAR antibodies and 14 days for animals infused with anti-LGI1 purified IgG antibodies.

The control and test solutions for NMDAR were provided by collaborators (Malviya et al., 2017). Monoclonal anti-NMDAR antibodies (rhuMab) were generated from patient samples and targeted a specific epitope in the GluN1 subunit of the NMDA receptor. The control preparation consisted of an isotype-matched recombinant human antibody with no reactivity against mouse brain epitopes.

The anti-LGI1 solution was obtained from patient sera with high-titre LGI1 antibodies. IgG was purified using agarose bead columns, dialyzed, filtered, and stored at –80°C until use. Specificity for LGI1 was confirmed using LGI1 knockout brain tissue and immunoabsorption assays. These purified patient-derived IgGs were then delivered intracerebroventricularly via

the same daily infusion protocol described above.

1.5. Perfusion and Histology

After any given experimental protocol animals were sacrificed. Animals that did not undergo any surgeries were sacrificed using CO_2. All remaining animals underwent cardiogenic perfusion for brain collection and processing. Animals were anesthetized with a Ketamine-Xylazine mixture (100 mg/kg and 10 mg/kg, respectively) until they exhibited no paw-pinch response. They were then perfused with 10 ml of PBS, followed by 10 ml of 4% PFA. After perfusion, the brain was carefully removed from the skull and incubated in 4% PFA at 4ºC.

Depending on the brain sectioning technique, a different protocol was applied:

- Brains processed using cryostat were then transferred to a 40% sucrose solution for 48 h at 4ªC. After, brains were embedded in an optimal cutting temperature (OCT) compound (Tissue-Tek OCT Compound, 4583) and frozen using liquid nitrogen for optimal preservation and storage. Finally, the tissue was sectioned using a cryostat, with section thickness ranging from 5 to 10 µm, depending on the histological analysis.
- Brains processed using a vibratome were then transferred to a PBS 1% solution until the tissue was sectioned. The section thickness was ranging from 25 µm to 40 µm depending on the histological analysis.

Cannula placement was also verified post mortem. Immediately before sacrifice, a small volume of methylene blue was infused intracerebroventricularly through the implanted cannula. Brains were then extracted and examined to visualize dye diffusion and cannula tracks. Only animals with correct placements confirmed by this anatomical verification were included in the analyses.

2. Behavioral Analysis

2.1. Coarse behavioral variables

To evaluate general performance and task engagement, we extracted several coarse behavioral variables during Stage 3 of training, when animals had already mastered the task rules and exhibited stable performance. These measures provided a quantitative description of task execution beyond memory-specific metrics.

- *Performance (% correct trials):* Calculated as the proportion of trials in which the animal poked the correct port within the reward time-window, divided by the total number of trials in the session.
- *Learning rate:* defined as the change in performance (%) across the early portion of the training session, capturing how quickly animals adapted to the new rewarded port location.
- *Reaction time:* measured as the mean time(s) from entrance to the trigger zone and the auditory cue starts, until the nose-poke at the correct port, restricted to correct trials. This variable quantified how quickly animals committed to a decision after the auditory cue.
- *Incorrect pokes within correct trials:* In correct trials, the number of additional pokes made in non-rewarded ports before reaching the correct port was counted. This measure was derived directly from the nose-poke event timestamps.
- *Inter-trial interval (ITI) errors:* The inter-trial interval (ITI) was defined as the period between the end of one trial and the onset of the next. All nose-poke events occurring outside of valid response windows — that is, during the ITI — were counted per session. This variable was used to track responses emitted outside of task contingencies.
- *Time to hit the trigger zone:* Defined as the time elapsed between the end of one trial (from the moment the animal pokes onto the correct port or when the time-window ends) and the next entry into the computer-controlled trigger zone, measured in seconds. This value was obtained from continuous tracking of the animal's position within the arena.

All variables were computed automatically using custom scripts (MATLAB/Python) that read and processed poke events, timestamps, and positional data exported from the behavioral control system. Continuous position signals (acquired at 80 Hz) were low-pass filtered (5-point zero-phase moving-average) and downsampled by a factor of two, yielding a final sampling rate of 40 Hz. Short tracking discontinuities were removed by smoothing. Spurious pokes (i.e. pokes on the same port <110 ms apart, unless the animal moved >4 cm from the port) were excluded from the analysis.

2.2. Memory strength using Memory Index

Memory recall strength was quantified using the Memory Index (MI), the MI is calculated from a polar spatial poke histogram, where each poke contributes a directionally weighted vector toward its port. The sum of these vectors, normalized by total pokes, yields a single vector representing poking bias. Projecting this vector onto the axis of the correct port gives the MI, ranging from +1 (all pokes at the correct port) to −1 (all pokes at the opposite port).

$$Memory\ Index(\theta_i, n_i) = \sum_{i=1}^{8} n_i = \frac{N_i}{N} cos(\theta_i) \quad ; \qquad \textit{(Equation 1)}$$

where:

- $i = 1,\ldots 8$ is the index of the ports in the maze
- $N_i \rightarrow$ is the number of pokes in port i
- $N = \sum_{i=1}^{8} N_i \rightarrow$ is the total number of pokes in that period
- $\theta_i \rightarrow$ is the angular position of port i expressed relative to a fixed convention.

The MI ranges from +1 (all pokes directed to the correct port) to –1 (all pokes directed to the opposite port), with 0 indicating no directional bias. We chose this vector-projection formulation because it preserves both directional bias and dispersion, and because it is invariant to overall poke count, enabling fair comparison across sessions.

Statistical significance of the MI was assessed through surrogate analyses designed to test whether observed recall exceeded what could be expected by chance**.** Specifically, for each session we generated a null distribution by randomly shuffling the identity of the rewarded port 1,500 times while keeping the animals' actual poke sequences unchanged. Shuffling the rewarded port while leaving the poke time-series unaltered preserves all temporal and spatial structure of the behavior, ensuring that the surrogate distribution controls for geometry- or sequence-driven biases. This procedure breaks the true link between past reward location and behavior, thereby providing a conservative baseline. The shuffled data were used to compute surrogate MI values, which formed a reference distribution against which the observed MI could be compared. From this surrogate distribution, we extracted the 1st and 99th percentiles to define a 99% confidence interval. An observed MI above this interval was interpreted as evidence that animals expressed a genuine directional bias toward the remembered location, rather than an artifact of arena geometry, uneven sampling, or random fluctuations. This procedure ensured that recall effects reflected memory-guided choices at the level of individual sessions.

This directional measure combines both the frequency of poking the correct port and the dispersion of pokes across neighboring ports, providing a single robust metric of memory strength and precision. The MI allows consistent comparison across sessions and animals.

To compute the memory strength using the MI for older memories poke histograms from each recall session were rotated such that the rewarded port from a given day (lag 0, lag 1, lag 2, etc.) was placed at 0, and the MI was recomputed relative to this reference. The rewarded port of the same day defined lag 0 (2h recall), while lag 1 corresponded to the port rewarded 24h earlier, lag 2 to the port rewarded 48h earlier, and so forth. For each lag, an MI was calculated using the same procedure described above, thereby yielding a "memory trace" across multiple days, and its statistical significance was assessed using the same

surrogate analysis as for lag 0 (1,500 shuffles of the rewarded port identity, 99% confidence interval).

2.3. Behavioral memory interference analysis.

To assess whether recall strength was influenced by overlapping memories from consecutive days, we examined how the angular distance between reward locations modulated the Memory Index (MI). Sessions were grouped according to the angular distance between the currently rewarded port (today, 2h memory) and the rewarded port from the previous day (24h memory). Angular distance was defined as the minimum circular distance between ports, i.e., $d = \min(|i-j|, 8-|i-j|)$ (where j is yesterday's port index), yielding values from 0 (same port as the previous day) to 4 (diametrically opposite ports in the maze). For each distance bin, the MI was computed as described in Section 2.2.

To evaluate whether apparent distance effects could arise by chance, two complementary surrogate controls were performed. Because the number of sessions per distance bin was uneven, surrogate MI distributions were computed separately for each distance category, ensuring matched confidence intervals. Each surrogate corresponds to a specific null hypothesis and was designed to rule out potential confounds that could generate spurious distance-dependent effects.

1. *Control 1 for the 2h memory.*

    - *Procedure*: For each shuffle, sessions were randomly reassigned to one of the five distance bins (0–4), while leaving the underlying poke data and port identity intact. This procedure preserved the distribution of pokes within each session but destroyed the true correspondence between the actual distance between 2 consecutive days. We generated 1,500 such surrogate datasets, recomputing the MI for each distance bin in every iteration.

    - *Null hypothesis*: This control tested the null hypothesis that there is no systematic relationship between the angular distance between the current and previous rewarded ports and recall strength. Under this hypothesis, any observed modulation of MI across distance bins would be equally likely if sessions were arbitrarily redistributed, and the empirical data should fall within the surrogate distribution. In other words, we test if the memory strength is genuinely modulated by the distance between two consecutive reward locations.

2. *Control 2 for the 24h memory.*

    - *Procedure*: For each session, the identity of the port rewarded on the previous day (lag 1) was randomly reassigned across sessions, while the current day's rewarded port and the actual poke distribution were preserved. After this shuffle, the angular distance between the reassigned "previous-day" port and the current rewarded port was recomputed, and sessions were re-binned according to this new distance, exactly as in the original analysis. In each surrogate iteration (1,500 total), MI values were then recomputed for each distance bin.

    - *Null hypothesis:* This control tested the null hypothesis that the apparent distance-dependent modulation of recall does not reflect genuine memory of the previous day, but instead emerges from structural factors inherent to the task. Specifically, it addresses the possibility that (i) any port located at a given angular distance from the current rewarded port would show a comparable decay in MI simply due to the spatial geometry of the arena, and/or (ii) such a decay arises from mathematical biases intrinsic to the definition of the MI. Since these two sources of artifact cannot be disentangled, the shuffle of the previous-day port identity controls for both simultaneously. Under this hypothesis, the surrogate datasets should reproduce the same distance-dependent modulation as the empirical data.

In both surrogate analyses, the 1st and 99th percentiles of the surrogate MI distributions were used to define a 99% confidence interval. Empirical MI values outside this range were considered significant, indicating that the observed distance-dependent modulation of recall could not be explained by the respective null model.

2.4. Generative models of memory interference.

Generative models were implemented to simulate poke distributions under alternative assumptions about memory use in the 8-port maze. Each model formalized a probability distribution over poke directions, consisting of a uniform exploration term and one or more memory-based components. The relative weights of these components specified the influence of each memory on simulated behavior.

2.4.1. Generative models definition

All models defined a probability distribution over the eight port directions θ_i $(i = 1,\ldots,8)$ as a mixture of von Mises–tuned components ("bumps") and a uniform baseline. The weights of these components capture the relative influence of each memory on behavior:

$$p(\theta_i|d) = \omega_0 U(\theta_i) + \omega_{2h} f(\theta_i;\ \mu_{2h},\ k_{2h}) + \omega_{24h} f(\theta_i;\ \mu_{24h}(d),\ k_{24h}) \quad \textit{(Equation 2)}$$

where:

- $U(\theta_i) = \frac{1}{8}$ → represents uniform exploration across the eight ports,
- $f(\theta_i;\ \mu_b,\ k_b) = \frac{e^{k.cos(\theta_i - \mu)}}{2\pi I_0(k)}$ → is the von Mises distribution centered at the angular location μ and concentration k.
- weights satisfy: $\omega_0 + \omega_{2h} + \omega_{24h}(d) = 1$ and $\omega \geq 0$

Angles were aligned so that the currently rewarded port was at $\mu_{2h} = 0$ and the previous-day (24h) rewarded port was positioned at $\mu_{24h}(d) = \mu_{2h} + d$, with angular separations

$$d \in \{0, \pi/4,\ \pi/2,\ 3\pi/4,\ \pi\}.$$

we implemented three variants:

1. *One-bump (2h-only) model*

$$\omega_{24h}(d) = 0\,, \qquad p(\theta) = \omega_0 U(\theta) + \omega_{2h} f(\theta;\ \mu_{2h},\ k_{2h}) \qquad \textit{(Equation 3)}$$

2. *Two-bump (2 h + 24 h) model*

$$p(\theta|d) = \omega_0 U(\theta) + \omega_{2h} f(\theta;\ \mu_{2h},\ k_{2h}) + \omega_{24h} f(\theta;\ \mu_{24h}(d),\ k_{24h}) \qquad \textit{(Equation 4)}$$

3. *Slope-modulated two-bump model*

In this case we modulate the weight of the 24h memory (bump).

$$\omega_{24h}(d) = \alpha + \beta\, d$$

To ensure valid probabilities, the distance-dependent weights $\omega_{24h}(d)$ were shifted so that their mean matched the baseline value α, and any values falling outside the range $[0,\ 1 - \omega_{2h}]$ were truncated to the nearest bound. These generative models were not intended to capture full behavioral complexity, but to serve as statistical baselines isolating the impact of the task's spatial structure alone.

2.4.2. Synthetic data generation and MI computation

For each distance d, model probabilities were evaluated at the eight port angles $\{\theta_i\}$ to obtain an 8-bin probability vector. Synthetic datasets were generated by sampling from the

mixture distribution and binning the samples into the eight port directions. Mixture weights were constrained to sum to 1, and concentration parameters k_{2h} and k_{24h} were shared across sessions. Using the same procedure as in Section 2.2, we computed:

- $MI_{2h}(d)$: Memory Index aligned to the 2 h port ($\mu_{2h} = 0$)
- $MI_{24h}(d)$: Memory Index aligned to the 24 h port ($\mu_{24h}(d)$)

Both indices were derived from the port probability (or sampled histogram) vectors exactly as described previously.

2.4.3. Parameter selection

Parameters were chosen by grid search over κ_{2h}, κ_{24h} (and β for the slope model), minimizing the discrepancy between model-generated and empirical MI curves across distances:

$$L = \sum_d |MI_{2h}^{data}(d) - MI_{2h}^{model}(d)| + \sum_d |MI_{24h}^{data}(d) - MI_{24h}^{model}(d)| \quad \textit{(Equation 5)}$$

In this way, the models serve as simple statistical baselines: if behavior resembles the uniform model, memory has little impact; if it is better captured by mixtures, then multiple traces are likely contributing simultaneously. We used an L1 objective to reduce sensitivity to outliers and ensure equal weighting of bins with different sample variance.

2.5. Generalized Linear Model - Hidden Markov Model (GLMHMM)

The poking dynamics of the animals suggested that non-linear relationships might underlie their task-solving strategies. To formally capture these dynamics and their temporal structure, we modeled the poking sequences using a Hidden Markov Model (HMM).

An HMM is a probabilistic model in which the system is assumed to transition between a set of unobservable (latent) states $z \in \{1, \ldots, K\}$ over time, according to a Markov process defined by the probability:

$$P(z_t | z_{t-1}) = T_{z_{t-1}, z_t} \quad \textit{(Equation 6)}$$

where T is the transition matrix defining the probability of switching between states. Each hidden or latent state encodes a distinct behavioral strategy which governs the probabilistic structure of observable choices. We applied a weak Dirichlet prior on transition probabilities without an explicit stickiness term, allowing the data to determine state persistence naturally.

The animals' poking decisions were modeled using a state-dependent Generalized Linear Model (GLM) with a circular (von Mises-like) link function. This GLM incorporates cosine-tuned responses to task-related and behavioral variables, allowing us to quantify their contributions to choice probabilities over discretized angular alternatives representing each port.

2.5.1. Regressors Design and assumptions

To ensure consistency and interpretability across sessions, all spatial variables with physical units (e.g., positions, distances) were normalized by the arena's radius R, such that all distances were expressed in units relative to this scale. The allocentric coordinate system was defined with the x-axis aligned in the direction of port 0.

Taking advantage of the circularity of the arena, all regressors (predictors) were reformulated in angular coordinates. This transformation enabled a more natural and accurate representation of spatial variables within the circular environment, making the model topology-aware and appropriate for circular data.

Each regressor was expressed in the following cosine-tuned form:

$$x_{t,a} = A_{a,t} cos(\phi_j - \psi_{a,t}) \qquad \textit{(Equation 7)}$$

where:

- ϕ_j is the angular location of port *j* being poked.
- $\psi_{a,t}$ is the angular direction associated with the regressor (e.g., heading, previous port)
- $A_{a,t}$ is its magnitude (amplitude)

This formulation ensures that each regressor modulates the log-probability of choosing a given port based on its angular proximity to behaviorally relevant reference directions. Cosine tuning respects the circular topology of port positions and enables smooth generalization across neighboring ports, unlike one-hot encodings which impose artificial discontinuities. Choice probabilities were computed using a softmax transformation across the eight ports, yielding a categorical distribution at each time step (see Eq. 8 for formal definition).

Based on this approximation, the probability of an observable being ϕ_j given the current state z, is defined as:

$$P(\phi_j, \psi, A, z) = \frac{1}{Z(k^z)} exp(\sum_a k_a^z A_a cos(\phi_j - \psi_a) \quad \textit{(Equation 8)}$$

where:

- ϕ_j is the angular location of port j ,
- ψ_a is the angular direction associated with regressor a (e.g., heading, previous port),
- A_a is the scalar amplitude of the regressor,
- k_a^z is the learned state-dependent weight (related to the precision of a von Mises)
- $Z(k^z)$ is the normalization constant ensuring a valid probability distribution over all ports.

The following regressors were included in the model:

1. *2h Memory*

This regressor captured the location of the correct port during the current session. It was defined as the allocentric angular position of the correct port, anchored to a consensus reference frame to best approximate the animal's internal representation of the goal location.

2. *24h Memory*

This regressor encoded the location of the correct port from the previously rewarded day. As in the previous one, it was defined as the *allocentric angular position* of the port that was rewarded in the immediately preceding day. This term was included to capture potential influences of longer-term spatial memory on current behavior.

3. *History Bias*

To account for the influence of recent behavioral history, this regressor was defined as the mean angular direction of pokes made during the previous trial. It served as a measure of short-term behavioral bias, reflecting a tendency to repeat previous actions regardless of reward outcomes.

4. *Non-Rewarded Win-Stay / Lose-Switch Strategy*

This regressor was constructed to capture poke-to-poke strategic adjustments based on reward feedback. If the previous poke resulted in a reward ("win"), the regressor took the form of a positive cosine bump centered on the current correct port—representing a tendency to repeat the rewarded/successful action. In contrast, if the previous poke was not rewarded ("loss"), the regressor was modeled as a negative cosine bump centered on the angular location of the previously chosen port—representing an avoidance of the unrewarded action. This formulation allowed the model to capture both the attraction to rewarded choices and the repulsion from non-rewarded ones in angular space.

5. *Euclidean Distance of the animal relative to the first choice port on the trial onset.*

This regressor measures the straight-line distance between the animal and the first choice port at the start of the trial. Positions were computed within the allocentric arena coordinates and normalized by the arena radius. This regressor models the animal's allocentric bias at trial onset, not the distance to the chosen port; therefore we use position-derived amplitude and angle to reflect natural movement constraints.

In this case the scalar amplitude of the regressor (A_t) was defined as:

$$A_t = \sqrt{(x_{animal})^2 + (y_{animal})^2} \qquad \textit{(Equation 9)}$$

and the associated angular direction (ψ_t) was given by:

$$\psi_t = Arg(x_{animal} + i\, y_{animal}) \qquad \textit{(Equation 10)}$$

where:

- x_{animal},y_{animal} are the cartesian coordinates position of the animal at the onset of the trial.

This measure captures spatial proximity, which may influence the likelihood of choosing nearby ports at trial onset.

6. *General Bias.*

General Bias was modeled as a cosine bump representing systematic spatial preference, estimated independently per state. This regressor captures consistent spatial preferences unrelated to task demands, such as a tendency to poke more frequently in ports farther from the entry door of the behavioral arena.

2.5.2. GLM–HMM training and inference.

We trained the GLM–HMM using a modified version of the *ssm* package (Linderman Lab, https://github.com/lindermanlab/ssm). The standard `InputDrivenObservations` class was replaced by a custom `InputDrivenVonMises` observation model, which computes choice probabilities from cosine-tuned regressors and applies a von Mises–like link function (see above). This allowed us to model the circular geometry of the 8-port maze.

- *Regularization.* To prevent overfitting, we implemented weak priors on the GLM weights k_a^z. Each state's active weights (defined by a Boolean mask) were penalized by (i) an optional Gaussian prior (ℓ_2, variance σ_2) and (ii) a hyperbolic-cosh approximation to an ℓ_1 penalty, controlled by ℓ_{1-T}. We used mild shrinkage without dominating the fit.

- *Training algorithm*. Models were trained via expectation–maximization (EM) using the routines in *ssm*. In the E-step, posterior state occupancies were computed with the forward–backward algorithm. In the M-step, transition probabilities were updated from expected counts, and GLM weights were optimized state-by-state using Newton–conjugate trust region minimization (`scipy.optimize.minimize`, method = "trust-ncg") with explicit gradient and Hessian. This adaptation ensured stable convergence of the circular von Mises GLM, which is not supported in the original *ssm* package.
- *Initialization and constraints*. Model parameters were initialized randomly at the start of training (weights and transition probabilities), and random seeds were fixed to ensure reproducibility.
- *Convergence*. Fits were run for up to 2000 EM iterations or until the relative change in log-likelihood fell below 10^{-5}.

2.5.3. Cross-validation and model selection.

For each candidate number of hidden states ($K = 1,\ldots,5$) models were trained with five-fold cross-validation, where data was held out to ensure independence between training and test sets. Cross-validation splits were defined at the session level, so that entire sessions were assigned to either the training or the test set, preserving within-session cohesion. Model performance was evaluated on the held-out folds using two complementary metrics: (i) average log-likelihood per poke, and (ii) predictive accuracy, defined as the fraction of choices correctly predicted by the model's posterior distribution. The optimal number of states was selected using an elbow criterion on cross-validated performance, while also requiring that the inferred states could be meaningfully interpreted. To ensure consistency

across folds, states were aligned using the Hungarian matching algorithm, based on cosine similarity of their regression weights. This procedure avoided arbitrary label switching and allowed direct comparison of state parameters across folds.

2.5.4. Per-state performance metrics.

Posterior probabilities of state occupancy were estimated for each poke using the forward–backward algorithm. These probabilities indicate, for each time point, the likelihood that behavior was generated by each latent state. For state-specific analyses, pokes were assigned to a latent state only when the corresponding posterior probability exceeded 0.8; trials not reaching this threshold were excluded from state-level summaries.

A threshold of 0.8 was chosen to ensure confident state assignments while retaining sufficient data. Similar HMM-based behavioral analyses have used slightly higher thresholds (e.g., Ashwood et al., 2022), supporting the use of this criterion as a balance between confidence and coverage.

To evaluate the reliability and explanatory capacity of each latent state, we computed two complementary metrics.

State-specific predictive accuracy. For each latent state, we identified pokes for which the posterior probability of belonging to that state was highest ("dominant state"). Predictive accuracy was then computed as the fraction of those pokes for which the model's predicted port (maximum probability under the GLM of that state) matched the animal's observed choice.

Variance explained. To assess how well each latent state captured the structure of poke behavior, we computed a circular analogue of explained variance. For pokes dominated by a given state, we compared the angular direction of the observed choice (θ_{true}) to that of the model's maximum-likelihood prediction (θ_{pred}) . Explained variance was defined as

$$EV = 1 - \frac{circvar(\theta_{true} - \theta_{pred})}{circvar(\theta_{true})} \qquad \textit{(Equation 11)}$$

where *circvar* denotes circular variance. This yields values near 1 when the model accounts for most of the angular variability, and values near 0 when predictions are no better than chance (used for Figure 7 d).

To determine which task variables drove predictive power within each state, we estimated the variance explained by individual regressors. For each regressor, we repeatedly shuffled its values (while keeping all others intact), refit the design matrix, and recomputed explained variance on state-dominant trials. The loss in explained variance relative to the unshuffled baseline quantifies the informational contribution of that regressor. This "regressor

shuffling" analysis provides a direct estimate of which features were most critical for supporting predictions in each state (used for Figure 7 e).

Temporal statistics of state usage. We quantified temporal statistics of state usage to characterize how animals engaged with each latent strategy.

- The *transition entropy* quantified the degree of randomness in state switching. In practice, we computed the entropy of the transition distribution from each state and then averaged across states. Low entropy indicates structured transitions (few preferred destinations), while high entropy reflects more random switching (used for Figure 10 a).

$$H = \frac{1}{K}\sum_{i=1}^{K} H_i \qquad \textit{(Equation 12)}$$

where H_i is the entropy of the transition distribution out of state i :

$$H_i = -\sum_{j=1}^{K} P(s_{t+1} = j \,|s_t = i)\, logP(s_{t+1} = j|s_t = i) \qquad \textit{(Equation 13)}$$

where:

- $P(s_{t+1} = j \,|s_t = i)$ denotes the empirical probability of transitioning from state i to state j, estimated from the inferred state sequence.
- K is the total number of latent states.

- The *dwell time* captured the typical duration of continues pokes within a state, computed as the average run length:

$$D_k = \frac{T_k}{N_k} \qquad \textit{(Equation 14)}$$

where the T_k is the total number of pokes emitted in state k and N_k is the number of times the animal entered that state k. (used for Figure 10 b)

- The *fraction of occupancy* of state *k* was defined as the fraction of pokes assigned to that state:

$$F_k = \frac{Number\ of\ pokes\ assigned\ to\ state\ k}{T} \qquad \textit{(Equation 15)}$$

where T is the total number of pokes. (used for Figure 10 c)

2.5.5. Global regressor importance

To assess the overall behavioral relevance of each regressor across states, we quantified their contribution to model performance using a permutation feature-importance approach. Two complementary analyses were carried out:

1. *Single-regressor models.* We trained models including only one predictor at a time, and measured their cross-validated log-likelihood and predictive accuracy. This provided an estimate of the total explanatory power of each regressor in isolation. (used for Figure 9)

2. *Permutation within the full model.* For the full multi-regressor model, we permuted the values of each regressor across trials while keeping all others fixed, and recomputed performance. The resulting drop in log-likelihood and accuracy was taken as that regressor's unique contribution, after accounting for redundancy with other predictors. (used for Figure 9)

Because some regressors are correlated (e.g., 2h and 24h memory), permutation importance quantifies unique variance explained but not total causal contribution.

RESULTS

Chapter 1: Dissociating Learning and Memory in the 8-Port Maze.

Chapter 1.1: Learning Performance Does Not Predict Memory Strength in the 8-Port Maze

- **Hypothesis**: We hypothesized that long-term memory strength—quantified by the Memory Index (MI)—is independent of classical learning performance metrics. If true, this would validate pooling behavioral sessions across animals and days without introducing confounds related to inter-session variability in learning.

To test this hypothesis, we focused on behavior during Stage 3 of the 8-port maze protocol, when animals had already mastered the basic rules of the task and exhibited stable performance across sessions. At this stage of behavioral proficiency, learning curves had plateaued on a session-by-session basis, trial structure was well understood, and engagement was stable. This ensured that animals were no longer in the process of acquiring task rules, allowing us to examine memory expression without confounding the influence of task rules' learning.

Within this stable-performance stage, we used recall sessions as a controlled context for measuring memory strength. In these sessions, we introduced a critical manipulation: the onset of reward and feedback was randomly delayed by 0 to 3 minutes. This temporal uncertainty prevented animals from anticipating when rewards would become available, maintaining engagement and encouraging exploratory behavior from the start of the session and across long experiments spanning several months. Crucially, reward and feedback were deliberately removed during this early phase. Here, feedback refers to any sensory or reward-based signal that normally follows a correct choice—specifically, the delivery of water and the cessation of the continuous tone that cues reward availability. In this phase, neither of these signals was provided, even after a correct choice. In the absence of feedback, animals could not actively confirm the success of their choices, effectively isolating recall behavior from re-learning processes. This ensured that early pokes reflected access to stored information rather than reinforcement-driven adjustments. Under these conditions, animals typically performed 3–20 trials while searching for the reward. These trials—executed under both spatial and temporal uncertainty—served as behavioral probes of memory. We quantified memory expression by analyzing the spatial distribution of pokes during this period.

As a summary measure of memory strength, we used the Memory Index (MI)—a validated, session-level metric originally described in Morales et al., 2020. The MI captures the

directional bias in poke behavior toward the port that had been rewarded in the previous session. Higher values indicate stronger memory expression, as animals preferentially return to the previously rewarded location. While not the only possible readout, the MI offers a simple, interpretable, and parametric indicator of recall strength, ranging from –1 to 1. We computed the MI separately for each session and animal (see Methods, section 2.2). Consistent with prior findings reported in *Morales et al. 2020*, animals showed a significant bias toward the port location learned most recently, indicating reliable memory recall.

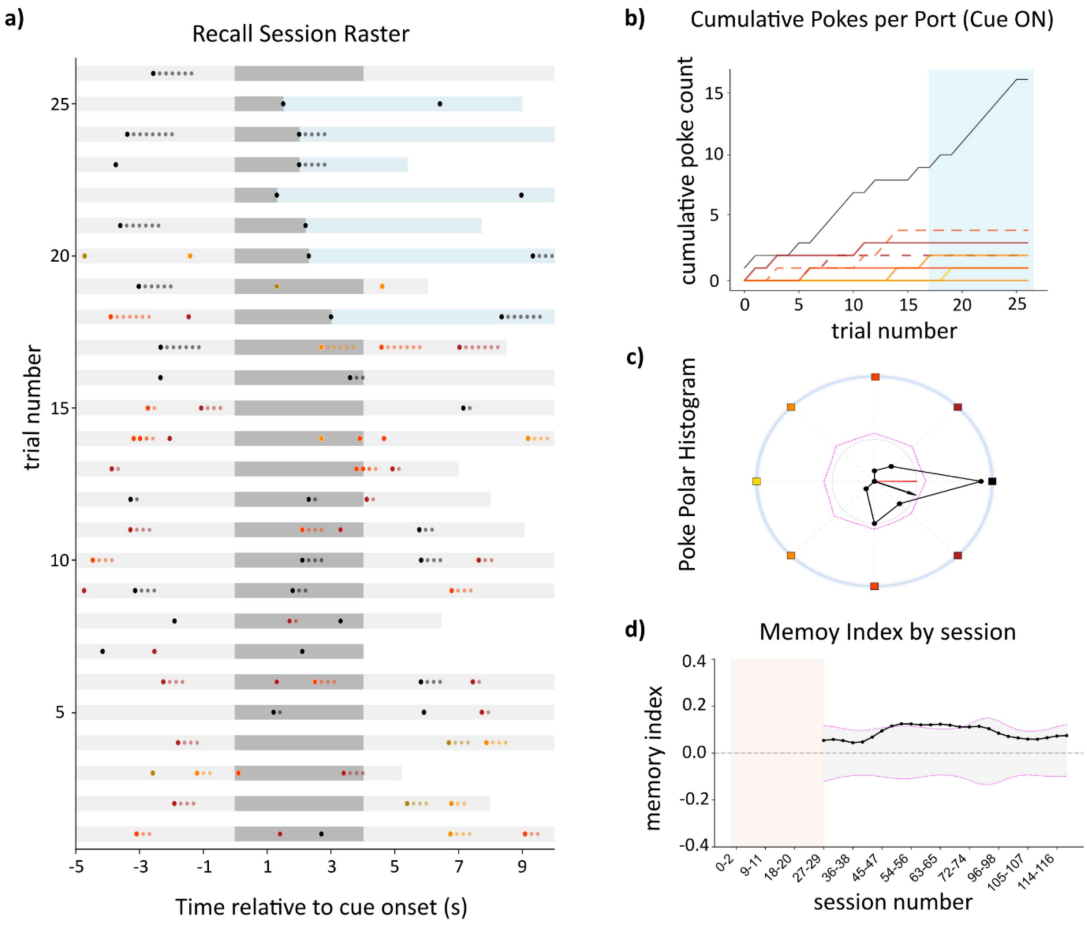

Figure 3: Recall sessions show stable memory index across training. (a) Example raster plot of recall trials from a single session, aligned to cue onset (time = 0 s, start of gray bar). Each row corresponds to a trial. Horizontal gray bars mark the cue ON period; light gray bars mark the trigger-zone phase. Black dots represent pokes into the currently rewarded port; colored dots represent pokes into other ports, color-coded by distance from the rewarded port. Small dots indicate licking events. Light blue shading denotes trials where water reward was delivered. x-axis: time (s).y-axis: trial number. (b) Cumulative poke counts by port for the same session. The black line indicates the rewarded port; colored lines represent non-rewarded ports. Blue shading indicates the rewarded period. x-axis: trial number. y-axis: cumulative poke count. (c) Polar histogram of pokes during the cue ON period for the same session. The rewarded port is shown in black; other ports are color-coded by distance. The dashed magenta circle shows the 95% confidence interval from surrogate controls. (d**)** Session-averaged memory index (MI) in blocks of

three sessions (black line). The magenta dashed band represents the 95% CI of the surrogate dataset (N = 1500). The beige shaded region marks the pretraining phase (stages 1–2.2, MI not computed). x-axis: session number (in blocks of 3). y-axis: MI

Does memory expression reflect prior behavioral learning quality?

To assess whether memory performance during recall reflects the quality of behavioral learning in the preceding session, we examined whether daily fluctuations in the Memory Index (MI) could be explained by specific behavioral variables measured during training. These included both coarse indicators—such as overall task *performance* and *rate of learning*—and more fine-grained features like reaction time, error rates, and engagement-related measures. By quantifying these variables during the late phase of training (i.e. stage 3)—when animals were proficient and task engagement was stable—we aimed to determine whether more efficient or structured learning translated into stronger memory expression the following session (recall).

We first examined overall task *performance*, a general indicator of how well the animals learned the rewarded location. Task performance was quantified as the percentage of correct choices made during the session, calculated by dividing the number of correct trials by the total number of trials and multiplying by 100. The rationale was that better task acquisition might translate into more accurate memory during recall (and the contrary should also be true). However, we observed no correlation between performance and subsequent MI values, indicating that overall *performance* during learning does not reliably predict memory strength (see Figure 3a).

It is possible that *performance* might not be sufficiently informative due to fluctuations across the entire session, making more detailed or temporally specific measures potentially better proxies. Thus, we computed the *rate of learning*, estimated by changes in performance during the early periods of the training session. The idea was to capture how quickly animals adapted to new locations of the rewarded port, under the assumption that faster adaptation might reflect more efficient learning and more trials to reinforce the rewarded location. Yet again, no significant association with MI emerged, suggesting that early-session learning rate offers no additional explanatory power for memory variability across sessions (see Figure 3b).

Because overall performance and learning rate failed to explain variability in recall, we next asked whether more implicit aspects of behavior (i.e., measures not directly tied to reward or task rules but potentially reflecting underlying cognitive or motor processes)—such as reaction time—might provide additional insight. We focused on *reaction time* on correct trials—a variable that, while constrained by the task's timing rules, is not itself decisive of success once the trial is deemed correct. The rationale was that faster responses might

reflect more efficient sensorimotor processing or greater certainty during learning, and thus stronger memory encoding. *Reaction time* was defined as the average duration it took animals to reach the correct port on correct trials, capturing the entire process from stimulus integration and decision-making to movement initiation. However, despite its stability across sessions, we found no significant correlation between *reaction time* and MI (see Figure 3c), suggesting that this measure also fails to account for variability in memory strength.

To further probe whether aspects of decision-making during learning might predict memory strength, we also examined the *number of incorrect pokes* occurring within correct trials. This metric was intended to capture exploratory behavior that preceded accurate decisions, offering insight into how confidently animals made their choices. Such behavior is often interpreted as a display of internal uncertainty or reduced decision confidence, both of which are relevant aspects of cognitive control and memory-guided behavior (Kepecs et al., 2008). Despite its theoretical relevance to decision-making precision, this variable showed no significant correlation with MI (see Figure 3d), indicating that the degree of within-trial exploration does not systematically account for differences in memory strength across sessions.

Having found no link between decision-related variables and memory strength, we next turned to behavioral variables associated with task engagement, to assess whether general attentional or motivational states during the learning phase might influence memory strength during recall. One such variable was the *number of inter-trial interval (ITI) errors* —pokes made outside the valid response window. This measure was used as a proxy for task engagement and behavioral inhibition, understood here as the animal's ability to suppress premature or context-inappropriate actions. In other words, fewer such errors were assumed to reflect behavior more strongly guided by task rules and reduced impulsivity. In line with theories linking cognitive control to memory encoding (e.g., Miller & Cohen, 2001; Gazzaley & Nobre, 2012), we hypothesized that higher engagement would support better consolidation. However, we found no significant correlation between *ITI errors* and subsequent memory performance, suggesting that variations in impulsivity or attentional control during the inter-trial period do not reliably predict long-term memory strength in this task (see Figure 3e).

Finally, we assessed the time to hit (latency) the trigger zone between trials—a measure intended to also reflect the animals' engagement with the task. Variability in this interval was taken as a potential indicator of fluctuating motivation or attentional readiness, both of which have been implicated in modulating memory processes (e.g., Mather & Sutherland, 2011; Aly & Turk-Browne, 2016). However, this measure also showed no significant association with memory strength, indicating that time taken to initiate a new trial during the learning phase does not reliably predict recall memory strength in this paradigm (see Figure 3f).

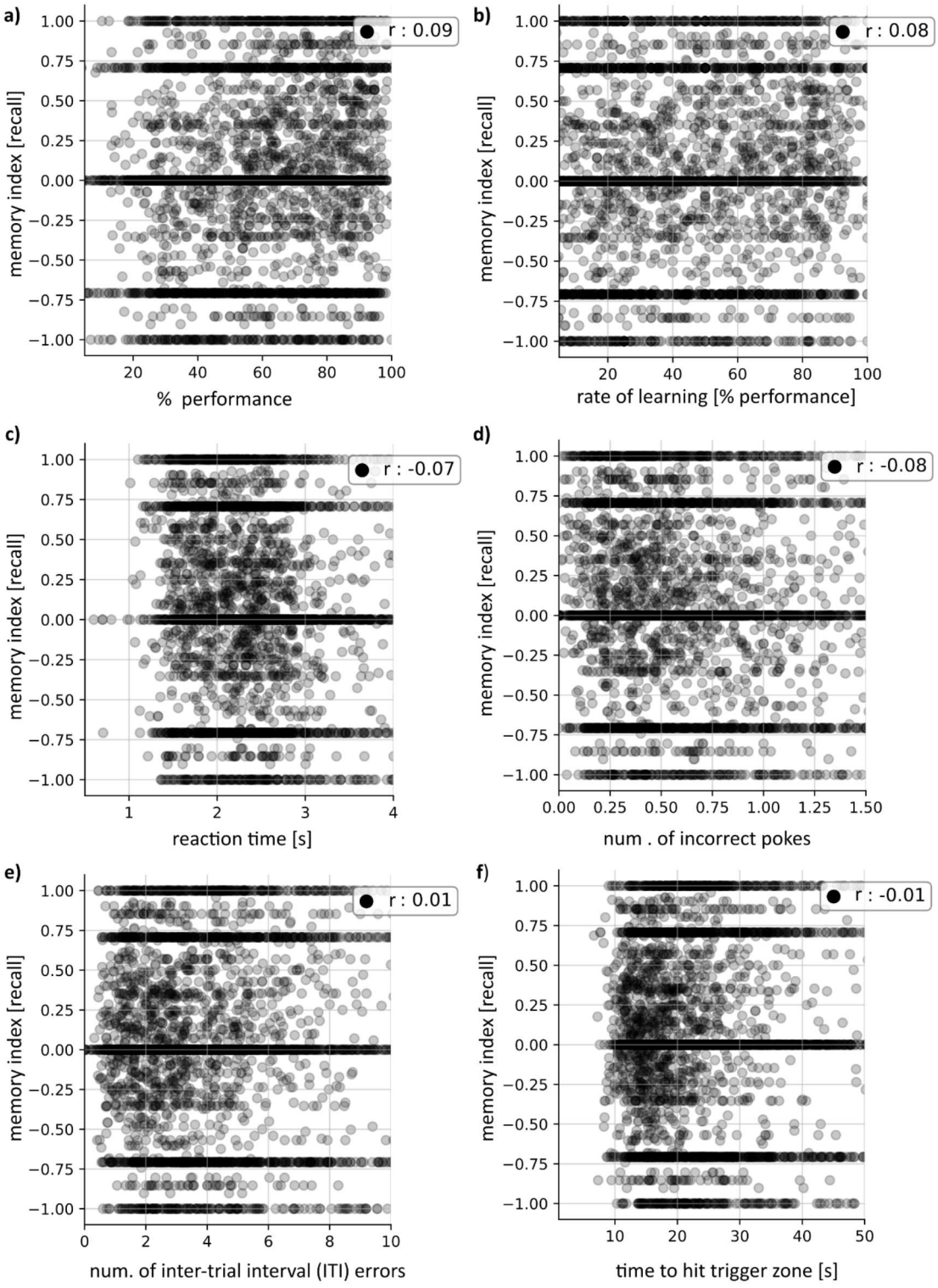

Figure 4. Behavioral performance metrics in training sessions do not correlate with memory strength in recall. Each point represents a single session. (a) Performance: Percentage of correct trials completed during the training session; no significant correlation (r = 0.09). (b) Rate of learning: Change in performance (%) across the early portion of the training session, capturing how quickly animals adapted to the new rewarded port location; no significant correlation (r =

0.08). (c) Reaction time: Mean time (s) from cue onset to nose-poke at the correct port, measured on correct trials; no significant correlation (r = –0.07). (d) Incorrect pokes: Number of port pokes occurring within trials scored as correct, used as an indicator of exploratory or uncertain decision-making; no significant correlation (r = –0.08). (e) Inter-trial interval (ITI) errors: Number of nose-pokes made during the ITI, outside the valid response window, used as a proxy for task engagement and inhibitory control; no significant correlation (r = 0.01). (f) Time to hit trigger zone: Time (s) taken to enter the trigger zone after the end of the response window, reflecting motivation and attentional readiness to start the next trial; no significant correlation (r = –0.01).

Conclusion: Our results demonstrate that classical learning performance metrics do not predict long-term memory strength, as quantified by the Memory Index (MI). Across multiple behavioral sessions, animals exhibited stable memory traces—enduring behavioral signatures of prior experiences—even when their performance levels varied. None of the tested behavioral variables—including *performance*, *reaction time*, or *number of errors*—showed a significant correlation with memory expression during recall. This suggests that variability in memory cannot be explained solely by task execution quality, motivation, or consistency during training. While these metrics capture important aspects of behavior, they fail to account for fluctuations in behavioral memory recall. Crucially, this dissociation supports the conclusion that differences in basic task performance do not confound the expression of memory. As a result, for the state of proficiency of the animals to solve the task, we can treat MI independently of learning level, and safely pool sessions across days and animals, increasing statistical power for subsequent analyses aimed at uncovering the structure and dynamics of memory-guided behavior.

Chapter 1.2: Animals display memory Recall for 2 and 24h in the 8-Port Maze Task.

Having established that traditional learning performance metrics do not predict memory strength (see Chapter 1), we aggregated behavioral data from different sessions, without factoring in the level of the performance during learning. Combining data across sessions allowed us to examine how memory expression varies with the age of the memory.

As in Morales et al. (2020), we asked whether individual memories from prior sessions could still be detected in our data. To do this, we leveraged the circular layout of the 8-port maze and aligned behavior relative to the rewarded port from the session of interest. For each memory age, we reorganized the data so that the rewarded port associated with the memory of interest was always placed at the center of the distribution. This was done by shifting poke counts from each session into a common reference frame, where the rewarded port was labeled as position 0 and neighboring ports were assigned positions relative to it (e.g.,[-3, -2, -1, 0, 1, 2, 3, 4]). This approach allowed us to average across sessions while preserving the spatial relationship between behavior and past rewards.

We quantified memory expression using the Memory Index (MI), a directional measure—that is, a metric that captures whether poke behavior is biased toward a specific spatial location (in this case, the previously rewarded port) rather than being uniformly distributed across all ports. To determine whether recall reflected true memory rather than chance, we compared the observed MI to a null distribution obtained by randomly shuffling the identity of the rewarded port across sessions. This procedure breaks any real association between poke behavior and reward history, providing a baseline for what would be expected if choices were unrelated to memory. For example, imagine that in a given session the actual rewarded port was port 2. In the shuffled control dataset, the same behavior might instead be analyzed as if port 4 had been the correct one. If the animal truly remembers the correct location, the directional bias in poke behavior (i.e., the MI) should be stronger toward port 2 than toward the shuffled control port (port 4). By repeating this shuffling many times (1500), we generate a distribution of MI values expected by chance. The observed MI was considered statistically significant if it exceeded the 99th percentile of this null distribution. The number of shuffles was determined by reaching stability of the confidence interval for increasing the number of surrogated data.

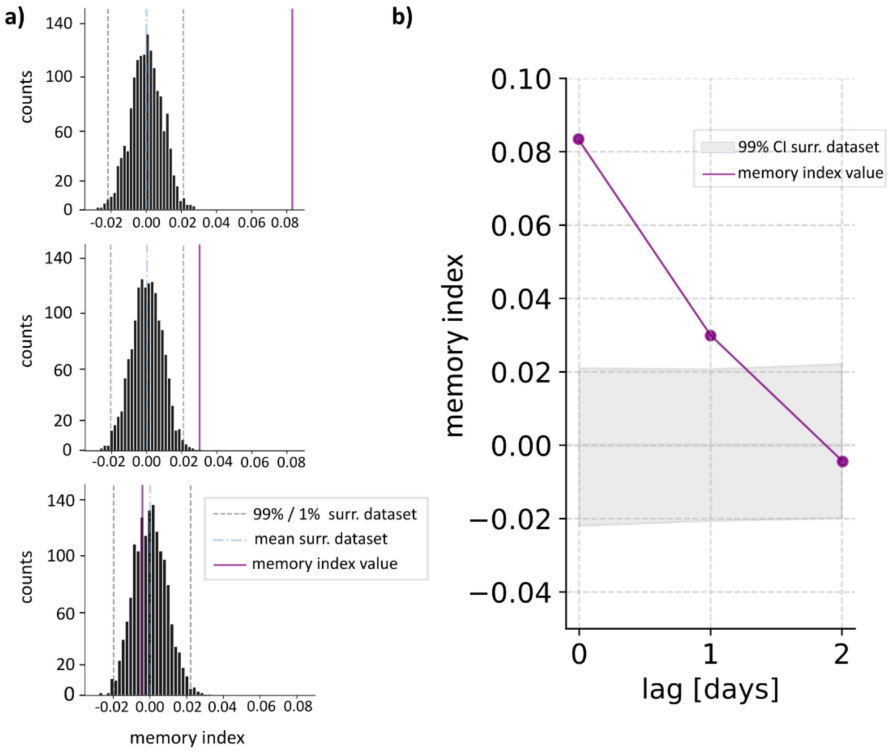

Figure 5. Animals are able to recall rewarded port locations after 2h and 24h delays in the 8-port maze task. (a) Histograms of surrogate memory index (MI) distributions for three memory lags (top to bottom: 0 (2h), 1 (24h), and 2 (48h) days). x-axis: MI value. y-axis: counts. Black bars are the surrogate MI distribution; blue dashed line is the surrogate mean for each distribution; gray dashed lines are the 1% and 99% surrogate percentiles; purple line is the observed MI for each lag.(b) Corrected memory index (observed MI − mean surrogate MI, mean surrogate ≈ 0)

across memory lags. x-axis: lag (days; 0 ≈ 2h, 1 ≈ 24h, 2 ≈ 48h). y-axis: memory index. Purple points and line indicate the observed MI; gray shaded region shows the 99% surrogate confidence interval (n = 50 mice, ~100 sessions/animal). The surrogate distribution was generated by randomly shuffling the identity of the rewarded port across sessions, thereby breaking the association between behavior and reward history. The observed MI exceeded the 99th percentile of the surrogate distribution at 2h and 24h , but not at 48h.

Consistent with prior findings, we found that animals reliably recalled rewarded port locations from both 2 and 24 hours earlier (see Figure 4). This indicates that discrete spatial memories remain behaviorally accessible across multiple time delays, even in a task that requires repeated exposure to changing reward locations.

Importantly, both 2-hour and 24-hour memory traces were detected within the same recall session. This suggests that behavior is shaped not by a single dominant memory, but by the simultaneous influence of multiple past experiences. In a task with repeated structure and overlapping spatial representations, this raises the possibility that memories from different days may interact during recall. In the next chapter, we test this hypothesis by examining whether the spatial relationship between rewarded ports across sessions affects memory expression—providing a behavioral window into how multiple memories may interact over time.

Chapter 2: Distance-Dependent Memory Interference in the recall of the 8-Port Maze Task

- **Hypothesis:** If memory traces from previous sessions persist, then their spatial proximity to the current target may interfere with recall, systematically modulating memory strength as a function of the distance between past and present rewards.

In Chapter 1, we demonstrated—using the Memory Index—that animals reliably recalled the locations of previously rewarded ports after both 2-hour and 24-hour delays. These results show that memory traces from prior sessions remain accessible well beyond the current session and produce a statistically detectable bias in poke behavior toward those past locations.

In our task, animals perform daily sessions in a near-identical environment, with only the rewarded port changing. Over time, this structure leads to repeated exposure to similar spatial contexts, increasing the likelihood that multiple memory traces coexist. If those traces of memory overlap, they might interact—potentially reinforcing or interfering with each other during recall.

This idea is rooted in the interference theory (Underwood, 1957), which proposes that memories formed in similar conditions may compete during retrieval. Our paradigm provides a natural setting to test this theory, because it inherently generates overlapping memory representations by design.

To ask whether such interference can be observed behaviorally, we tested if memory strength depends on the spatial relationship between the rewarded port in the current session and the one from the previous day. If interference plays a role, we would expect recall strength to be modulated when the two ports are at different distances in space. To do so, sessions were grouped by the angular distance between consecutive rewarded ports—ranging from 0 (same port) to 4 (diametrically opposite in the maze)—and computed the Memory Index (MI) for each grouped distance.

Our results show that MI values decreased as the distance between consecutive rewarded ports increased (see Figure 5 b), with the strongest recall observed when the same port was rewarded on both days. This distance-dependent decrease was evident at both 2h and 24h delays. To evaluate whether this distance-dependent pattern reflects a true memory effect—or arises from geometric biases (i.e., systematic preferences in poke distribution driven by the maze's spatial symmetry) or the definition of the MI—, we generated two sets of surrogate data as control, each designed to test a specific null hypothesis (see Methods 2.2).

The first control tested whether the distance-dependent decrease observed in the 2h MI could arise by chance. To do this, we preserved the memory content of each session but randomly reassigned sessions to different distance bins, breaking any systematic link between spatial proximity and recall. If distance has no effect, the MI should remain inside the 99 percentile of the shuffled data. A consistent deviation from this control would indicate that spatial distance between the reward of two consecutive sessions influences memory strength.

In our data, MI values at several distances diverged from the shuffled mean value, supporting the notion that spatial distance modulates recall. However, this modulation did not follow a simple, monotonic decay with distance. Instead, we observed a structured pattern: at short distances (e.g., 0 and 1), MI values were elevated relative to the control, suggesting that spatial proximity may facilitate recall. At larger separations (e.g., distance 3), MI values dropped below the shuffled mean value, indicating that reward locations farther apart in space may interfere with recall. This interaction did not resemble the surrogate data, where MI values remained flat across all distances—showing no systematic variation. These results reveal a spatially structured interference profile: close proximity between past and current reward locations may enhance memory expression, whereas greater spatial separation appears to impair it.

Crucially, however, this result alone does not demonstrate that the modulation is driven by memory of the previously rewarded port. Due to the circular geometry of the task, ports located at the same angular distance from the current (2h) goal location—such as ports at positions ±1—could produce similar effects on the MI simply because of their spatial configuration. That is, the observed pattern could arise from geometric symmetry in how the MI is computed, rather than from memory of the prior reward location.

To rule out this geometric confound, we next examined a second control that directly tests whether the 24h memory trace contributes to the observed pattern. Here, we shuffled the identity of the 24h rewarded port across sessions while preserving the animals' actual poke behavior and the currently rewarded port. This manipulation breaks the link to the true past reward location while maintaining the overall spatial structure and poke statistics. If the decrease in MI remains present under these conditions, it would suggest that the effect arises from the mathematical computation of the index rather than from memory of the prior reward location. If not, it supports the interpretation that animals recall a specific reward location from the previous day, and that this 24h memory contributes to the distance-dependent modulation observed in the 2h MI—consistent with the idea that older memories interfere with the recall of more recent ones. In our data, several distances in the 24h MI diverged from the control baseline (see Figure 5 b), though this effect was only marginally significant. Still, the direction and consistency of the deviations support the interpretation that animals recall specific past reward locations, and that older memory traces continue to influence 2h recall. This persistent influence aligns with interference

theory, which predicts that overlapping memories—especially those formed in highly similar contexts—can compete during retrieval, thereby shaping the strength and accuracy of current recall.

Together, these results not only demonstrate that memory traces from both 2h and 24h delays are simultaneously expressed during recall, but also reveal that their spatial relationship modulates behavioral accuracy. The decrease in memory strength as a function of distance—particularly when the previously rewarded location is far from the current memory target—suggests a form of interference between memories from consecutive sessions. However, the overall statistical support for this effect remains relatively weak, raising the question of whether this reflects meaningful behavioral structure or simply limited data. To gain a deeper understanding of the origin of this effect, we turned to generative modeling (Methods Section 2.4). Rather than attributing the effect to complex memory strategies, we asked whether minimal assumptions—such as recalling only the current port—could account for the observed MI modulation. If successful, these models would suggest that the interference-like pattern emerges from simple memory interactions. If not, it would imply that the observed structure reflects a meaningful aspect of memory dynamics that cannot be explained by simple models based on a handful of assumptions.

We first considered a minimal generative model—the “one-bump” model—to test whether the observed MI pattern could be explained without involving memory interference. This model assumes that animals recall only the currently rewarded port, and that any apparent influence from previous sessions reflects either noise or structural artifacts in the data. To simulate this, we generated synthetic poke distributions composed of a uniform exploratory baseline and a single directional bias—a ‘bump’—centered on the correct port for the current session. Crucially, no contribution from 24h memory is assumed. If this model could replicate the distance-dependent modulation observed in the data, it would suggest that a single memory component is sufficient to explain the effect. Despite its simplicity, this model provides a useful baseline for evaluating whether the observed structure in recall behavior requires additional memory traces.

However, even after optimal parameter selection (see Methods), the one-bump model failed to reproduce the structure observed in the real data. Not only did it not recover the drop in 2h MI at large separations, but it also failed to produce any systematic decrease in memory strength across distances. This result indicates that the pattern seen in the data cannot be explained by the 2h memory strength alone. At least one additional memory trace is required to capture the observed modulation in recall strength—consistent with the idea that memories from previous sessions persist and actively interfere with the encoding or retrieval of new ones. (see Figure 5 c,d,f).

We next addressed whether explicitly including a memory trace from the previous day could better capture the structure observed in the data. To test this, we extended the model to

include two bump components: one centered on the currently rewarded port and another aligned with the port rewarded in the previous session. This *two-bump* model reflects the idea that recall behavior may be shaped by the combined influence of recent and older memory traces that persist over time. By modeling memory as a mixture of multiple overlapping representations, we aimed to test whether their interaction alone could account for the distance-dependent modulation seen in the data.

Adding the second bump substantially improved the model's ability to reproduce the overall decrease in MI across distances. Compared to the *one-bump* version, the *two-bump* model more closely resembled the structure of the real data, supporting the idea that memory from the previous session plays a role in current recall behavior. However, while the model captured the general shape of the MI decrease, it underestimated the magnitude of the effect—particularly in the 2h MI curve.

This discrepancy raised the possibility that the influence of the 24h memory trace may itself depend on spatial proximity. To test this, we modified the *two-bump model* by scaling the amplitude of the older memory component as a function of distance (see Figure 5 i,j,k). This scaling could be biologically substantiated by a time decay modulated by distance interference. This refinement allowed the strength of interference to increase with spatial separation, while preserving the overall structure of the model. With this adjustment, the model better captured the steep drop in MI at larger distances, especially in the 24h data. Nonetheless, even with slope tuning, it failed to fully replicate the magnitude of the observed effect, suggesting that simple models, even when incorporating distance-dependent memory strength, are insufficient to fully explain the interference patterns observed in recall behavior.

Conclusion: Our results reveal a trend suggesting that memory strength decreases as the spatial distance between successive rewarded locations increases. Although this effect is only marginally significant, it is consistent with the hypothesis that distant memories interfere with the retrieval or even integration of current ones, pointing toward interference-like dynamics in spatial memory.

To strengthen this interpretation, we systematically evaluated potential confounds by testing alternative null hypotheses and comparing observed patterns against generative models. These complementary analyses confirmed that the distance-dependent modulation could not be explained by geometric artifacts or by models lacking contributions from past experiences.

Together, Chapters 1 and 2 establish the utility of the Memory Index (MI) as a robust measure of long-term memory that is independent of learning performance and sensitive to spatial structure across sessions. These findings consolidate the MI as a meaningful behavioral readout for tracking memory recall across days.

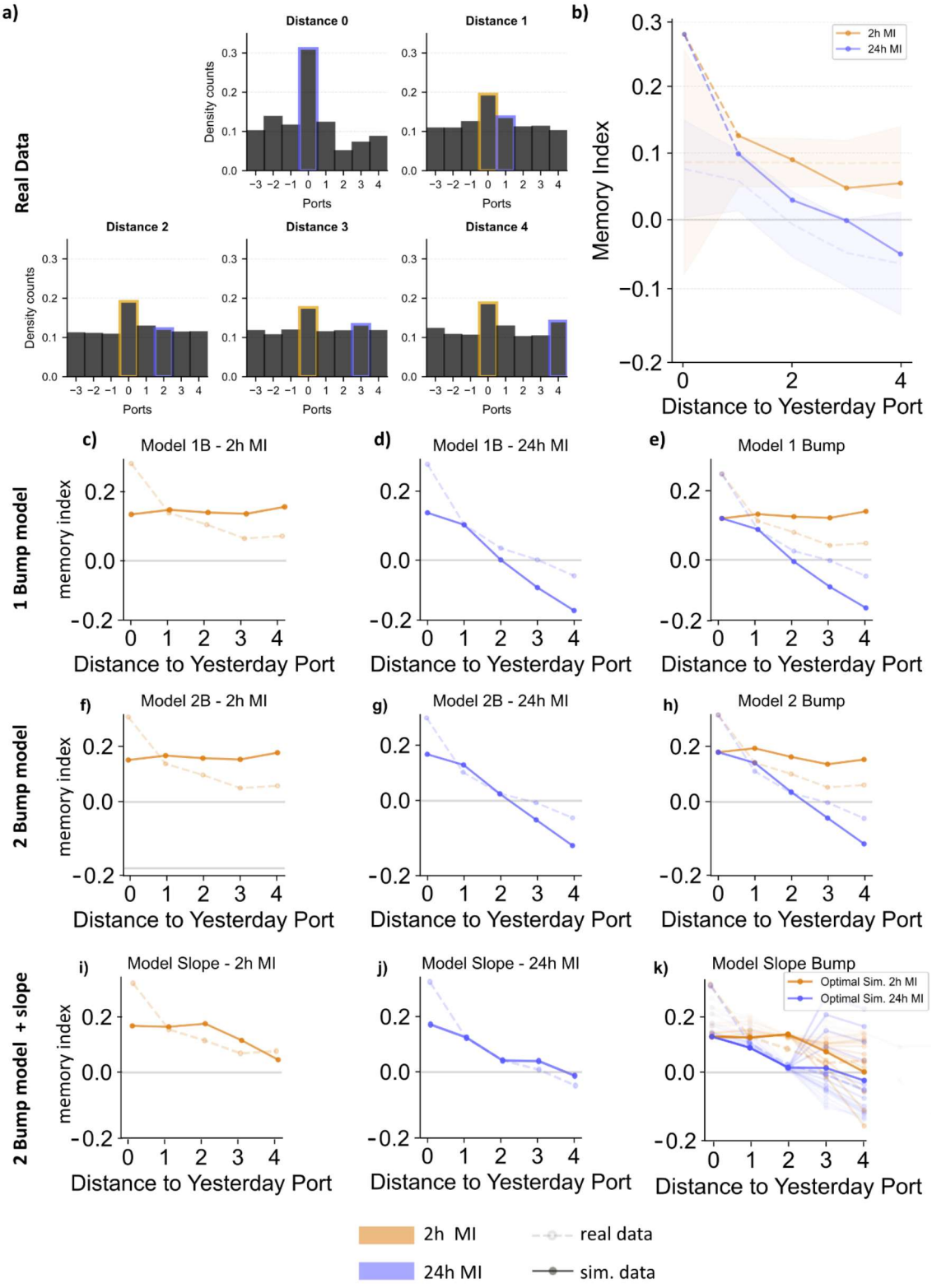

Figure 6: Memory strength decreases with spatial distance between rewarded ports and is not reproduced by simple bump generative models. **Top row (Real Data)**: (a) Normalized poke histograms grouped by the distance (0–4) between the current and previous rewarded ports.

x-axis: port position relative to the currently correct port (0) (2h ago). y-axis**:** normalized poke counts. Orange bar indicates the currently correct (2h) port; blue bar indicates the port rewarded 24h earlier. (b) Memory index (MI) as a function of distance for 2h (orange) and 24h (blue) lags. Shaded regions show 99% confidence intervals(CIs) from surrogate controls (N = 1500). Two complementary shuffles were used: (i) random reassignment of sessions to distance bins, preserving 2h memory content but breaking the link to spatial distance; and (ii) randomization of the 24h rewarded port identity, preserving poke behavior and spatial geometry but disrupting the true 24h memory trace. Both curves decline with distance, more strongly at 24h. For the 2h condition, MI values at distances 0 and 1 exceeded the 99% CI of the bin-shuffled surrogate distribution. For the 24h condition, MI values at distances 0 and 3 diverged from the 24 h-shuffled surrogate baseline. **Middle Rows: One-bump and Two-bump Generative Models.** x-axis: distance (0–4). y-axis: memory index (MI); (c) MI values for 2h memory as a function of port distance, computed from the one-bump model (1B, solid orange) and overlaid with real data (dashed orange). The model produces a flat MI curve and fails to reproduce the observed decline across distances. (d) Same as (c) but for 24h memory (1B model in solid blue, real data dashed blue). The model fails to capture the sharp decrease in MI observed in the data. (e) Combined overlay of real and model data for both timescales. The 1B model lacks the distance-dependent modulation present in the data. (f) 2h MI values generated from the two-bump model (2B). The model produces a decline with distance, but it underestimates the depth of the modulation seen in the real data. (g) 24h MI values from the 2B model closely follow the real data trend, capturing the overall shape and magnitude of the decline. (h) Combined overlay of real and model data from the two-bump simulation. The 2B model improves the fit compared to 1B, especially for 24h MI, but does not fully reproduce the structure of the 2h MI curve. **Bottom Row: Slope-Modulated Two-Bump Model. (i–k)** Two-bump model with the 24h component scaled linearly by distance. x-axis: distance (0–4). y-axis: memory index (MI). This slope modulation improves the fit relative to the fixed-amplitude model, particularly improving the fit for 2h MI at larger distances. (k) Overlay of model predictions (solid lines) and empirical data (faint lines) for each variant and timescale (2h and 24h). The slope-modulated model best matches the empirical trend but still underestimates the magnitude of the decline, particularly for 2h MI.

However, while the MI provides a reliable and interpretable session-level measure, its aggregate nature limits its ability to capture the fine-grained dynamics of behavior on a trial-by-trial basis. This becomes particularly problematic when attempting to disentangle the strategies animals use to navigate the task or to trace how interference unfolds over time in a particular session. Critically, the MI cannot distinguish whether reductions in recall reflect memory decay, uncertainty, exploratory behavior, or competition between memories—all of which may result in similar average patterns.

To address these limitations, we turn to high-throughput computational modeling approaches that analyze behavior at the level of individual pokes. Such models—commonly used in decision-making research—allow us to infer latent cognitive variables and to trace how they evolve over time. By doing so, we gain access to the internal dynamics that shape behavior, enabling a more mechanistic understanding of how animals resolve memory interference and adapt to structured environments. Crucially, analyzing behavior at the trial

level offers a clearer window into how different strategies unfold over time, revealing the hidden dynamics that shape memory-guided decisions. In the context of this work, it also provides a structured way to relate behavioral signatures to underlying neural activity, helping to uncover how memory-related information is represented and dynamically accessed within neural circuits.

Chapter 3: Modeling Latent Memory States and Strategy Dynamics in Recall Behavior

In the preceding chapters, we established two key findings. First, we demonstrated that traditional learning performance metrics—such as performance or reaction time—do not predict the strength of memory expression during recall. This dissociation validated our use of the Memory Index (MI) as a learning-independent measure of memory-guided behavior (Chapter 1). Second, we showed that memory strength is systematically modulated by the spatial distance between previously rewarded locations, suggesting that memories from prior sessions interact with newly created ones (Chapter 2). While these results underscore the robustness and utility of the MI, they also present limitations. Importantly, it should be recalled that throughout this thesis we focus exclusively on the no-reward epoch of the recall sessions—i.e., on poking behavior that occurs in the absence of water delivery or feedback.

By design, the MI is an aggregate measure: it compresses behavior over entire sessions into a single scalar value. As such, it cannot resolve how recall unfolds over time within a session or whether different strategies are deployed across trials. Critically, a drop in MI could arise from several indistinguishable underlying processes—such as memory decay, exploratory behavior, increased uncertainty, or even disengagement—that the MI cannot distinguish. To disentangle these latent contributions, we need a finer-grained approach.

In this chapter, we move beyond summary statistics and turn to computational modeling of behavior at the single-trial level. Our goal is to uncover the internal states—that is, what the animal might be thinking or trying to do—that govern decision-making during recall, and how these states fluctuate across time. Rather than assuming that animals rely on a single behavioral strategy throughout the session, we ask whether their choices reflect a dynamic mixture of strategies—such as memory-guided search, uncertainty-driven exploration, or disengagement. Traditionally, variability in behavioral tasks like ours has often been interpreted as noise or as behavior unrelated to task structure. However, recent studies of working memory in animals have shown that such variability is often not random, but instead reflects transitions between distinct internal states, each characterized by different sensitivities to task-relevant information (Ashwood et al., 2022). For example, an animal might begin a session by persistently revisiting the port that was rewarded 2h ago (suggesting active recall), but later shift to scattered poking across ports (suggesting reduced confidence or task disengagement). These states are not directly observable, but their influence can be inferred from the animal's pattern of choices over time.

To formalize our hypotheses, we used a Generalized Linear Model embedded in a Hidden Markov Model (GLM-HMM)—a hybrid framework in which two modeling components

capture different aspects of behavior. The Hidden Markov Model (HMM) governs the sequence of internal states that an animal might occupy across pokes. These states are considered latent, meaning they cannot be directly observed, but they change over time following transition rules that the model learns from the data. At any given moment, the animal is assumed to be in one of these discrete states—each representing a distinct behavioral strategy. The second component, the Generalized Linear Model (GLM), is used to describe how the animal behaves when it is in one of these states. In practice, this means that within each state, the model estimates how different task features—such as the location of the previously rewarded port—affect the probability of poking a particular port. These task features are represented as regressors—variables that encode the structure of the environment and recent experience. For example, one regressor might indicate whether a given port matches the previously rewarded location, while another might capture spatial proximity to it. The GLM assigns a weight to each of these regressors, determining how strongly they influence decisions within that state.

To train the model on our data, from a broader set of candidate variables, we selected a small subset of task-relevant regressors—variables designed to reflect the environmental cues and task features that animals might plausibly use to guide their decisions. These regressors serve as the input to the GLM, determining how choice probabilities vary within each latent state. Specifically, we included:

- Reward Location (2h memory): captures the precision of the animal to poke the port that was rewarded 2h ago.
- Older Reward Location (24h memory): captures the precision of the animal to poke the port that was rewarded on the previous day.
- History Bias: captures short-term repetition of poking behavior—specifically, whether the animal pokes in the same area of the arena than on the immediately previous trial.
- Non-rewarded Win-Stay / Lose-Switch: Captures whether the animal tends to repeat a poke after a correct choice, or switch after an error—even though no external feedback is given. This regressor describes mouse behavior based on our experimental hypothesis of repeating or alternating the poking location based on some sensory feedback that we cannot measure.
- Euclidean Distance to the first choice (port) on trial onset: captures spatial proximity to the animals' first choice on trial onset. This regressor tests whether the animal's position in the arena at trial onset influences where it decides to poke—that is, whether animals tend to choose the port that is physically nearest to them when the trial begins.
- General Bias: Captures whether the animal has a consistent preference for a certain range of locations inside the arena—specifically this regressor measures the tendency to poke farther from or closer to the behavioral box's door.

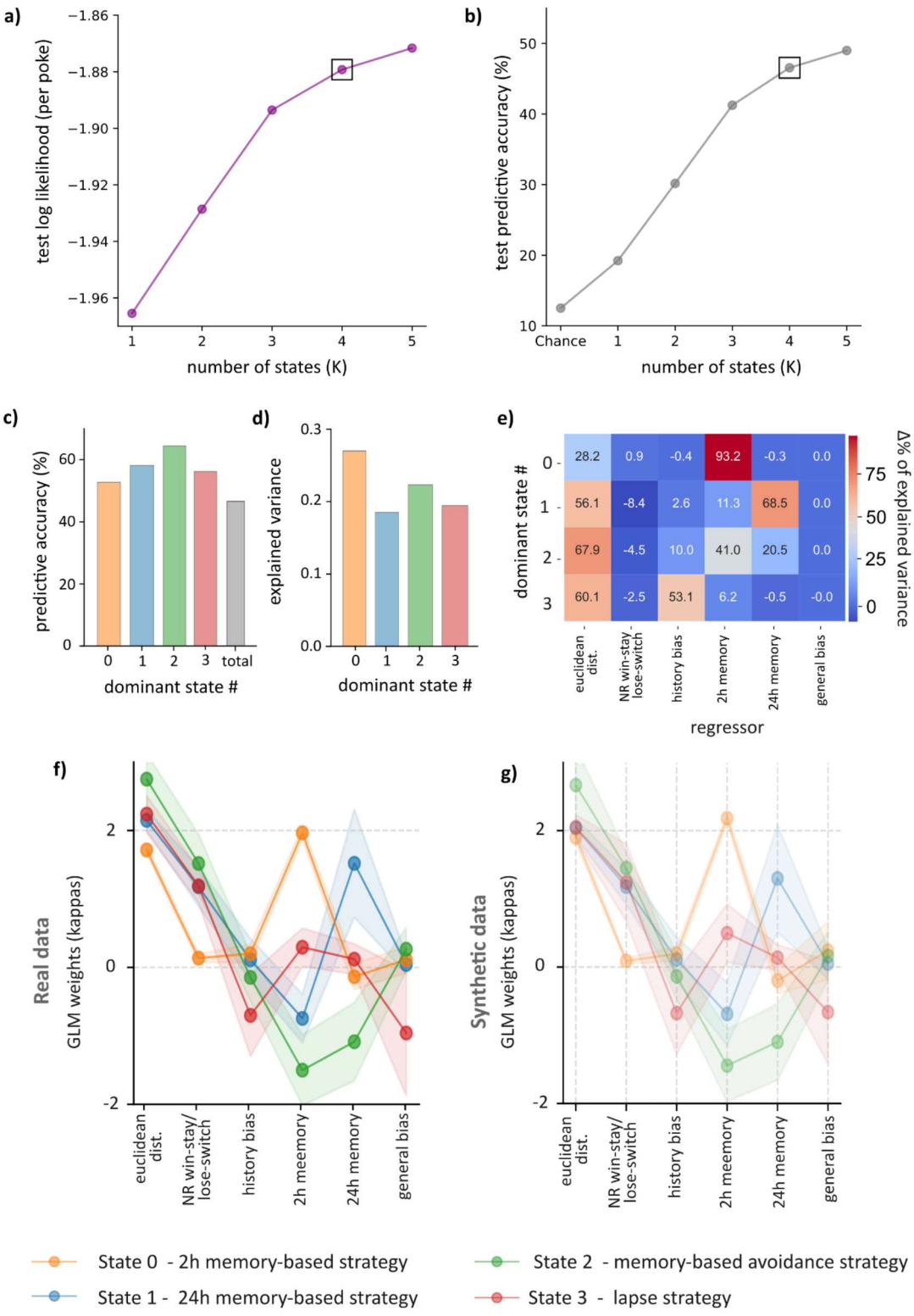

Figure 7: GLM-HMM modeling reveals four distinct latent strategies underlying recall behavior.
(a) Model log-likelihood per poke, evaluated on held-out data, vs. number of states for models

with 1–5 hidden states; Performance increased sharply from K = 1 to K = 4 and plateaued thereafter, indicating that recall behavior is best explained by a four-state model. (b) Predictive accuracy on held-out data vs. number of states (K). K = 4 yielded the highest accuracy before plateauing, confirming that additional states do not meaningfully improve generalization. (c) State-specific predictive accuracy (poke locations with a posterior ≥ 0.8 are taken as the prediction) shows that all states predict behavior above chance (12.5%); State 0 achieved the highest accuracy (~64%) and State 1 the lowest (~53%). (d) Explained variance for each state; State 0 accounts for the largest share of predictable variability (~0.28), whereas State 1 explains the least (~0.18). (e) Heatmap of the dominant regressor per state. Columns: task regressors (Euclidean distance, NR win-stay/lose-switch, history bias, 2h memory, 24h memory, general bias). Rows: latent states (0–3). Numbers indicate the percentage of explained variance attributed to each regressor, with color coding reflecting magnitude. State 0 is dominated by 2h memory (~93.2%), State 1 by 24h memory (~68.5%), State 2 by Euclidean distance (~67.9%), and State 3 by spatial and history bias variables. (f) GLM weight profiles for models fit to real data. x-axis: task regressors (Euclidean distance, NR win-stay/lose-switch, history bias, 2h memory, 24h memory, general bias). y-axis: GLM weights (κ). Each line shows the weight profile for one latent state. Shaded areas indicate the SEM across folds of the 5-fold cross-validation, with each fold consisting of distinct data chunks. State 0 shows a large positive weight for 2h memory; State 1 shows a positive weight for 24h memory and a negative weight for 2h memory; State 2 shows a positive weight for Euclidean distance and negative weights for both memory regressors; State 3 shows moderate-high weights for spatial bias regressors and near-zero weights for memory regressors. (g) GLM weight profiles for models fit to synthetic data generated from the fitted model recapitulate the state-specific patterns observed in real data, confirming that the inferred strategies are robust, reproducible, and recoverable.

These features allow the model to detect patterns in how different types of information—such as memory of past rewards, recent actions, or spatial proximity—influence behavior on a trial-by-trial basis, and how those influences change over the course of a session.

In the context of our task, this framework allows us to move beyond assumptions of static behavior and instead ask whether animals' trial-by-trial choices reflect stable use of memory or other behavioral modes not captured by session-level summaries. Rather than assuming that memory expression is uniform across time, the model allows us to infer *when* memory is being used, *how* it interacts with other strategies, and *whether* these dynamics are consistent across animals and sessions. This trial-level analysis offers a powerful complement to the session-averaged results from Chapters 1 and 2, and sets the foundation for linking behavioral states to neural activity in future work.

Chapter 3.1. Model Overview and Fit Quality.

To determine how many latent behavioral states best explain recall behavior, we trained GLM-HMMs with one to five hidden states and assessed their fit using five-fold cross-validation (see Methods). Model performance, measured via log-likelihood on held-out data, improved consistently with model complexity, suggesting that recall behavior cannot be explained by a single fixed strategy. The most substantial improvements occurred when increasing from one to four states, with performance plateauing thereafter (see Figure 7 a).

This pattern was mirrored in the model's predictive accuracy on held-out data (see Figure 7 b). Accuracy increased steadily from the single-state model (near chance) to K = 4, after which it plateaued. The convergence of both log-likelihood and accuracy-based metrics provides strong evidence that a four-state model offers the optimal balance between predictive power and parsimony.

Notably, the single-state model (K = 1)—equivalent to a stationary GLM applied uniformly throughout the session—performed markedly worse. Its poor generalization to held-out data confirms that recall behavior involves systematic switches in strategy across pokes. This result supports the view that memory-guided behavior during recall is structured, heterogeneous, and best understood through a latent-state framework.

Having established that a four-state model provides the best overall fit to the data, we next asked whether these latent states reflect meaningful differences in how animals make decisions during recall. If the states capture distinct behavioral strategies, they should not only improve model fit, but also differ in how accurately they predict choices and in which task variables they rely on.

Chapter 3.2. Predictive Accuracy and Variance Explained by Latent States.

To assess whether the latent states inferred by the GLM-HMM reflect meaningful structure in recall behavior, we first grouped pokes according to their most likely latent state (the state with posterior probability ≥ 0.8), and then evaluated the model's predictive accuracy for poke location within each state. This allowed us to test whether the strategies uncovered by the model are not only internally consistent, but also predictive—i.e., whether each state supports reliable, state-specific predictions of the animal's choices. All state-specific accuracies were well above chance level (12.5%), ranging from 53% in State 0 to 64% in State 2 (Figure 7 c). This confirms that, despite variability in strategy, each state captures structured components of behavior that allow for reliable prediction.

We next quantified how much variance in poking behavior was explained by the model within each state (see Methods 2.5.4). The results revealed marked differences across states: State 0 accounted for the largest share of predictable variability, while State 1 explained the least (Figure 7 d). This contrast suggests that State 0 reflects a well-structured, goal-directed strategy—one in which behavior closely follows task regressors. In contrast, State 1 appears to reflect a noisier or less task-driven strategy, where behavior is either more stochastic or guided by factors not included in the model.

While state-specific accuracy confirms that each latent state captures structured components of behavior, accuracy alone cannot reveal which task variables drive those predictions. We therefore complemented accuracy analyses with explained variance derived from regressor shuffling, which directly tests the informational contribution of each predictor to behavior within a given state. We measured the change in variance produced by shuffling each GLM regressor, one at a time, restricting the analysis to pokes for which one of the hidden states had posterior probability exceeding 0.8 (See methods), and attributing that poke to that dominant state (Figure 7 e). This revealed distinct, state-specific profiles: in State 0, 2h Memory dominated, accounting for 93.2% of the explained variance; in State 3, Euclidean Distance had the strongest effect ($\Delta \sim 60.1\%$), consistent with a spatial heuristic strategy; in State 1, 24h Memory contributed most ($\sim 68.5\%$), with other predictors showing minimal or negative influence, indicative of a narrower, possibly interference-driven strategy; and State 2 displayed a more distributed structure, with moderate contributions from both 2h ($\Delta \sim 41.0\%$) and 24h ($\Delta \sim 20.5\%$) memory regressors alongside Euclidean Distance ($\Delta \sim 67.9\%$). These findings confirm that each latent state is supported by a distinct set of behavioral features, with some (e.g., State 0 and State 3) exhibiting strong, focused structure, and others (e.g., State 1 and State 2) reflecting more diffuse or context-specific strategies.

To better understand the internal structure of each strategy, we next examined the GLM weight profiles that define how task variables influence behavior in each state.

Chapter 3.3. GLM Weight Profiles and Interpretation of Latent Strategies.

To better understand how task variables shape behavior under each strategy, we examined the GLM weight profiles associated with each latent state. These weights modulate the parameters of a von Mises-like choice distribution in a circular GLM-HMM framework, and reflect how each predictor contributes to directional bias in poke behavior when a given state is active. While they are not directly interpretable as marginal effects—as in a standard linear GLM—they still provide a compact summary of the behavioral structure encoded by each strategy.

Crucially, these weights complement the regressor importance analysis presented earlier: while the latter quantifies how much each variable contributes to prediction, the GLM weights reveal the directional structure through which task variables shape behavior in each latent state. One consistent finding was that Euclidean Distance carried relatively high weights across all states, indicating that it shaped behavior independently of which strategy was being used. Because Euclidean Distance does not differentiate the states, our interpretations below focus instead on the relative dominance—or absence—of the other task variables. It's important to note that the state labels used in this model are operational and derived from its regression structure, rather than from direct behavioral measurements.

Each state exhibited a distinct pattern of regressor influence, consistent with the use of different decision strategies (see Figure 7 g):

- State 0: 2h memory-based strategy. This state was dominated by a strong positive weight on the 2h memory regressor, with minimal contributions from other variables. This aligns closely with the results of the regressor importance analysis (see Section 2), where 2h Memory emerged as the most influential predictor in this state. Together, these findings support the interpretation that State 0 reflects a task engaged and goal-directed state, in which behavior is primarily driven by recall of the currently relevant (2h) reward location.
- State 1: 24h memory-based strategy. This state was marked by a strong positive weight on the 24h Memory regressor, accompanied by a negative weight on 2h Memory. This interpretation is further supported by the regressor importance analysis, where the 24h Memory regressor accounted for the largest proportion of explained variance in this state, and where other predictors had minimal or even negative contributions (see Figure 7 e), reinforcing the idea that this strategy reflects systematic recall of and foraging at the previous day's reward location. Together, these results suggest a consistent but suboptimal strategy in which animals rely primarily on long-term memory traces from the previous day—reflecting structured, memory-driven interference rather than random exploration (While this strategy is suboptimal given the reward contingencies of our task, it could be considered rational in natural environments, where food sources often remain stable from one day to the next).
- State 2: Memory-based avoidance strategy. This state featured strong positive weights on the Euclidean Distance regressor, coupled with consistently negative weights on both 2h and 24h Memory. The active avoidance of memory is further supported by the regressor importance analysis, where memory variables are important to explain variance, confirming that actually, animals use that information to express behavioral poking. These findings suggest a strategy in which animals actively bypass mnemonic guidance and rely instead

on spatial heuristics—possibly to resolve internal conflict or ambiguity when multiple memory traces are in competition.

- State 3: Lapse or disengaged state. This state showed near-zero weights on memory regressors and moderate-to-high weights on spatial features. Regressor importance analysis further revealed a substantial contribution of history bias (~53% of explained variance; Figure 7 e). Notably, however, the GLM weights for history bias were negative (Figure 7 f), indicating that animals in this state were not following recent choices but actively avoiding them. Taken together, these patterns suggest that behavior in this state is largely decoupled from mnemonic guidance, instead reflecting a disengaged mode dominated by simple heuristics—such as preferring closer ports—combined with an avoidance of immediately preceding actions. This structure is consistent with transient lapses of task engagement, where choices seem to be task-variable agnostic rather than by memory or reward contingencies.

To confirm that these weight patterns are not artifacts of overfitting or dataset-specific idiosyncrasies, we used the fitted GLM-HMM from the real dataset to generate synthetic behavioral data. In practice, this involved sampling choices directly from the model's learned parameters—effectively creating a "fake" dataset whose statistical structure mirrors that of the real data. We then re-fit the GLM-HMM to this synthetic dataset and examined the recovered κ parameters (GLM weights). These recovered GLM weights closely matched those from the original dataset (see Figure 7 g), reproducing the same dissociation between memory-based, avoidance, and lapse-like strategies. This demonstrates that the model captures a reliable and generalizable structure in behavior, and that the latent strategies can be recovered even from simulated data.

The consistency of GLM weight profiles across cross-validation folds—and their recovery in synthetic data—confirms that the model captures robust and reproducible behavioral strategies. But while these weights describe how each latent state uses task variables in theory, they do not tell us whether animals actually behave in ways that match those predictions. To test this, we next examined whether choices made under each strategy are consistent with the behavioral patterns implied by the model's internal structure.

Chapter 3.4. Behavioral Expression of Task Variables Across Latent States

Having established that each latent state can account for a distinct internal strategy, we next asked whether these strategies are actually expressed in observable behavior. While the GLM-HMM defines each state through a set of regressor weights, these parameters alone do not confirm that animals behave in accordance with the model's interpretation. Moreover, because the task variables may be measured on different scales, the size of their weights is

not directly comparable. As a result, looking only at the raw weight values can give a misleading picture of their relative importance.

To test this, we aligned poke counts relative to each regressor's preferred direction and computed normalized poke histograms separately for pokes strongly assigned to each state (posterior ≥ 0.8; see Figure 8). For example, if the model predicted that the 2h Memory variable pointed to the "north" location, we rotated all pokes so that "north" was always at the center of the plot, then counted how often the animal poked in each direction around it. We repeated this for each regressor and compared the resulting patterns across states. This analysis reveals how behavior is structured with respect to task-relevant variables under each inferred strategy.

Figure 8. Behavioral alignment to task variables varies systematically across latent states. Normalized poke distributions aligned to the preferred direction of each task regressor, computed from pokes strongly assigned to each latent state (posterior ≥ 0.8). Each row corresponds to one latent state, each column to a task regressor. x-axis: port position relative to the regressor's preferred direction (centered at 0). y-axis: normalized poke counts. In State 0, distributions peak sharply around the 2h Memory direction. In State 1, alignment is broader, with pronounced structure relative to the 24h Memory and History Bias regressors. In State 2, distributions are consistently offset from the 2h and 24h Memory directions, with peaks often opposite to the

predicted goal. In State 3, distributions show little consistent alignment across regressors, with only Euclidean Distance exhibiting a modest peak.

In State 0, poke distributions sharply peaked around the predicted 2h Memory direction, consistent with a precise, goal-directed policy. In State 1, alignment was broader but still structured, with clear modulation by 24h Memory, supporting the idea of memory-driven interference. In State 2, poke distributions showed consistent misalignment with both 2h and 24h Memory directions, with peaks often opposite to the predicted goal. This anti-alignment, combined with weak structure elsewhere, supports the interpretation of a memory-avoidance strategy, potentially deployed when competing mnemonic signals create internal conflict. In State 3, poke behavior lacked consistent alignment with any regressor. Although Euclidean Distance showed the strongest peak among all states, the overall pattern was noisy and unstructured—suggesting a disengaged or exploratory mode, in which decisions are decoupled from task-relevant variables.

Taken together, these analyses validate the interpretations given in Section 3.3 based on the GLM weights and explained variance, showing that each latent state is defined by a distinct configuration of task variables and highlighting the diversity of strategies that animals deploy during recall. Yet this raises a broader question: does the model truly need all of these variables to explain behavior, or could a smaller subset capture most of the predictive structure? To address this, and to finalize the characterization of the model itself, we next quantified the global importance and redundancy of each regressor across the full four-state model. This step completes the model-centric analysis, after which we shift our focus to what the model reveals about the organization and dynamics of recall behavior.

Chapter 3.5. Global Importance and Redundancy of Behavioral Regressors.

While the previous section clarified how task variables guide behavior within each strategy, it did not quantify which predictors are most critical for explaining poking behavior overall. Identifying these key predictors is essential for understanding which aspects of the task animals rely on most, and for separating indispensable variables from those that are secondary, redundant, or negligible. To make this distinction, we applied a permutation feature importance analysis (inspired by Musall et al., 2019)—a widely used approach that estimates the behavioral relevance of each regressor by measuring how much model performance deteriorates when its values are randomly shuffled.

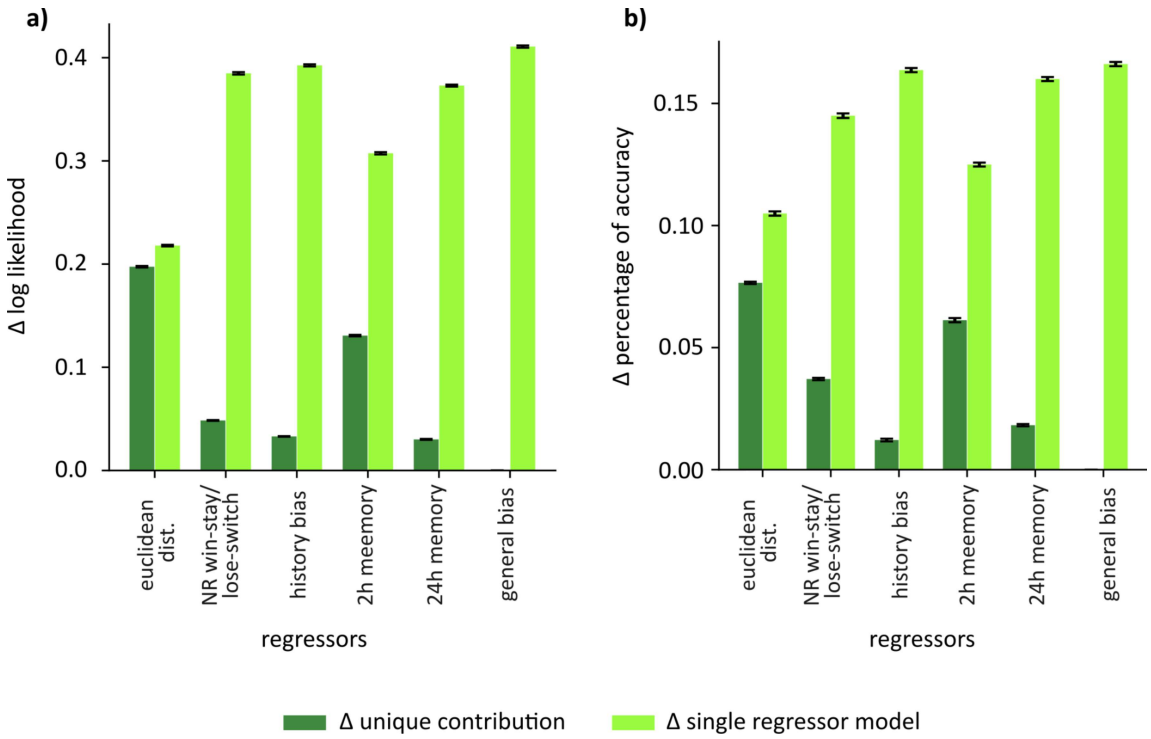

Figure 9: Model explanatory power is driven primarily by 2h Memory and Euclidean Distance. (a) Change in cross-validated log-likelihood (ΔLL) and (b) change in predictive accuracy (ΔAcc) for each regressor. Light green bars: performance of single-regressor models, reflecting the total explanatory power of each variable. Dark green bars: drop in performance when shuffling each regressor within the full model, reflecting its unique contribution after accounting for redundancy. x-axis: task regressors. y-axis: ΔLL (a) or ΔAcc (b). 2h Memory and Euclidean Distance performed strongly in isolation (cvLL ≈ 0.40; cvAcc ≈ 0.14–0.16) and retained substantial unique contributions (ΔLL ≈ 0.20 for Distance; ΔAcc ≈ 0.08), indicating complementary, non-redundant explanatory value. In contrast, 24h Memory, History Bias, and General Bias performed well in isolation but contributed little unique information once other predictors were included, and the Non-rewarded Win-Stay/Lose-Switch regressor showed minimal influence. Error bars indicate SEM across cross-validation folds.

We assessed our quantification by using two complementary metrics: the cross-validated performance of single-regressor models (cvLL and cvAcc), and the drop in performance when permuting each regressor within the full model (ΔLL and ΔAcc). The former captures the total explanatory power of a variable in isolation; the latter estimates its unique contribution, after accounting for redundancy with other predictors.

Among all regressors, 2h Memory and Euclidean Distance emerged as the most behaviorally informative, both in isolation and within the full model. Each showed strong predictive power when used alone (cvLL ≈ 0.40; cvAcc ≈ 0.14–0.16) and retained substantial unique contributions in the full model (ΔLL ≈ 0.20 for Distance; ΔAcc ≈ 0.08). While 2h Memory reflects the use of relevant mnemonic information and task-engaged and goal-directed

behavior, Euclidean Distance captures a spatial heuristic that is not explained by other task variables. These findings indicate that recall behavior is shaped by the interplay between internal memory states and external spatial contextual information—each offering non-redundant explanatory value.

In contrast, other variables such as General Bias, History Bias, and 24h Memory also performed well in isolation but showed limited unique contributions (ΔLL and ΔAcc < 0.05), suggesting that much of their explanatory power overlaps with more dominant predictors. The Non-rewarded Win-Stay/Lose-Switch regressor showed the weakest influence overall.

Together, these results reveal a layered structure of behavioral control. While several regressors encode overlapping information, 2h Memory and Euclidean Distance stand out as complementary, high-impact variables that jointly anchor animals' recall-based poking behavior. These findings underscore the value of decomposing behavior into latent components to uncover how memory and spatial heuristics jointly shape recall.

Until now, these results complete a model-centric view of recall-based poking behavior. We have shown that the GLM-HMM captures structured, computationally distinct strategies; that these strategies are expressed behaviorally and rely on task-specific as well as task-agnostic navigational variables; and that, in particular, two core variables—2h Memory and Euclidean Distance—anchor most of the model's predictive power.

This concludes our characterization of the model itself: its fit, structure, behavioral signatures, and explanatory capacity. Having validated the model and its components, and our interpretation of its hidden states, we now turn to a different question—how these latent strategies are deployed across time and individuals.

Chapter 3.6. Temporal Dynamics of State Usage

To better understand how these four strategies unfold over time, we analyzed the temporal structure of latent state expression—specifically, how frequently each strategy is used (occupancy), how long it persists when active (dwell time), and how predictably animals transition between them (transition entropy). These metrics reveal how memory-guided behavior is organized—not as a continuous drift, but as a sequence of discrete cognitive states. By characterizing the dynamics of these states, we gain a framework for dissecting behavioral variability and for linking internal strategy switches to underlying neural mechanisms, in the future.

The results, summarized in Figure 10, revealed consistent structure in strategy dynamics. The 2h memory-based state showed both the highest occupancy and the longest dwell time, suggesting that this goal-directed recall strategy is dominant and temporally stable. Once engaged, animals tended to remain in this state across multiple pokes, reflecting sustained

use of task-relevant memory. In contrast, the 24h memory-based and lapse/disengaged states were expressed more transiently, with shorter dwell times and higher transition entropy—indicative of more unstable or stochastic dynamics. The memory-avoidance state fell in between: less stable and frequent than the 2h state, but more structured and recurring than the lapse state.

To further characterize the stability of these strategies, we examined the average transition matrix. As expected under a sticky HMM, self-transitions dominated the dynamics, indicating that once a strategy is engaged, animals tend to remain in that state for multiple pokes and trials. This is important because stability distinguishes genuine, persistent strategies from momentary noise, quantifies their control timescale, and reveals how manipulations reorganize decision policies even when mean accuracy looks similar.

We also computed the entropy of each model's transition matrix as a measure of how predictable state-to-state transitions were. Low-entropy states—such as the 2h memory-based state—exhibited strong persistence, with animals frequently remaining in the same state across pokes. In contrast, the 24h memory-based and lapse states had higher entropy, reflecting more variable and less structured switching patterns. These differences offer insight into how reliably each strategy is deployed and may serve as markers of underlying cognitive stability or volatility.

These results highlight that among the different strategies, the 2h memory-based state is not only the most predictive and goal-directed, but also the most stable and dominant across time. Its high occupancy, long dwell times, and low transition entropy suggest that animals preferentially engage this strategy when memory is preserved and relevant. In contrast, other states—such as the 24h memory or lapse-like strategies—appear more transient and volatile, suggesting they may reflect moments of interference or disengagement rather than sustained behavioral modes.

While these findings reveal how recall behavior is structured over time, they reflect average patterns across animals and sessions. Yet cognitive strategies are unlikely to be uniformly deployed: animals may differ in how consistently they use each strategy, how this use changes over time, or how they adjust behavior within trials. We next turn to this finer-grained structure—examining how latent strategies vary across individuals, evolve during sessions, and unfold from moment to moment.

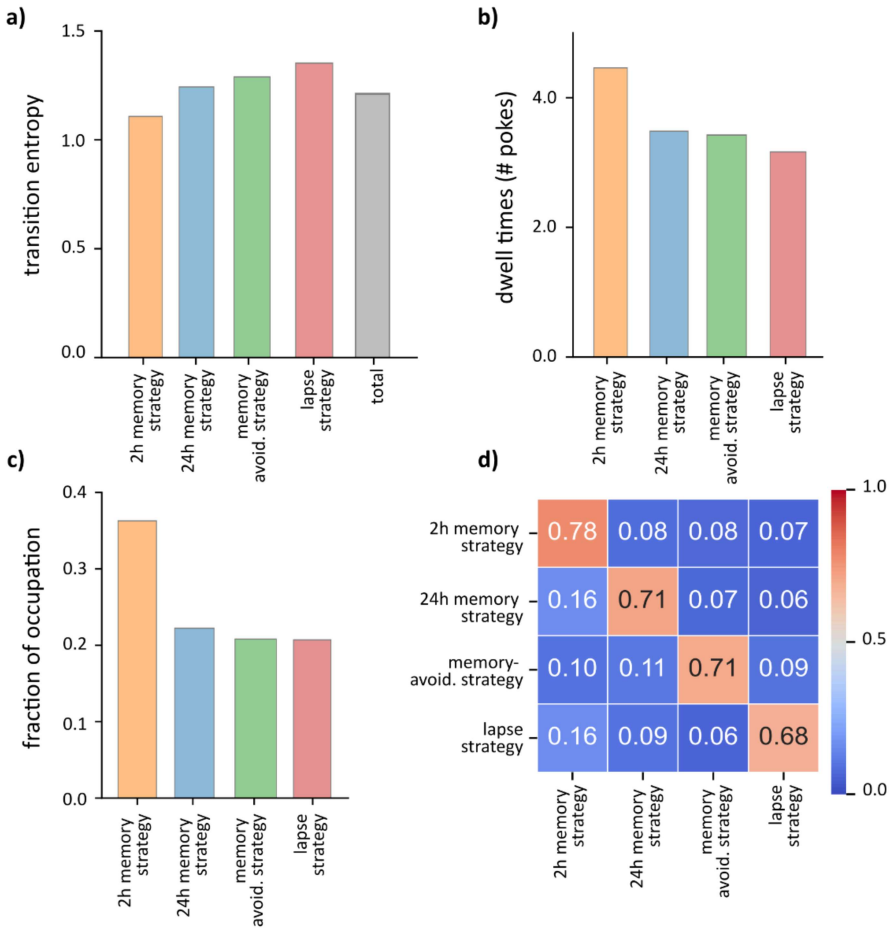

Figure 10: The 2h memory-based strategy dominates and persists longer than alternative strategies. (a) Transition entropy for each strategy, where the y-axis indicates entropy and the x-axis lists latent states. The 2h memory-based state showed the lowest entropy, while the 24h memory and lapse states exhibited higher, more variable transitions. (b) Dwell times, shown as the average number of consecutive pokes on the y-axis across strategies on the x-axis, were longest for the 2h memory-based state and shorter for the 24h memory and lapse states. (c) Fraction of occupation, plotted as the proportion of pokes on the y-axis for each strategy on the x-axis, revealed that the 2h memory-based state dominated across sessions, followed by memory-avoidance and the less frequent 24h memory and lapse states. (d) The average transition matrix (color scale: 0–1) displays the probability of switching between strategies; strong diagonal elements indicate dominant self-transitions, with the 2h memory-based state being the most stable.

Chapter 3.7. From Aggregated-Level Structure to Subject- and Trial-Specific Dynamics

While the previous sections established that recall behavior is structured around a discrete set of latent strategies, it remains unclear how these strategies are expressed across

individuals, sessions, and trials. At the population level, the *2h memory-based strategy* dominates—but do all animals rely on it to the same extent? Are alternative strategies—such as those reflecting interference or disengagement—expressed uniformly, or do they fluctuate with internal state or task demands among different animals? Here, we address these questions by examining how strategy expression varies across subjects, unfolds over time, and adapts within trials.

We began by quantifying how often each animal occupied each latent state. As shown in Figure 11 a, although the *2h memory-based strategy* was the most prevalent overall, its usage varied substantially across subjects. Some animals relied on it in nearly half their pokes, while others showed much lower engagement levels. This heterogeneity suggests that, even though goal-directed recall is dominant at the population level, not all animals adopt it equally. In contrast, memory avoidance and lapse strategies were used more sparsely and with less individual variability. These initial differences suggested that, despite a dominant population-level state, animals may differ in how they balance structured and alternative strategies.

To understand the structure of this variation, we examined pairwise correlations between state occupancies (Figure 11 c). Notably, animals that frequently engaged the *2h memory strategy* were less likely to use the *24h memory strategy*, suggesting a trade-off between reliance on recent versus older memory traces. A similarly strong anti-correlation emerged between *2h memory* and *memory-avoidance* strategies, while *24h memory* and *avoidance* were positively correlated—suggesting that animals either engage in goal-directed recall or co-express other strategies. These reciprocal patterns reveal stable, subject-specific preferences for strategy use, confirming that the latent states reflect consistent behavioral modes across animals—rather than just population-level averages. Together, these patterns point to a core axis of individual variation: some subjects consistently engage in goal-directed recall, while others more frequently default to alternative strategies (Note: the model-based analysis in this chapter is limited to the no-reward epoch of the recall sessions; animals do consistently visit the reward port, which matches the 2h memory, once reward becomes available and they receive their first reward.).

While strategy usage varied markedly across individuals, it remained unclear whether these differences reflect fixed, subject-specific preferences—or whether they arise from dynamic fluctuations in internal state that play out across time. In other words, do animals consistently rely on the same strategy throughout a session, or do they shift between strategies as memory demands, motivation, or engagement change?

To address this, we examined how both strategy usage and posterior probabilities evolved over the course of a session. We aligned individual pokes to the beginning of each session

and tracked changes in latent state expression across poke indexes (see Figure 11 b). This revealed systematic, time-dependent dynamics in how strategies were expressed.

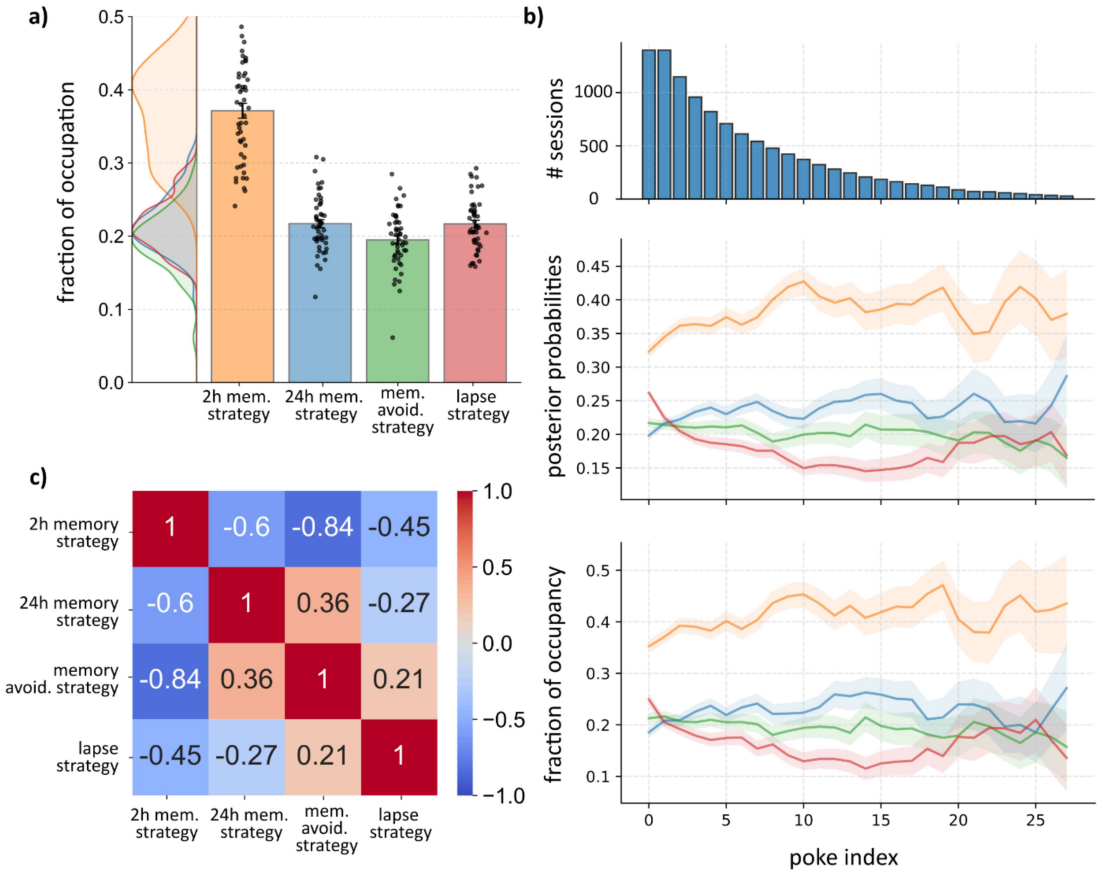

Figure 11: Strategy expression varies across individuals and evolves systematically within sessions. (a) Fraction of occupation for each latent strategy across animals. x-axis: latent strategies; y-axis: fraction of pokes. The 2h memory-based strategy dominated at the population level, but its use varied widely across individuals, ranging from nearly half of pokes in some animals to much less in others. Memory-avoidance and lapse strategies were less frequent and showed lower inter-individual variability. Bar plots show the distribution across animals; dots represent individual subjects. (b) Within-session dynamics. Top: distribution of session lengths (x-axis: number of pokes; y-axis: number of sessions). Middle: posterior probabilities of each state across pokes (x-axis: poke index; y-axis: posterior probability). Bottom: fraction of occupation over the session course (x-axis: poke index; y-axis: fraction of pokes). The 2h memory strategy was consistently expressed from early pokes, whereas the 24h memory and lapse strategies increased toward the session's end; the memory-avoidance strategy remained stable. (c) Pairwise correlations between state occupancies across animals (color scale: correlation coefficient –1 to 1). The 2h memory strategy was negatively correlated with 24h memory (–0.60), memory-avoidance (–0.84), and lapse (–0.45), while 24h memory and memory-avoidance were positively correlated (0.36). Lapse showed weaker negative correlations with the other strategies.

The *2h memory-based* strategy remained relatively stable, maintaining a high and consistent expression throughout the beginning of the session. In contrast, the *24h memory-based* strategy increased gradually, particularly toward the session's end—suggesting that

interference from long-term memory traces becomes more influential over time, probably due to the loss of confidence of obtaining the reward in the port of that day. The *lapse strategy* also showed a modest increase near session boundaries. Meanwhile, the *memory-avoidance* strategy remained stable across the session, suggesting it reflects a tonic, context-insensitive strategy. Statistical trend analysis confirmed that these changes were significant in the end-aligned data, supporting the view that recall behavior becomes increasingly focused on relevant memory as sessions progress.

These results indicate that recall strategies are not used uniformly across time. While goal-directed recall remains consistently expressed, the growing prevalence of 24h memory and lapse strategies suggests that internal state dynamics—such as interference, fatigue, or disengagement—increasingly shape strategy selection. Rather than a simple decay in memory-based control, these dynamics reflect a redistribution of behavioral modes, with some strategies gaining influence while others remain stable—or even decline—at higher trial counts.

To further dissect temporal dynamics, we asked whether strategy use remains fixed within trials or adapts as behavior unfolds. Specifically, we tested whether animals commit to a single internal state per trial, or switch strategies across pokes. To test this, we grouped pokes by their position within each trial (e.g., first, second, third) and computed the probability of each latent state at each timepoint, stratified by trial length (Figure 12).

The results revealed structured, time-dependent shifts in strategy expression. Single-poke trials were strongly dominated by the 2h memory-based strategy, suggesting that these one-shot retrievals are often guided by accurate, goal-directed recall. In contrast, multi-poke trials showed more variable patterns. In 2-poke trials, 2h memory was less prominent in the first poke but increased on the second, suggesting a corrective switch toward memory-based behavior. In longer trials, initial pokes were increasingly associated with 24h memory-based or memory-avoidance strategies, while 2h memory expression was weaker and more diffuse—but increased reliably as the trial progressed, eventually even reaching the occupancy levels observed in single-poke trials. Again consistent with the animals tending to correct mistakes in earlier pokes.

This pattern indicates that animals do not rigidly commit to a single latent strategy per trial. Instead, they flexibly adjust their internal state as the trial unfolds. In multi-poke trials, initial pokes often reflect more variable or suboptimal strategies—such as 24h memory-based or memory-avoidance —but subsequent pokes show a gradual increase in the occupancy of the 2h memory-based state. This progression suggests a dynamic engagement of goal-directed recall, potentially reflecting delayed memory access, an internal resolution of competing strategies or as previously stated, a corrective switch toward memory-based behavior. The shift was particularly evident after the second poke, with significant increases in 2h memory occupancy across longer trials ($p < 0.05$, mixed ANOVA). These results highlight the flexibility

of strategy use within trials and suggest that animals dynamically refine their behavior as uncertainty resolves.

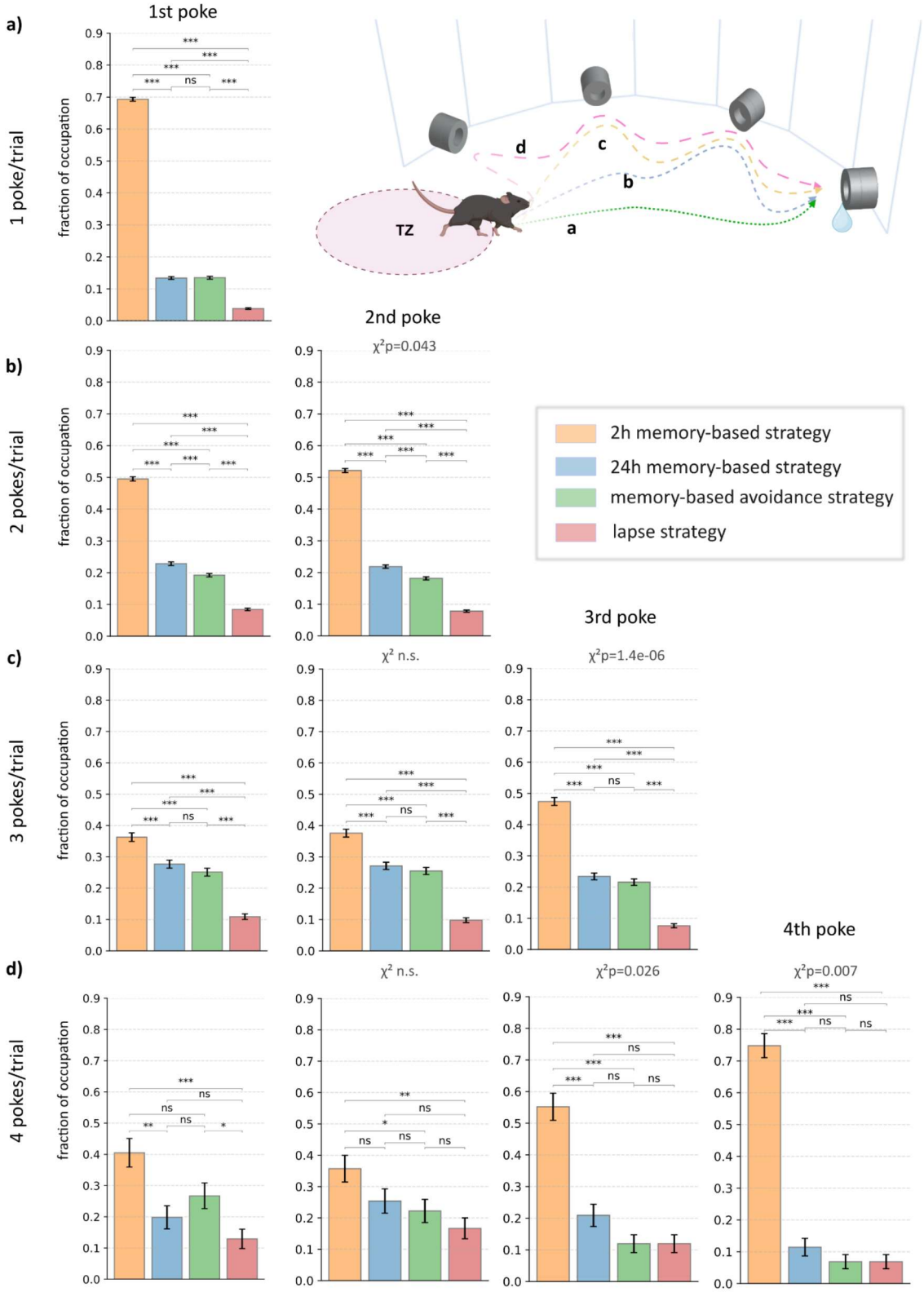

Figure 12: Animals flexibly switch strategies within trials, converging on 2h memory-based strategy. (a–d) Fraction of pokes occupied by each latent strategy, grouped by poke position and stratified by trial length (1–4 pokes). x-axis: poke position within the trial; y-axis: fraction of pokes. Bars show mean ± SEM across pokes; colors denote latent states inferred by the GLM-HMM (orange = 2h memory-based strategy; blue = 24h memory-based strategy; green = memory-avoidance strategy; red = lapse strategy). Asterisks above bars indicate pairwise *t*-tests between strategies within a given poke (*p < 0.05, **p < 0.01, ***p < 0.001). Results of χ^2 tests comparing overall distributions between successive pokes are reported in the panels. Single-poke trials were dominated by the 2h memory-based strategy. In 2-poke trials, engagement of 2h memory increased significantly on the second poke. In 3- and 4-poke trials, initial pokes showed greater reliance on 24h memory and avoidance strategies, but later pokes revealed significant shifts toward 2h memory, ultimately reaching levels comparable to single-poke trials.

Chapter 3.8. Spatial Modulation of Latent Strategies: A Model-Based View of Memory Interference

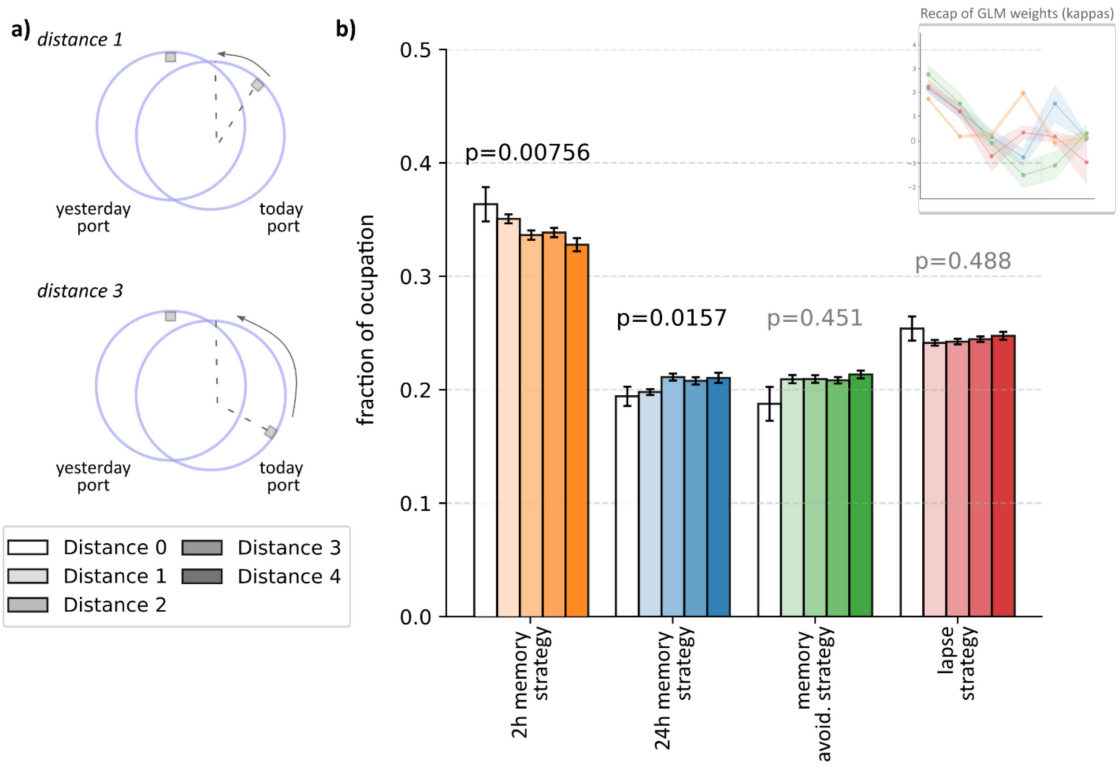

Figure 13: Spatial distance between reward locations modulates latent strategy expression. (a) Schematic illustration of angular distances between the reward port from the previous day (yesterday port) and the currently rewarded port (today port). Distance 1 indicates a nearby separation, while Distance 3 represents a larger angular separation. (b) Fraction of pokes dominated by each latent strategy (posterior probability ≥ 0.8) as a function of distance between the 2h and 24h reward ports. x-axis: distance (0–4); y-axis: fraction of pokes. Colors denote the four GLM-HMM states (orange = 2h memory strategy; blue = 24h memory strategy; green = memory-avoidance strategy; red = lapse strategy). Bars show mean ± SEM across sessions. The 2h memory strategy was significantly more prevalent when the previous reward location was close

and declined with increasing distance (p = 0.00756, repeated-measures ANOVA). The 24h memory strategy increased modestly but significantly at larger distances (p = 0.0157). Memory-avoidance and lapse strategies showed no significant modulation by distance (p = 0.451 and p = 0.488, respectively). Inset: Recap of GLM weight profiles (κ values) for each state, illustrating their temporal modulation across pokes.

We have demonstrated that animals dynamically switch between internal strategies over the course of a session. However, these transitions were modeled independently of a key variable highlighted in Chapter 2: the spatial distance between the current and previous reward locations. There, we observed that memory strength—as measured by the Memory Index (MI)—systematically attenuates with increasing distance between today and yesterday reward ports. This pattern was consistent with interference between overlapping memory traces, but our interpretation was limited by a potential confound: because the MI is computed over spatial distributions, it could be sensitive to geometric artifacts inherent to the maze layout.

Here, we revisit the question of memory interference using a different lens. Rather than relying on the MI, we take advantage of the trial-level resolution of the GLM-HMM to ask whether the expression of latent behavioral strategies is modulated by the distance between current and past reward locations. This approach provides a conceptually orthogonal test of interference—one that is independent of the MI's spatial structure and its potential geometric artifacts. To do this, we first grouped pokes according to their most likely latent state (posterior probability ≥ 0.8), and then examined how these state-dominant pokes were distributed across sessions with different angular distances between the currently rewarded port and the one rewarded 24 hours earlier. Crucially, the model itself was not re-fit; instead, we used the original state assignments to ask whether the fraction of occupation of each strategy varies systematically with the spatial relationship between past and present goals (see Figure 13 b).

This analysis revealed that the expression of memory-based strategies is not spatially uniform. One strategy, associated with recall of the 2h reward location, was used more frequently when the previously rewarded port was nearby, and declined as spatial distance increased. In contrast, the strategy associated with 24h memory recall showed a modest but significant increase at longer distances. A test of repeated-measures one-way ANOVA, which revealed that both strategies were significantly modulated by spatial proximity to the previous reward.

These results are important for two reasons. First, they demonstrate that the spatial interference effect described in Chapter 2 generalizes beyond the Memory Index: it is not simply a byproduct of the way recall is quantified. Instead, it is reflected in how the animal's internal strategy allocation shifts as a function of the spatial interplay between current and past reward memories. Second, this shift occurs independently of any geometric constraints

imposed by the MI computation itself. The GLM-HMM captures strategy changes based solely on patterns of choice behavior—making no reference to angular distributions or symmetry—therefore providing a model-based validation of the interference hypothesis.

Together, these results indicate that the GLM-HMM not only captures internal strategy dynamics, but also reflects how spatial overlap between past and current experiences shapes cognitive state expression. This model-based signature of interference complements the behavioral results from Chapter 2 and shows that memory competition is not simply expressed as performance degradation, but rather as a shift in internal strategy.

Chapter 4: Autoantibody-Induced Disruption of Latent Strategy Dynamics during Memory Recall in the anti-NMDAR Encephalitis Model.

In Chapter 3, we showed that recall behavior in healthy animals is governed by a limited set of latent cognitive strategies, expressed dynamically across trials and time. These strategies reflect the animal's internal state and support, for example, memory-guided navigation, including recent memory recall, interference from past experiences, and the use of spatial heuristics.

Although these strategies are inferred from behavior, their expression must depend on underlying neural mechanisms that support the encoding, maintenance, and retrieval of memory. Seminal clinical cases —such as that of patient H.M.— demonstrated that the hippocampus is essential for the formation and retrieval of declarative memories, enabling the use of stored experiences to guide behavior (Scoville & Milner, 1957). In rodents, the hippocampus has been shown to support spatial recall and memory-guided navigation through stable yet flexible patterns of neuronal activation over time, called place cells, the strongest candidates of memory cells in the brain (O'Keefe & Dostrovsky, 1971). Altogether, these findings make mice a suitable mammalian model for studying memory-guided behavior.

One mechanism thought to support these hippocampal functions — namely the encoding, stabilization, retrieval, and updating of memory representations — is NMDA receptor–dependent synaptic plasticity, which plays a central role in modifying synaptic strength during learning. NMDA receptors (NMDARs) are critical for long-term potentiation and for stabilizing memory traces over time. Disruption of NMDAR has been shown to impair recall, degrade spatial specificity, and interfere with the updating of internal neural representations (Morris et al., 1986).

To directly test whether the expression of latent strategies during memory recall depends on NMDA receptor–mediated plasticity, we turned to a well-established mouse model of anti-NMDAR encephalitis. This model mimics a rare autoimmune disorder in humans, in which the immune system mistakenly produces antibodies that target NMDA receptors in the brain. In patients, these antibodies primarily bind to the GluN1 subunit of the NMDA receptor, a critical component involved in synaptic signaling and plasticity. This leads to a broad range of cognitive and neuropsychiatric symptoms, including memory loss, disorientation, hallucinations, and motor disturbances—highlighting the importance of NMDAR function for memory.

In this clinical condition, the antibodies cause NMDA receptors to be internalized and removed from the neural surface, effectively silencing their function without inducing cell death. The resulting reduction in receptor availability disrupts synaptic transmission and impairs the brain's ability to adaptively modify its connections—an essential process for learning and memory consolidation.

To model this condition in rodents, we infused patient-derived monoclonal anti-NMDAR antibodies directly into the lateral ventricles (intracerebroventricular infusion), producing a controlled and targeted disruption of NMDA receptor function. Previous studies have shown that this approach reduces surface expression of NMDA receptors by up to 60%, particularly in the hippocampus and prefrontal cortex, regions that are crucial for memory encoding, spatial navigation, and flexible decision-making. Importantly, this manipulation does not cause tissue damage or inflammation, allowing us to isolate the effects of NMDAR loss without confounding injury (Planagumà et al., 2015). This mouse model captures the post-acute, recovery phase of the disease—when patients are no longer in the severe stage but still present marked memory impairments.

To adjust this murine model of anti-NMDAR encephalitis with our behavioral paradigm, we implemented a chronic delivery protocol, with daily antibody infusions over a 24-day period. This approach ensured sustained disruption of NMDAR function, allowing us to track how changes in synaptic plasticity influence the expression and dynamics of internal memory strategies. This design provides a controlled framework to test whether the structured behavioral strategies observed in healthy animals depend on intact NMDA receptors–mediated plasticity (see Figure 14 a for a schematic timeline; full details in Methods section 1.4.1).

- **Hypothesis:** If recall behavior relies on structured internal strategies dependent on memory related neuronal substrates, then autoantibody-induced disruption of NMDA receptors should alter the expression of these strategies, due to its role in formation and maintenance of memories. Specifically, we hypothesized that anti-NMDAR antibodies would affect the frequency, flexibility, and modulation of memory-guided strategies—leading to changes in latent state dynamics during recall, even if overall task performance (e.g., accuracy) remains unaffected.

Chapter 4.1. Altered State Occupancy in the anti-NMDAR Animal Model.

To assess whether synaptic disruption alters the deployment of recall strategies, we compared the frequency of latent state occupancy across two matched time windows: the 24-day period immediately preceding antibody infusion (baseline), and the 24 days during which animals received daily anti-NMDAR treatment. This within-subject design allowed us to track changes in strategy use over time while controlling for inter-animal variability

As shown in Figure 14 b, anti-NMDAR animals exhibited clear deviations from their pre-infusion strategy profiles. In the control condition, the 2h memory-guided state was the most frequently expressed overall, while other strategies—24h memory, memory-avoidance, and lapse—were expressed at comparably lower levels. Animals receiving anti-NMDAR antibody infusions showed a selective increase in the 2h memory state, accompanied by a reduction in non-goal-directed states, particularly lapses and memory-avoidance.

The changes in the fraction of occupation suggest that anti-NMDAR antibodies do not simply impair memory recall, but rather alter the balance of internal strategies over time—biasing behavior toward shorter-term memory use while suppressing alternative strategies. To examine whether this shift was expressed consistently across animals, we next assessed state usage at the subject level.

Chapter 4.2. Subject-Level Expression of Latent Strategies in the anti-NMDAR Animal Model.

To determine whether anti-NMDAR exposure alters not only the overall prevalence of recall strategies but also their expression at the individual level, we quantified the fraction of time each subject spent in each latent state across conditions (see Figure 14 d). This analysis revealed that, compared to their baseline (pre-infusion) period, animals receiving anti-NMDAR antibodies showed a consistent increase in the use of the 2h memory strategy across subjects ($p < 0.05$). In contrast, the occupancy of non-goal-directed states—particularly lapse and memory-based avoidance—was reduced ($p < 0.05$). These effects were robust at the individual level, suggesting that the presence of anti-NMDA antibodies reshapes strategy allocation in a systematic manner rather than introducing random variability (see Figure 14 e).

To assess whether antibody exposure altered the relational structure of strategy use, we compared pairwise correlations between state occupancies in the NMDA group and their corresponding controls. In the control condition, 2h memory occupancy was strongly negatively correlated with both memory-avoidance and lapse strategies, and also showed a weaker negative correlation with 24h memory. A positive correlation emerged between 24h memory and memory-avoidance, while other associations remained weak. Following anti-NMDAR infusion, this structure was preserved in part but with notable changes. Negative correlations between 2h memory and alternative states (24h memory, memory-avoidance, lapse) remained present, but were generally stronger than in controls. At the same time, correlations among the non-goal-directed strategies (24h memory, avoidance, lapse) increased slightly, with positive associations emerging more clearly (Figure 14 c). These results indicate that anti-NMDAR treatment reshaped the correlation structure of strategy deployment across animals, reinforcing the antagonism between goal-directed recall and alternative strategies while clustering non-memory-based modes together.

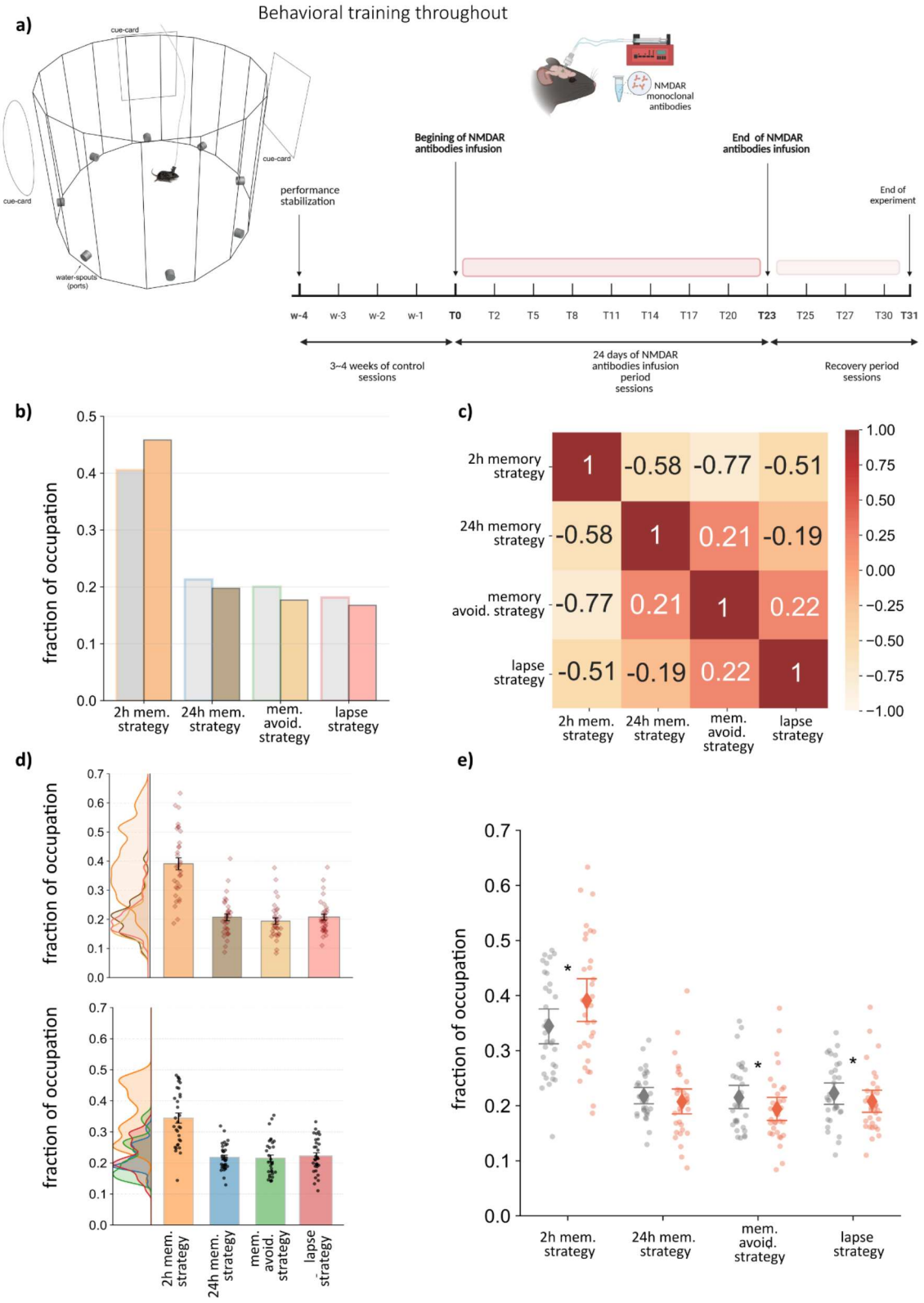

Figure 14: Anti-NMDAR antibodies bias recall behavior toward shorter-term memory strategies.

(a) Experimental design. A cohort of n = 35 mice performed baseline recall sessions, followed by 24 days of daily intracerebroventricular infusion of patient-derived anti-NMDAR antibodies, and a recovery period after infusion ended. (b) Fraction of occupation of latent strategies during antibody infusion. x-axis: strategy identity; y-axis: fraction of occupation. The 2h memory strategy dominated overall, while 24h memory, memory-avoidance, and lapse states were less frequent.(c) Correlation matrix of state occupancies across animals. Values represent Pearson correlations. The 2h memory strategy was strongly negatively correlated with alternative states, while 24h memory, memory-avoidance, and lapse strategies showed positive associations.(d) Fraction of occupation across baseline (top) and antibody infusion (bottom) periods. Bars show mean ± SEM; dots represent individual animals. (e) Subject-level comparison of strategy occupancy between baseline (gray) and antibody infusion (orange). Each dot corresponds to an animal; error bars indicate mean ± SEM. Compared to baseline, anti-NMDAR exposure increased the prevalence of the 2h memory strategy and reduced lapse and avoidance strategies ($p < 0.05$, paired tests).

While these results demonstrate that anti-NMDAR exposure alters strategy allocation across subjects, they leave open an important question: How do these effects unfold over time? Given the chronic nature of the manipulation—daily antibody infusions over 24 days—it is critical to ask whether the observed changes emerge gradually, stabilize early, or fluctuate throughout the treatment period. To address this, we next examined the evolution of state usage across sessions in anti-NMDAR animals.

Chapter 4.3. Temporal Dynamics of Strategy Use across Sessions in the anti-NMDAR Animal Model.

Given the chronic nature of the manipulation, we next examined how strategy use evolved across sessions during the 24-day antibody infusion period. Figure 15 a shows the trajectory of each latent state, aligned to the first infusion (Session 1).

Once the infusion started, in comparison with the baseline (Day 0), animals exhibited a sustained increase in the expression of the 2h memory strategy during the early phase of treatment, with levels rising within the first four days. Around session 10, we observed a sharp drop in 2h memory occupancy, accompanied by a rise in lapse state expression. Although the underlying mechanism remains unclear, this transient shift suggests a temporary disruption in memory-guided behavior, potentially reflecting instability in strategy deployment during ongoing synaptic perturbation, although further, more detailed analyses are needed.

Following the drop, 2h memory strategy usage stabilized. During the second half of the infusion period, the 2h memory state regained prominence, while non-goal-directed states progressively declined. Notably, once antibody infusions ceased, the 2h memory strategy again dropped to values prior to the infusion of the antibodies, and alternative states increased. This pattern suggests that the behavioral effects of anti-NMDAR antibodies are

closely linked to the presence of circulating antibodies, and are rapidly reversible once the infusion of antibodies ends, as previously reported by *Planagumá et al. 2015.*

To determine whether animals still adjusted their memory strategies based on spatial separation between rewards, we next examined whether strategy use remained sensitive to spatial interference following synaptic disruption.

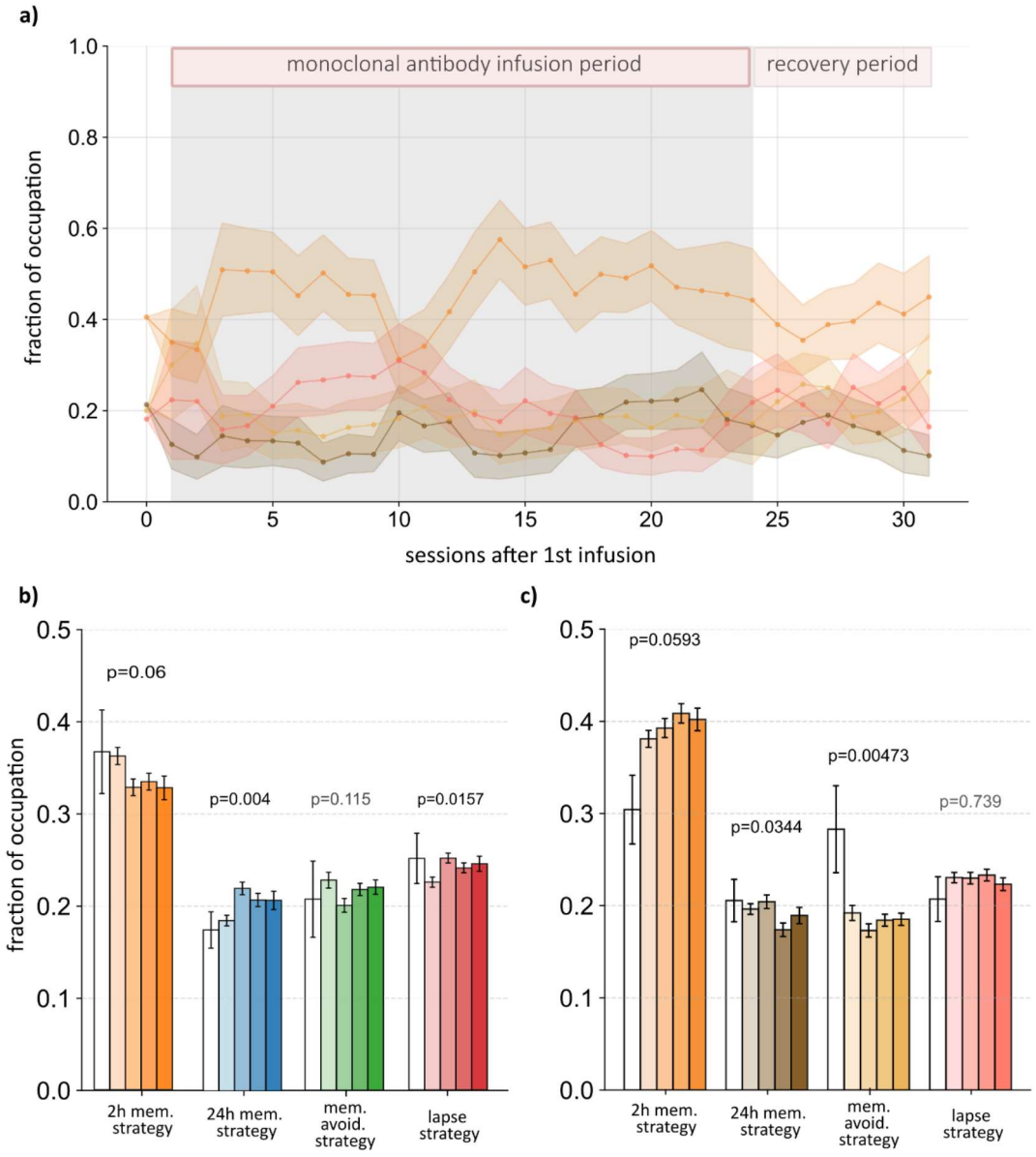

Figure 15: Anti-NMDAR antibodies reverse interference relationship with port distance. (a) Time course of latent-state occupancy aligned to the first infusion session. x-axis: session number after first infusion; y-axis: fraction of occupation. Colored lines denote the four latent strategies (see color key in the figure); shaded bands show SEM across animals. The pale-red band marks the 24-day infusion period; the lighter band marks recovery. Fraction of occupation by each latent

strategy (posterior probability ≥ 0.8) as a function of the angular distance (0–4) between the currently rewarded port (2h ago) and the port rewarded 24h earlier, computed from the 24 control sessions preceding antibody infusion. x-axis: latent strategies; y-axis: fraction of occupation. Bar hue indicates the four GLM-HMM strategies (orange = 2h memory, blue = 24h memory, green = memory-avoidance, red = lapse); saturation encodes spatial distance (light = distance 0, dark = distance 4). Bars show mean ± SEM across sessions; *p*-values reflect repeated-measures ANOVA across distances for each strategy. (c) Same analysis as in (b), but for the 24 sessions during antibody infusion. Conventions as in (b). Reported *p*-values indicate repeated-measures ANOVA across distances within each strategy.

Chapter 4.4: Memory-Based Strategies Lose Spatial Sensitivity in anti-NMDAR Animal Model.

In Chapter 3, we showed that the expression of memory-based strategies in control animals varies systematically with the spatial distance between the current and previous reward locations—a model-derived signature of memory interference. Specifically, animals were more likely to rely on recent (2h) memory when the previous reward occurred nearby, but shifted toward 24h memory strategies as the spatial separation increased. This shift reflects a growing interference between memory traces at larger distances, which reduces the reliability of short-term recall and leads animals to rely more heavily on longer-term strategies.

This distance-dependent modulation suggests that in untreated animals, memory strategies are flexibly adjusted in response to interference. To test whether this capacity is preserved following synaptic disruption, we next examined whether strategy use in anti-NMDAR animals remained sensitive to the spatial separation between consecutive days reward locations. Specifically, we grouped trials according to the angular distance between the currently rewarded port and the port rewarded 24 hours earlier, and quantified the relative occupancy of each latent state as a function of this distance Figure 15 c and 15 d. As in previous analyses, control data corresponds to the same animals during the 24-day baseline period preceding anti-NMDAR antibody infusion. If anti-NMDAR treatment disrupts the brain's ability to resolve memory interference, this distance-dependent pattern should be also disrupted.

Consistent with this prediction, anti-NMDAR treatment suppressed the spatial modulation of memory strategy use. As in previous analyses, control data shown here correspond to the same animals during the 24-day period preceding antibody infusion. In the pre-infusion condition, animals showed a clear pattern of distance-dependent interference: the probability of using the 2h memory strategy declined with increasing spatial distance, while 24h memory use rose accordingly—consistent with flexible adjustment of recall strategies on the distance between consecutive reward locations. This distance-dependent pattern was disrupted following anti-NMDAR treatment. The 24h memory strategy no longer increased with spatial separation but instead remained flat, suggesting a loss of sensitivity to

contextual interference. Strikingly, the 2h memory strategy exhibited a trend toward increased use with distance—an inverted pattern compared to controls, although this effect is clearly recognized on the output analysis, did not reach statistical significance.

These findings suggest that in anti-NMDAR animals, the contribution of longer-term (24h) memory to behavior is diminished, likely due to impaired consolidation. Moreover, the use of memory strategies no longer varied with the spatial separation between consecutive rewards, pointing to a breakdown in the interference-sensitive dynamics that normally shape recall behavior.

Chapter 5: Autoantibody-Induced Disruption of Latent Strategy Dynamics during Memory Recall in the anti-LGI1 Encephalitis Model.

In the previous chapter, we showed that the presence of autoantibodies targeting NMDA receptors disrupts the flexible deployment of latent cognitive strategies during memory recall, without necessarily impairing overall task performance. The anti-NMDAR encephalitis is one example of a wider class of autoimmune encephalitides, in which circulating antibodies disrupt synaptic function by targeting specific neuronal surface proteins.

In this chapter, we turn our attention to another member of this clinical family: LGI1 antibody–mediated encephalitis, the second most prevalent autoimmune encephalitis after anti-NMDAR encephalitis (Lancaster & Dalmau, 2012), a distinct but related autoimmune disorder. Like anti-NMDAR encephalitis, LGI1 encephalitis involves the production of pathogenic antibodies that influence synaptic signaling—yet it targets a different molecular pathway. Specifically, the Leucine-rich Glioma-Inactivated 1 (LGI1) is a synaptic protein that plays a critical role in stabilizing excitatory synapses, particularly within the hippocampus and medial temporal lobe—regions essential for memory formation and consolidation. LGI1 regulates synaptic plasticity by modulating postsynaptic excitability and maintaining synaptic strength over time, primarily through its interactions with Kv1 potassium channels and the synaptic adhesion protein ADAM22(Fukata et al., 2006). Disruption of LGI1 function has been shown to impair long-term potentiation (LTP), reduce dendritic spine stability, and influence hippocampus-dependent memory tasks (Ohkawa et al., 2013). These properties make LGI1 encephalitis a particularly compelling model for testing how synaptic dysfunction impacts the long-term organization and flexibility of memory-guided behavior.

To assess whether LGI1-dependent synaptic disruption alters the expression of latent recall strategies, we used the same experimental protocol as in Chapter 4, applying a previously established mouse model of anti-LGI1 encephalitis based on chronic intracerebroventricular infusion of patient-derived antibodies (Petit-Pedrol et al., 2018). Mice received daily infusions over a 14-day period, producing a sustained disruption of LGI1 function in hippocampal circuits. This manipulation has been shown to reduce synaptic LGI1 expression, weaken excitatory transmission, and impair hippocampus-dependent spatial memory, making it well-suited for probing strategy use during memory recall. (Petit-Pedrol et al., 2018). As in the case of the murine model of anti-NMDAR encephalitis, this model also captures the post-acute phase of the disease, when the acute symptoms have resolved but cognitive deficits persist.

- **Hypothesis:** If memory-guided behavior relies on intact synaptic signaling within hippocampal and temporal lobe circuits, then autoimmune disruption of LGI1 function should impair the flexible deployment of internal cognitive strategies—albeit through mechanisms that differ from those observed in anti-NMDAR encephalitis. Specifically, we hypothesized that anti-LGI1 antibodies would lead to disorganized or unstable expression of memory strategies, reduce sensitivity to interference, and increase reliance on non-goal-directed behavior.

Chapter 5.1. Altered Latent State Occupancy in the LGI1 Model.

To assess how LGI1 disruption impacts recall strategies, we compared latent state occupancy across two matched 14-day periods: baseline (pre-infusion) and the antibody infusion phase. As shown in Figure 16 b, anti-LGI1 animals exhibited a notable increase in 2h memory strategy use, accompanied by a subtler rise in 24h memory. In contrast, memory-based avoidance and lapse states both showed a modest decrease. This pattern differs from the anti-NMDAR model, where the redistribution of strategy use was more pronounced and skewed more strongly toward recent memory, at the expense of lapse and memory-based avoidance strategies. These results suggest that while LGI1-related synaptic disruption does alter the balance of internal strategies, the resulting profile is distinct and comparatively more balanced than that observed following NMDAR disruption.

Chapter 5.2. Subject-Level Expression of Latent Strategies.

At the individual level, animals receiving anti-LGI1 antibodies exhibited similar levels of variability in strategy use as control animals. While there was a slight trend toward increased use of both 2h and 24h memory strategies, and a modest decrease in lapse, none of these differences reached statistical significance. Memory-based avoidance occupancy remained stable across conditions. These results suggest that under the present conditions, LGI1-related synaptic disruption does not produce strong or consistent changes in individual recall strategy expression, and any observed trends should be interpreted with caution.

To further examine the structure of individual differences, we computed pairwise correlations between state occupancies in the LGI1 group and their corresponding controls. In the control condition, 2h memory was strongly negatively correlated with both 24h memory and memory-avoidance, and showed weaker negative associations with lapse. Positive correlations were observed between 24h memory and avoidance, while 24h memory was negatively correlated with lapse. Following LGI1 infusion, the negative correlations of 2h memory with 24h memory and avoidance were preserved, but associations with lapse weakened. At the same time, the negative correlation between 24h memory and lapse was reduced, while memory-avoidance became more strongly anti-correlated with lapse. Together, these subject-level correlations indicate that LGI1

exposure alters the relational structure of the strategies that animals most rely on during recall.

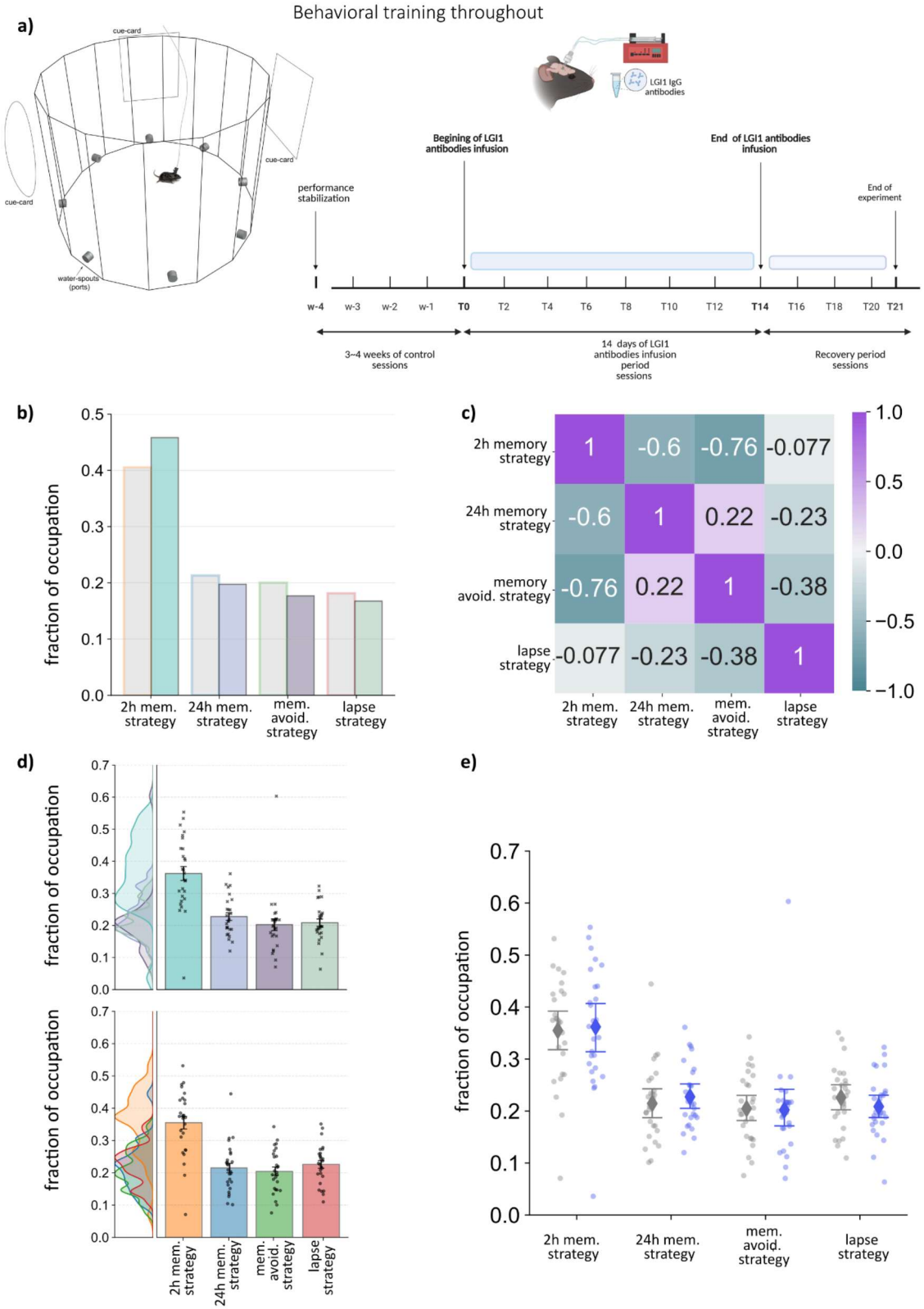

Figure 16: Altered latent strategy expression in the anti-LGI1 model. (a) Experimental design. Mice underwent baseline training and recall sessions, followed by 14 days of daily intracerebroventricular infusion of patient-derived anti-LGI1 antibodies, and a recovery period after infusion ceased. (b) Fraction of occupation of latent strategies during antibody infusion. x-axis: strategies; y-axis: fraction of occupation. The 2h memory strategy dominated, followed by lower levels of 24h memory, memory-avoidance, and lapse.
(c) Correlation matrix of state occupancies across animals in the LGI1 group. Negative correlations were observed between the 2h memory strategy and both 24h memory and memory-avoidance; associations with lapse were weaker. Positive correlations emerged between 24h memory and avoidance. (d) Fraction of occupation of each strategy across baseline (top) and antibody infusion (bottom) periods. x-axis: strategies; y-axis: fraction of occupation. Individual animals are shown as dots; bars represent group means ± SEM. (e) Subject-level comparison of state occupancy between baseline (gray) and infusion (blue). Each dot corresponds to one animal; error bars represent mean ± SEM. n= 30 animals.

Chapter 5.3. Temporal Dynamics of Strategy Use Across Sessions.

To assess how recall strategies evolved over time, we tracked state usage across sessions during and after the 14-day anti-LGI1 antibody infusion period. As shown in Figure 17 a, 2h memory increased early during antibody-infusion and remained stable, while 24h memory and memory-based avoidance declined modestly and plateaued. Notably, lapse occupancy showed a transient rise near the end of the infusion period, followed by a decline back to near pre-infusion levels once treatment ceased. This late fluctuation is reminiscent of the pattern observed in the anti-NMDAR model, where lapse rates also increased transiently before normalizing. Although the effects were milder here, the pattern suggests that the behavioral impact of LGI1 antibodies may also be closely tied to the presence of circulating antibodies, with potential for recovery once exposure ends. However, it remains possible that a longer post-infusion window would be necessary to observe full stabilization or compensatory shifts.

Chapter 5.4. Loss of Spatial Sensitivity in Memory-Based Strategies.

We next assessed whether LGI1-treated animals retained the ability to flexibly adjust memory strategies in response to the spatial distance between consecutive rewarded locations. In the baseline (pre-infusion) condition, animals exhibited a partial expression of interference-sensitive modulation: 2h memory use declined with increasing spatial distance, and memory-based avoidance increased, suggesting some adjustment away from recent memory under high-interference conditions. However, unlike what was observed in Chapter 3, 24h memory strategy use remained stable across distances, indicating an absence of compensatory engagement of longer-term memory strategies.

Following LGI1 antibody infusion animals exhibited a different and non-monotonic pattern of modulation. Specifically, 2h memory strategy usage increased at larger distances, but showed a transient dip at intermediate separations (Distance 1.0–2.0) (Figure 17 c). Interestingly, this dip was accompanied by a slight increase in 24h memory, suggesting that some degree of adaptive switching between shorter- and longer-term strategies remained intact, even if the overall structure was altered. Additionally, memory-based avoidance decreased significantly with distance, while lapse usage remained stable. These results indicate that spatial modulation was not abolished, but followed a different and less structured form compared to baseline, highlighting that LGI1-related synaptic disruption reshapes—rather than eliminates—interference-sensitive memory dynamics.

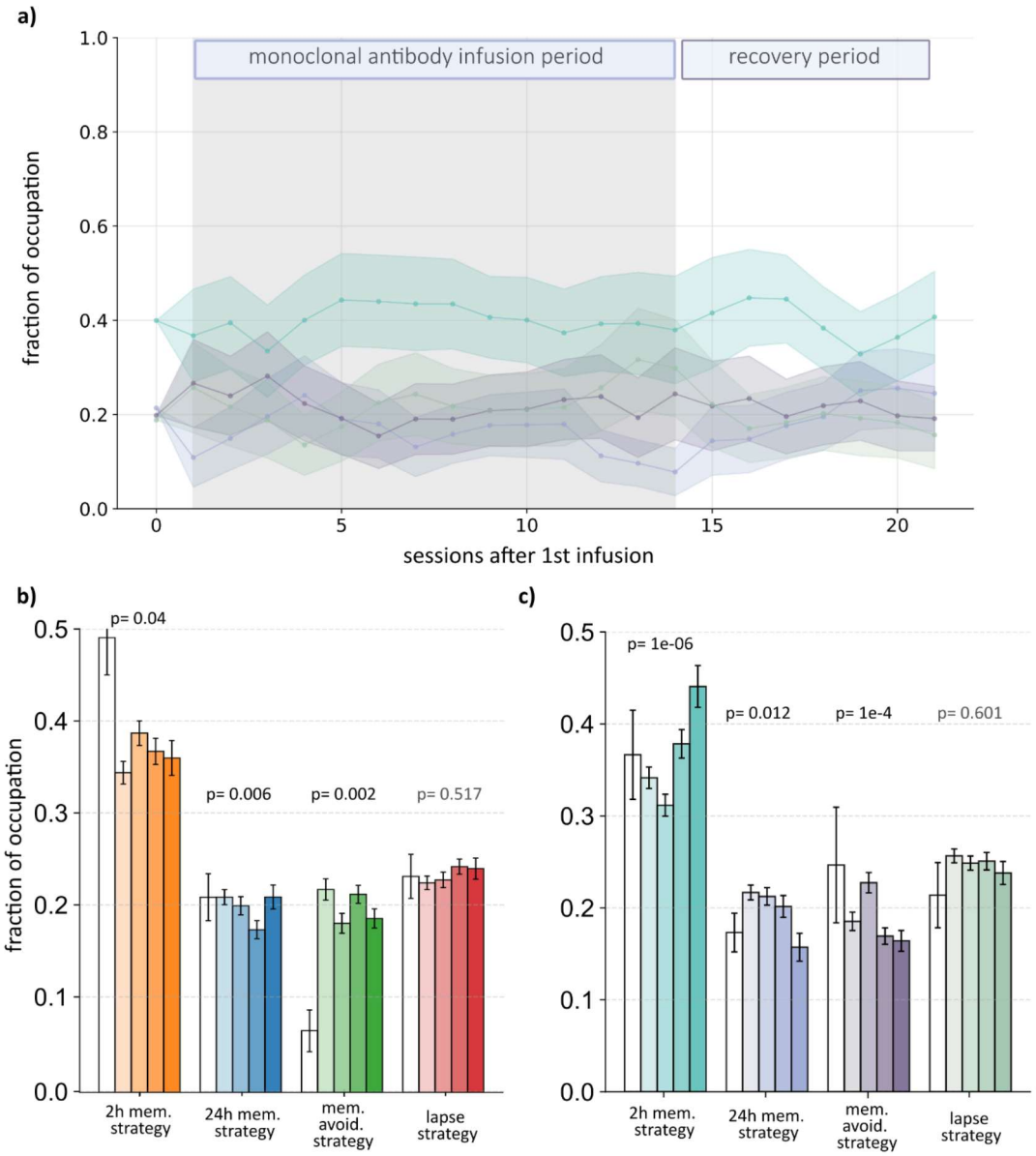

Figure 17: Anti-LGI1 antibodies induce a non-monotonic pattern of spatial interference in memory strategy use. (a) Time course of latent-state occupancy aligned to the first infusion session. x-axis: session number after first infusion; y-axis: fraction of occupation. Colored lines denote the four latent strategies (see color key in the figure); shaded bands show SEM across animals. The pale-blue band marks the 14-day infusion period; the lighter band marks recovery. Fraction of occupation by each latent strategy (posterior probability ≥ 0.8) as a function of the angular distance (0–4) between the currently rewarded port (2h ago) and the port rewarded 24h earlier, computed from the 14 control sessions preceding antibody infusion. x-axis: latent strategies; y-axis: fraction of occupation. Bar hue indicates the four GLM-HMM strategies (orange = 2h memory, blue = 24h memory, green = memory-avoidance, red = lapse); saturation encodes spatial distance (light = distance 0, dark = distance 4). Bars show mean ± SEM across sessions; *p*-values reflect repeated-measures ANOVA across distances for each strategy. (c) Same analysis as in (b), but for the 14 sessions during antibody infusion. Conventions as in (b). Reported *p*-values indicate repeated-measures ANOVA across distances within each strategy.

Conclusion: Our results show that the dynamics of latent strategies during memory recall are systematically reorganized under autoimmune synaptopathies (anti-NMDAR and anti-LGI1), even in the absence of overt performance deficits. In controls, recall behavior was explained by a limited set of latent strategies, dominated by a recent-memory (2h) guided strategy, sensitivity to cross-day spatial interference, and predictable modulation by the distance between rewarded locations. In the autoimmune models, this internal architecture was altered: both groups showed increased reliance on the 2h memory strategy and reduced engagement of non-goal-directed states, but displayed distinct profiles of spatial sensitivity (loss and/or reorganization of distance-based modulation). Together, these findings indicate that synaptic dysfunction caused by autoantibodies impacts not only memory strength but also the selection and deployment of internal strategies that support recall. The GLM-HMM framework reveals these changes as reallocations of latent states and provides an analytical bridge linking synaptic pathology to the structure of behavior during memory retrieval.

DISCUSSION

Memory is one of the fundamental cognitive functions, enabling organisms to retain information about past experiences and use it to guide future behavior. From navigation and foraging to social interaction and decision making, long-term memory underlies virtually every aspect of adaptive behavior. Understanding how such memories are formed, stabilized, and accessed over time remains a core objective in neuroscience.

Despite major advances in identifying the neural substrates of memory—from synaptic plasticity to hippocampal circuit dynamics—many core questions remain unanswered. In particular, we still lack a clear understanding of how individual memories' evolving in time (i.e. memory traces) are organized, how they maintain, enhance, or decay across time —especially under conditions where multiple memories overlap and potentially interfere with one another, and under what conditions they can be retrieved or disrupted. These aspects demonstrate the complexity of memory research, because memory expression in behavior is often entangled with ongoing learning, motivation, and task structure. As a result, isolating memory-guided decisions and understanding the cognitive strategies that support them continues to be a major challenge.

Because memory operates across multiple scales—from molecular plasticity to behavior—isolating its mechanisms requires narrowing broad theoretical challenges into tractable, well-defined questions. This strategy allows us to probe critical components of memory while still addressing the larger issues of how memories are formed, stabilized, and expressed. In this study, we focused on three such questions: (1) Does successful recall depend on how well information was learned during training? (2) Can multiple memories coexist and compete for behavioral expression? (3) What internal strategies govern memory retrieval, and how are they affected by disruptions in synaptic function?

To address these questions, we employed a high-throughput, repeated spatial memory-based task in which mice learn and retrieve daily-changing reward locations within an 8-port maze. This task structure provides several key advantages. First, it separates the learning and recall phases temporally—across hours and across days—, allowing us to assess memory independently of ongoing learning. Second, it introduces controlled uncertainty in the recall phase—by withholding immediate feedback—thus encouraging exploratory behavior that is sensitive to internal memory representations and not driven by immediate reinforcement. Third, the repeated and self-paced nature of the task yields rich datasets that are well-suited to both generative modeling and latent-state inference. These features make the paradigm particularly useful in uncovering the temporal and spatial dynamics of long-term memory expression.

In the following sections of this discussion, we revisit the key findings of the study and interpret them within this broader framework. The narrative is organized chapter by chapter, beginning with the relationship between learning and recall, and culminating with how memory strategies are affected by synaptic perturbation.

Chapter 1: Dissociating Learning and Memory in the 8-Port Maze.

Chapter 1.1: Learning Performance Does Not Predict Memory Strength in the 8-Port Maze

It is intuitive—almost self-evident—to assume that successful memory recall should depend on how well information was initially learned. In spatial tasks like ours, this would imply that the better an animal performs during the learning (training) phase, the stronger its memory expression should be during recall. This assumption underlies many behavioral paradigms in neuroscience, where observable performance during acquisition is often taken as a proxy for memory strength (e.g., Squire & Kandel, 2000). However, unlike classical forced-choice paradigms, our task allows self-paced movement and flexible strategies, making it less obvious whether training performance reflects memory encoding. When this link has been examined more directly, it is typically through neural markers of memory—such as hippocampal replay—rather than through behavioral recall performance. From that perspective, learning and memory appear tightly linked, but often through different types of readouts.

Yet, in the context of our 8-port maze task, we find that this assumption does not hold at the behavioral level. Despite the logic behind it, none of the variables we typically associate with learning—such as overall performance, rate of learning, reaction time, inter-trial interval (ITI) errors, or general engagement—systematically predicted how well animals remembered a rewarded location in the subsequent recall session. To test this directly, we performed a series of analyses comparing behavioral metrics from the training session to memory strength as measured by the Memory Index (MI), a validated session-level measure that quantifies directional memory-guided behavior. Across all animals and sessions, none of these metrics showed a significant correlation with MI, even after controlling for trial count, latency measures, and engagement-related variables. In other words, although we expected that learning and memory should go hand-in-hand, our data reveal a clear disconnect: the behavioral features that signal acquisition success during training do not predict the strength of memory expression measured behaviorally the next session.

This lack of correlation held true whether we examined coarse indicators of engagement or more nuanced measures of cognitive processing. Importantly, this null result is not trivial. Many of these variables have been shown in other contexts to combine learning, motivation, and attentional states (e.g., Heath et al., 2015; Birrell and Brown, 2000). That they failed to predict memory strength here suggests that recall in the 8-port maze is decoupled from

these dimensions of acquisition behavior—consistent with the MI's purpose of isolating recall during a no-feedback period.

Taken together, this discrepancy raises two important possibilities. First, it may be that memory recall—at least as measured by our Memory Index (MI)—is shaped by factors that are independent of overt classical learning behavior, such as offline consolidation, latent encoding mechanisms, or memory access dynamics. Second, and perhaps more critically, it may suggest that what we label as "learning performance" in this task does not fully capture the underlying memory formation process in this task structure.

In fact, accumulating evidence suggests that in goal-directed spatial tasks—especially those without punishment or error correction—memory encoding can occur rapidly, sometimes after just a single or few encounters with a rewarded location, even in the absence of measurable learning curves. For instance, recent studies have shown that hippocampal and entorhinal representations of goal locations can form during the very first successful navigation to a target (e.g., Alabi et al., 2022; Pfeiffer et al., 2022). In these paradigms, neural representations of reward-predictive locations emerge before behavioral consistency is observed. This aligns with the idea that in freely moving environments like ours—where there is no fixed trial structure, in the sense that animals are not forced to answer—the moments when an animal encounters a reward may drive rapid encoding, while behavioral performance across training may reflect other variables such as strategy shifts, motivation, or exploratory biases. In this view, it may not be that memory and learning are dissociated, but rather that our current behavioral metrics fail to capture when and how learning truly occurs, particularly if learning happens in short, goal-directed episodes that are behaviorally subtle or too sudden to be registered.

In summary, while it would be reasonable to expect that stronger learning performance leads to stronger memory recall, our data challenge this assumption. At least in this paradigm, standard behavioral readouts of learning fail to predict how well memory is later expressed. In the absence of behavioral variables that reliably correlate with memory strength, we adopt a conservative stance: that, under our current task design and metrics, learning and recall appear decoupled at the behavioral level when analyzed on a session by session level. This dissociation reinforces the value of the Memory Index (MI) as a robust and interpretable measure of long-term memory, relatively unaffected by session-level fluctuations in acquisition performance. Crucially, this allows us to pool data across sessions and animals without concern for acquisition-level confounds—thereby increasing both statistical power and interpretability in downstream analyses.

With this dissociation established, we next turn to the long-term temporal dynamics of memory in this task: Can animals recall rewarded locations not just after short delays, but even after a full day? And when multiple past experiences are available, are they recalled in

parallel or do they compete for behavioral control? These questions guide the analyses presented in Chapter 1.2.

Chapter 1.2: Memory Recall at 2h and 24h Delays in the 8-Port Maze Task

Building on the foundation laid in Chapter 1.1, we next asked whether animals can retrieve memories of past rewarded locations across different temporal delays. Specifically, we tested whether port locations rewarded 2, 24 or 48 hours prior to the recall session could still influence behavior, as quantified by the Memory Index (MI) computed on the no-feedback trials of the recall session.

This analysis served two complementary goals. First, it allowed us to determine whether memory traces in this task persist beyond short-term intervals—addressing the temporal boundaries of recall. Second, it asked whether multiple past memories can be expressed within the same session, potentially revealing how temporally distinct experiences are integrated or kept separate in guiding behavior. If both recent and older reward locations exert influence, it would suggest that the task taps into a form of long-term memory capable of supporting the retrieval of multiple, coexisting representations.

To address the question of whether animals can recall rewarded locations after different temporal delays, we analyzed poke behavior aligned to each past reward location and computed the Memory Index (MI) relative to the corresponding target. As expected, the magnitude of the MI was stronger at 2h than at 24h, but both were clearly above chance, indicating that animals retained and expressed directional memory traces across both timescales. However, no significant 48h memory trace was found. These findings, which replicate earlier observations reported in Morales et al. 2020, provide direct evidence that the task captures long-term memory recall spanning at least a full day.

Crucially, the expression of memory for both time points (2h and 24h) within the same session suggests that recall is not governed by a unitary, context-dependent representation of the current goal. This effect is not an artifact of the MI-recentering procedure, since the MI is computed separately for each target and tested against its own surrogate distribution. Rather, animals appear capable of accessing multiple memory traces in parallel—each corresponding to a distinct past episode. This opens the possibility that recall in this task reflects a competitive or integrative process in which multiple representations are retrieved and weighed during decision making.

The simultaneous expression of memories from different time lags challenge more classical interpretations that emphasize sequential consolidation and reduced reliance on hippocampal representations for older memories (e.g., Marr, 1971; McClelland et al., 1995).

Under such frameworks, earlier traces are typically assumed to be either overwritten or rendered inaccessible once new episodes are encoded—an assumption that does not align with the behavioral patterns observed in our task.

Instead, our findings are consistent with more recent theoretical frameworks that support the coexistence of multiple memory traces. Frameworks such as multiple trace theory (Nadel & Moscovitch, 1997) and contextual binding models (e.g., Yonelinas et al., 2019) posit that distinct episodes can be maintained in parallel and retrieved flexibly depending on context or task demands. From this perspective, the simultaneous recall of 2h- and 24h-old memories in our task reflects a memory system that preserves past experiences across time, allowing multiple representations to remain accessible and guide behavior.

These findings also underscore the sensitivity of the MI to latent structure in behavior. Rather than simply averaging across noisy decisions, the re-centered MI analysis reveals how individual past experiences leave detectable, directional imprints on exploratory behavior—even in the absence of immediate reinforcement. This sensitivity is particularly valuable in tasks like ours, where memory must be inferred under conditions of uncertainty.

In sum, this chapter demonstrates that animals not only recall rewarded locations after delays of 2 and 24 hours, but also express those memories concurrently within the same session. These results validate the 8-port task as a tool to study long-term memory expression and suggest that temporally distinct traces can coexist and guide behavior in parallel, suggesting a retrieval system that does not strictly prioritize the most recent memory. This lays the groundwork for the next set of analyses, where we ask whether and how these overlapping memory traces interact—and what cognitive rules govern their competition and integration.

Chapter 2: Distance-Dependent Memory Interference in the Recall of the 8-Port Maze Task

A longstanding idea in memory research is that retrieval is not solely determined by the strength or accessibility of an individual trace, but also by interference from other, competing representations. Classical interference theories (e.g., Underwood, 1957) and more recent computational frameworks (e.g., Mensink & Raaijmakers, 1988; Norman & O'Reilly, 2003) propose that when multiple memories overlap in features—such as temporal proximity, semantic content, or spatial location—they may compete for retrieval resources. This competition can lead to partial suppression, distortion, or reduced access to one or more of the memory traces.

Such competition effects in interference theory have been well documented in human episodic memory tasks, particularly when items are highly similar or contextually confusable. Given that spatial similarity provides a natural analog of representational overlap in the hippocampus, similar interference processes might also arise in spatial domains. How these principles extend to animal behavior—and specifically to spatial memory under naturalistic recall—remains, however, less understood.

In the context of our task, where we previously established that animals can retrieve both 2h- and 24h-old memory traces within the same session, this raises a natural question: do these coexisting memory traces interact, and if so, how? In particular, does spatial similarity between memories from different days lead to interference during recall, as predicted by classical theories of memory competition? If memory traces are not fully independent but partially overlapping in their representations, we might expect competition between them.

To examine whether spatial proximity between memories modulates recall performance, we grouped recall sessions based on the distance between the rewarded port on the current day (2h memory) and the one rewarded the previous day (24h memory). For each distance bin, we computed the Memory Index (MI) and compared the resulting values to surrogate control distributions. These controls were essential to ensure that any observed effects were not attributable to maze geometry or intrinsic biases in the MI computation.

Our analysis revealed a non-monotonic, distance-dependent pattern in MI values. The recall curve showed: a strong peak at distance 0, no modulation at distance 2, and significant declines at distances 3 and 4. This enhancement likely reflects repeated reinstatement of the same spatial memory across consecutive days *whenever* the same port is selected, which strengthens the trace through retrieval. Such repetition may facilitate memory trace strengthening not only through consolidation but also through retrieval itself, a phenomenon well documented in memory research (e.g., Karpicke & Roediger, 2008; Antony et al., 2017; Soderstrom & Bjork, 2015). Under this view, the act of recalling a memory strengthens it

further, increasing the likelihood of successful retrieval when past and current targets coincide.

Interestingly, MI values declined as the spatial distance between reward locations increased, but this modulation of recall strength by distance did not follow a simple monotonic trend. At intermediate distances—particularly at distance 2—recall performance showed no modulation relative to distance. This suggests a transitional regime in which the memory traces are neither close enough to reinforce each other nor sufficiently distinct to trigger strong competition. At greater separations (distances 3 and 4), memory strength decreased significantly, with MI values approaching or even falling below the levels expected from a distance-insensitive control. The observed decline in MI therefore reflects increasing interference, likely due to representational competition: older memory traces modestly reduce recall strength, consistent with representational competition, enough to prevent reinforcement. In other words, spatially dissimilar memories compete for expression, weakening recall under conditions of high separation in our task.

Together, these results reveal a non-linear profile of memory interaction, shaped by the spatial relationship between overlapping traces and modulated by both cooperative and competitive dynamics. However, the behavioral data alone cannot determine whether the observed distance-dependent recall pattern reflects genuine memory interactions or instead arises from structural aspects of the task—such as geometric biases or artifacts in the Memory Index computation. To disentangle these possibilities, we turned to generative modeling as a complementary approach-one that allowed us to test whether the observed modulation in recall could emerge from minimal memory assumptions, such as encoding only the current goal, or whether it required structured interference between coexisting memory traces.

Based on the complexity of the different memory lags' interactions, model-based simulations serve as a test of interpretability: if the empirical pattern can be reproduced without invoking prior reward locations, it would suggest a trivial explanation. Conversely, if the distance-dependent modulation can only be captured by models incorporating multiple memory components, it would support the conclusion that spatial competition between memories actively shapes recall behavior.

We began with the most parsimonious hypothesis: that recall behavior could be explained by a single memory trace centered on the current (2h) reward port. To test this, we implemented a baseline "one-bump" model, which failed to account for the decline in MI at intermediate or large spatial separations. This result suggests that relying solely on the most recent memory is insufficient to explain the structured pattern of recall across distances. In contrast, a "two-bump" model—incorporating both the current and previous (24h) reward locations—better captured the overall shape of the MI curve. This improvement supports the

idea that animals retain overlapping memory representations across days, and that these traces jointly influence recall behavior.

However, while the two-bump model provided a better fit than the one-bump version, it still failed to fully capture the magnitude of the interference effect—particularly at larger spatial separations. This discrepancy suggests that the influence of past memories is not merely additive: instead of simply combining the effects of recent and older traces, the memory system appears to weigh them unevenly depending on how spatially similar or distinct they are. When the memories are far apart, their representations may compete more directly for expression, leading to stronger interference than predicted by linear models. This was implemented as a simple monotonic weight that reduced the influence of the 24h memory with increasing distance. Incorporating this distance-dependent weighting improved the fit, suggesting that the expression of older memories is dynamically weighted by their spatial relationship to the current memory. From a cognitive perspective, this could indicate that recall is not simply a result of multiple memories competing for expression, but is guided by a selective process that weighs their influence based on how similar they are to the current goal.

Altogether, these findings suggest that the observed interference may not be unique to this paradigm, but may reflect a broader principle of memory systems—namely, that the expression of a memory trace is shaped not only by its recency or strength, but also by its contextual similarity to competing episodes. This dynamic weighting—shaped by spatial proximity—could serve as an adaptive mechanism, allowing animals to prioritize relevant experiences while filtering out conflicting or redundant ones.

Such a mechanism may be particularly valuable in naturalistic environments, where similar experiences often recur with partial overlap. In our task, this overlap—when recent and past reward locations are spatially close—appears to *facilitate it,* suggesting that representational similarity can facilitate retrieval rather than interfere with it. This pattern is consistent with a form of cooperative interaction between memory traces—possibly through cumulative reinforcement or overlapping encoding pathways.

From a computational perspective, this behavior aligns with models such as the Complementary Learning Systems framework (McClelland et al., 1995; Norman & O'Reilly, 2003) and the Context Maintenance and Retrieval model (Polyn et al., 2009), which propose that memory traces are not strictly segregated but interact based on contextual similarity. In these models, overlapping memories can reinforce each other when they converge on shared representations—particularly in the absence of strong contextual shifts—while interference tends to arise when memories are more distinct and thus compete more directly for retrieval.

At the systems level, the spatial gradient of recall strength observed here may reflect hippocampal mechanisms that regulate overlap and competition between memory traces. When reward locations are close in space, shared place cell ensembles may be co-activated, leading to facilitation through generalization or the reactivation of overlapping codes. In contrast, when locations are far apart, distinct representations may interfere more directly—especially in the absence of strong contextual segmentation. In our task, the uniform design of the maze and the lack of salient contextual cues across ports may contribute to this representational overlap, despite spatial separation. These dynamics resonate with empirical findings showing that hippocampal networks can flexibly integrate or separate memories depending on their spatial and temporal similarity (e.g., Leutgeb et al., 2007; Libby et al., 2019), while also highlighting that greater distinctiveness does not always guarantee reduced interference, particularly when representational boundaries are weak or ambiguous.

Our results highlight a key constraint in memory-guided behavior: the trade-off between integrating relevant past experiences and avoiding interference from competing ones. Consolidation processes have been shown to balance these opposing demands, allowing related memories to overlap while still protecting them from becoming confusable (Tompary & Davachi, 2017). Consistent with this, hippocampal differentiation can reduce interference by separating neural codes for similar experiences (Favila et al., 2016). In our task, this arbitration appears to be guided by the spatial relationship between past reward locations—memories of nearby locations are more likely to reinforce each other, whereas those farther apart tend to interfere. Similar mechanisms have also been proposed in other domains, including temporal proximity between experiences and broader contextual boundaries (Schiller & Eichenbaum, 2015). Together, these findings support the view that animals rely on internal heuristics to arbitrate among past representations, selectively expressing those most relevant to the current goal.

More broadly, this chapter illustrates how a simple, model-agnostic behavioral measure—the Memory Index—can uncover latent interactions between memory traces that are not apparent from accuracy alone. Rather than reflecting storage and retrieval alone, the patterns we observe point to processes of selection and weighting among past experiences, consistent with theories that emphasize competition and integration between memory traces (Norman & O'Reilly, 2003)

These insights raise deeper questions: How are memory weights dynamically learned or modulated over time? Can animals suppress interference through learning, repetition, or attentional strategies? And what neural processes govern the competition or cooperation between memories with varying degrees of similarity? Addressing these questions requires moving beyond session-level summaries and into finer-grained analyses of behavior and memory dynamics. This motivates the poke-level modeling introduced in the next chapter,

where we investigate the computational principles and neural correlates that govern memory-guided decision making under uncertainty.

Chapter 3: Latent Memory States and Behavioral Strategy Dynamics During Recall

In Chapters 1 and 2, we established that recall behavior in the 8-port maze is both robust and interference-sensitive: animals can retrieve memories of spatial goals across multiple timescales, yet their performance is shaped by the similarity of competing memory traces. These findings, however, relied on session-averaged measures—specifically, the Memory Index (MI)—which, while useful, can obscure the rich temporal structure of behavior that unfolds within sessions. A critical limitation of this measure is that it cannot distinguish between true memory failures and fluctuations in strategy or engagement: two sessions with identical MI scores could reflect entirely different scenarios. In Chapter 3, we addressed this limitation by using a GLM-HMM framework to model behavior at the single-poke level. This modeling strategy builds on recent systems neuroscience work revealing latent cognitive states in animals task responses (e.g., Ashwood et al., 2022; Wiltschko et al., 2015), reframing behavioral variability not as noise, but as transitions between distinct internal strategies with differential sensitivity to task variables or motivational states. Could the poke-to-poke variability observed in our task during recall reflect a similar process? Our observations suggest it does. Animals switch between multiple mnemonic and non-mnemonic strategies, and these transitions are neither random nor trivial—they reveal structured, interpretable patterns that reflect how animals resolve uncertainty, arbitrate between memory traces, and sometimes disengage from the task altogether.

The latent behavioral strategies inferred by the GLM-HMM did more than increase predictive accuracy—they provided a structured framework for capturing variability in how animals approached the recall task. However, this structure did not emerge spontaneously: it depended critically on model design and selection. To identify the appropriate number of latent states, we trained models with one to five hidden states and evaluated performance using cross-validated log-likelihood and predictive accuracy on held-out data. Both metrics improved with increasing model complexity, but exhibited a clear elbow at four states—indicating a point of diminishing returns. The emergence of a plateau suggested that a four-state solution offered the best trade-off between explanatory power and parsimony. With this model in hand, we then asked a central question: do these latent states reflect genuinely distinct behavioral strategies—not only in terms of how well they predict choices, but also in how they differentially use task-relevant information?

To test whether the inferred states reflect meaningful behavioral patterns, we measured how well the model predicted choices within each state. This analysis allowed us to ask whether the behavioral strategies captured by the model are not only internally consistent, but also predictive—i.e., whether knowing the current state helps explain what the animal is likely to do next. We found that all four latent states supported port predicted choices well above the

12.5% chance level (corresponding to the 1/8 probability of selecting any port at random), indicating that each state captured structured, non-random patterns of behavior. We interpret each state as reflecting a distinct behavioral strategy, defined by differential reliance on task-relevant information:

The state with the highest predictive accuracy was strongly driven by recent (2h) memory cues. Based on this pattern, we interpret the underlying strategy as one of precise, goal-directed recall guided by recent experience. Another state showed a stronger dependence on 24h memory and was less predictive overall; we interpret this as reflecting a structured but suboptimal recall strategy, possibly shaped by interference from outdated information. A third state relied more heavily on spatial proximity than on mnemonic cues, consistent with a spatial-heuristic or memory-avoidance strategy—potentially used when memory signals are weak or ambiguous—suggesting a conflict-resolution mechanism. The fourth state had the lowest predictive structure and minimal influence from task variables; we interpret this as reflecting task disengagement or a default, exploratory mode. While these interpretations are grounded in the regression profiles and predictive patterns of each state, they remain inferences about the underlying cognitive processes. Taken together, the results suggest that recall behavior is best captured not by a single strategy, but by a dynamic mixture of strategies that animals flexibly switch among as uncertainty, interference, and task demands change.

Crucially, these model-derived strategies were not only internally consistent—they were also reflected in systematic patterns of behavior across the task. When we examined poke distributions aligned to task variables, each latent state exhibited spatial patterns that closely mirrored its computational profile. The *2h-memory strategy* drove sharply focused pokes toward the current goal; the *24h-memory strategy* reflected broader alignment with outdated targets; and the *memory-avoidance strategy* showed systematic misalignment with both memory cues, consistent with an evasive policy. Even the *lapse strategy*, while generally unstructured, revealed a reproducible bias toward the port closest to the exit door of the behavioral box—a pattern suggestive of a fallback or conflict-resolution policy. These converging lines of evidence—from model structure to observable choice patterns—reinforce that the latent states identified by the GLM-HMM capture real, strategy-specific shifts in behavior, not merely statistical noise.

These behavioral patterns may reflect broader principles of cognitive control. Rather than reflecting fixed modes of behavior, the latent states uncovered by the GLM-HMM likely capture how animals arbitrate between conflicting sources of information: balancing recent memories, interference from outdated experiences, spatial heuristics, and disengagement. This dynamic arbitration aligns with theoretical frameworks from reinforcement learning and decision-making under uncertainty, which propose that agents flexibly switch between internal policies based on confidence, expected reward, or contextual reliability (Yu & Dayan, 2005; Daw et al., 2006). The identification of lapse-like states echoes observations of

fluctuating cognitive engagement in both rodents and primates (Ebitz & Hayden, 2016; Pisupati et al., 2021), suggesting that disengagement is not random but governed by internal computations. That these states were inferred purely from behavior—without access to neural signals or external feedback—underscores the strength of this modeling approach. Rather than treating behavioral variability as noise, this framework offers a structured, interpretable window into possible latent cognitive processes.

The next analyses extend the model-centric view by examining how task variables contribute globally to behavior, and how internal strategies unfold across time, trials, and individuals. Permutation-based importance metrics, obtained by shuffling individual regressors while keeping others fixed, revealed two predictors—2h memory and Euclidean distance—as particularly influential, underscoring a central theme: memory-guided recall is not governed by a single computational mechanism, but emerges from the interplay between internal memory access and external spatial heuristics. While previous work has emphasized the role of spatial context in shaping hippocampal and navigational codes (e.g., Fuhs & Touretzky, 2007; McNaughton et al., 2006), our findings suggest that such spatial variables also provide behaviorally distinct signals—especially when memory is uncertain or conflicting.

Interestingly, other regressors—such as 24h memory and general bias—were behaviorally informative when considered in isolation, but contributed less explained unique variance in the full model. This pattern is consistent with a hierarchical structure in behavioral control: certain variables (e.g., 2h memory, distance) serve as core anchors, while others act as fallback cues when higher-confidence signals are weak or ambiguous. This interpretation is consistent with broader computational theories proposing that agents prioritize reliable predictors and resort to simpler heuristics when uncertainty increases (Collins & Frank, 2013; Gershman et al., 2015). At the same time, the fact that many of these variables retain strong predictive power in isolation reinforces the idea that all included regressors are behaviorally meaningful. Rather than being superfluous, each captures a distinct aspect of choice behavior, confirming their value in explaining different components of recall behavior.

Having established that all variables contribute to recall behavior at the global level, we next examined how internal strategies are deployed over time. These analyses revealed that the *2h-memory strategy* is not only the most predictive, but also the most stable. Its high dwell time and low transition entropy suggest that, once engaged, animals tend to remain in this state—potentially reflecting a reliable internal mode of confident memory access. By contrast, the *24h-memory* and *lapse strategies* were shorter-lived and more volatile, consistent with transient engagement or uncertainty-driven shifts in behavior.

Importantly, these dynamics varied substantially across animals. While all individuals engaged the *2h-memory strategy* to some degree, there was clear variability in how much they relied on it relative to alternative strategies. Animals with higher occupancy of the 24h-memory or memory-avoidance states tended, on average, to rely less on the 2h-memory

strategy less frequently—though it remained dominant overall—indicating a graded trade-off rather than a categorical shift in behavioral mode. This pattern is consistent with theoretical accounts of arbitration between competing cognitive systems, where agents adjust their reliance on different strategies depending on internal confidence or contextual reliability (Collins & Frank, 2013; Gershman et al., 2015). Correlation analyses confirmed this interpretation: a strong anti-correlation between the *2h strategy* and both *24h-memory* and memory-*avoidance strategies* revealed a dynamic balance between precise goal recall and fallback strategies under uncertainty. Meanwhile, a modest positive correlation between *24h-memory* and memory-*avoidance strategies* suggested a subgroup of animals prone to switching between suboptimal recall modes—potentially reflecting difficulties in trace access or strategic resolution. These individual differences in strategy balance align with prior findings of stable cognitive styles and decision-making profiles in rodents and non-human primates (Pope et al., 2021; Cinotti et al., 2019; Ebitz et al., 2019). The *lapse state*, in contrast, was more uniformly distributed across animals and showed weak correlations with all others—supporting the view that it reflects a general disengaged or default mode, rather than a structured recall strategy, in line with prior work suggesting that behavioral lapses arise from latent internal states rather than random noise (Pisupati et al., 2021; Ebitz & Hayden, 2016).

Finally, our within-session and within-trial analyses reveal that strategy use is not fixed, but dynamically adapts to evolving task demands and behavioral unfolding. Animals did not commit to a single internal state per trial; instead, longer trials exhibited a temporal gradient, with goal-directed recall becoming increasingly dominant as the trial progressed. This suggests that behaviors initially driven by exploration or uncertainty can transition into structured recall as the animal accumulates evidence or resolves internal conflict. Such within-trial refinements are reminiscent of adaptive control processes, where agents integrate incoming information over time to disambiguate latent task states and refine their responses (Nassar et al., 2010; Wilson et al., 2014). This temporal evolution highlights that even apparently noisy or ambiguous choices may reflect structured internal dynamics unfolding beneath the surface.

In sum, the results presented in Chapter 3 reveal that recall behavior is structured not by a single cognitive strategy, but by a dynamic interplay of internal patterns of behavior that unfold over time and vary across individuals. Rather than being random or noisy, this variability reflects flexible cognitive control—animals switch between memory-guided, heuristic, and disengaged strategies depending on task demands, spatial context, and internal uncertainty. These findings provide a nuanced picture of recall as an adaptive process, supported by latent cognitive states that may guide behavior in real time.

Beyond validating the existence of distinct strategies, the GLM-HMM also offers a richer view of how interference shapes recall. The model revealed that when reward locations from consecutive days were spatially close, animals relied more consistently on precise,

2h-memory recall, whereas larger separations shifted behavior toward reliance on older 24h traces. This pattern parallels the interference effects detected with the Memory Index (Chapter 2), but manifests here as a redistribution of strategy use rather than a drop in overall recall strength. In this light, memory interference is not only a degradation of recall strength, but a rebalancing of the internal policies animals use to resolve competing mnemonic signals.

While these strategies are inferred solely from behavior, they must ultimately depend on underlying neural mechanisms—particularly those supporting memory encoding, consolidation, and retrieval. In the chapters that follow, we shift from descriptive behavioral modeling to direct experimental tests, asking whether the integrity and flexibility of latent memory-based strategies depend on intact synaptic signaling. Using two mouse models of autoimmune encephalitis—anti-NMDAR and anti-LGI1—we examine how targeted disruption of specific molecular pathways alters the deployment and dynamics of internal memory strategies. These models allow us to directly test whether different forms of synaptic dysfunction lead to distinct breakdowns or reorganizations of memory-guided behavior, offering a path toward identifying the neural mechanisms that support strategic flexibility during recall.

Chapter 4: Autoantibody-Induced Disruption of Latent Strategy Dynamics in the Anti-NMDAR Encephalitis Model.

Having established that recall behavior in healthy animals is mainly governed by a small number of latent strategies—dynamically deployed across trials and shaped by internal memory based processes—we next asked whether these internal dynamics depend on the state of synaptic plasticity. In our lab we have used a well-characterized mouse model of anti-NMDAR encephalitis, in which circulating autoantibodies target the NMDA receptor, leading to synaptic disruption without cell death. We have used this model because anti-NMDAR encephalitis in patients involves autoantibody-driven internalization of NMDA receptors and pronounced, yet typically reversible, memory impairment. Because NMDA receptors support activity-dependent synaptic plasticity and contribute to stabilizing and updating memory traces, reducing NMDAR function makes this model well suited to test whether perturbing this mechanism alters the deployment and flexibility of internal strategies during recall—beyond what is detectable through conventional performance metrics. This well-established mouse model captures the post-acute, recovery phase of the disease—when overt symptoms have subsided but significant memory impairments persist.

Our hypothesis was that memory-guided behavior—particularly the use of flexible strategies—would be altered by NMDA receptor disruption. We found that animals continued to express memory-guided behavior after anti-NMDAR infusion, but the balance of latent strategies was significantly altered. Specifically, there was a marked increase in the use of the 2h memory-guided strategy and a reduction in the expression of lapse and memory-avoidance states. This shift suggests a narrowing of the effective strategy space: animals defaulted toward recent-memory recall while expressing fewer fallback or exploratory strategies. Such a pattern may reflect a compensatory bias toward the most reliable memory-based available strategy—favoring 2h memory when flexibility is compromised. Rather than abolishing recall, anti-NMDAR treatment appeared to constrain the flexibility with which animals navigated between internal strategies—prioritizing the most reliable and recent trace while reducing behavioral diversity.

While overall strategy frequencies shifted during the presence of anti-NMDAR antibodies, their temporal dynamics revealed a structured and reproducible pattern. Shortly after infusion onset, 2h–strategy use increased, followed by a transient decline (accompanied by more lapses) and then recovery. This initial surge was followed by a transient drop—coinciding with a rise in lapse expression—and a subsequent recovery. After infusions ended, 2h strategy use returned to pre-treatment levels. The timing and reversibility of these effects argue against non-specific behavioral drift or model-detection artifacts. Instead, they

reflect a dynamic reorganization of internal strategy allocation triggered by NMDAR disruption. Notably, the immediate increase in recent-memory use suggests that underlying memory traces remained accessible, while their regulation became less stable. Rather than indicating overt memory loss, our findings point to a more subtle disruption in the flexible control of memory-based strategies during early stages of synaptic impairment.

One of the clearest indicators of impaired flexibility came from the spatial interference analysis. In healthy animals, the use of memory-based strategies varies systematically with the spatial distance between rewards—reflecting adaptive arbitration between competing memory traces. After NMDAR disruption, this modulation was strongly reduced: 24h–memory use no longer increased with distance, and the 2h strategy exhibited a flattened—or occasionally inverted—trend. One plausible explanation is reduced stabilization of the 24h trace under diminished NMDAR-mediated plasticity. With weaker long-term competition, interference signals diminish and behavior defaults toward a simpler, distance-insensitive policy. In this view, the absence of interference is not a sign of efficient recall, but a consequence of reduced competition between memory traces—reflecting a loss of the strategic dynamics that normally guide flexible behavior. While more subtle than the frank amnesia reported in clinical cases of anti-NMDAR encephalitis (Dalmau et al., 2011), this pattern may represent an early-stage disruption in the ability to coordinate and deploy multiple memory representations.

These findings suggest that, in our task, an intact NMDAR function is not strictly required to express memory-guided behavior once learned, but is essential for flexible deployment and modulation of internal strategies. Beyond impaired stabilization, NMDAR disruption may also degrade internal models of task structure, reducing the ability to infer—based on sensory inputs—when different strategies should be engaged. Both interpretations are consistent with extensive literature linking NMDAR signaling in hippocampus and prefrontal cortex to the stability of memory representations and the cognitive flexibility required to navigate between them (Malenka & Nicoll, 1993; Bannerman et al., 2008).

Importantly, the behavioral effects of anti-NMDAR antibodies were reversible: after infusions ended, strategy use gradually returned to baseline. This mirrors previous reports in both animal models and patients, where symptoms subside as circulating antibody levels decline (Planagumà et al., 2015; Dalmau et al., 2008). While neuronal integrity was not directly assessed here, behavioral recovery suggests a functional—and potentially reversible—disruption of synaptic signaling rather than lasting structural damage.

Together, these results show that while NMDAR signaling is not required for the existence of recall strategies, it is critical for their diversity, modulation, and context-dependent expression. When NMDAR function is compromised, animals do not simply forget—recent memories remain accessible—but flexibility in selecting and adjusting strategies is reduced. This highlights a fundamental role for synaptic plasticity in enabling cognitive flexibility,

allowing the brain to deploy the most appropriate policy for a given context. That these effects emerge in latent-state modeling—but are muted in traditional performance measures—highlights the value of fine-grained behavioral analysis for probing circuit function..

In sum, anti-NMDAR antibodies selectively reshape the internal dynamics of memory-guided behavior. The resulting profile—a bias toward recent memory, loss of interference sensitivity, and temporary fluctuations in strategy allocation shortly after antibody infusion—constitutes a behavioral signature of impaired synaptic plasticity. These findings link molecular-level disruption to strategy-level behavior and emphasize the role of NMDA receptor function in supporting flexible recall.

Chapter 5: Distinct Strategy Disruption in the Anti-LGI1 Encephalitis Model

In Chapter 4, we showed that disruption of NMDA receptor function alters the internal architecture of recall behavior—biasing animals toward recent memory strategies while suppressing flexibility and sensitivity to interference. Here, we asked whether a mechanistically distinct autoimmune encephalitis—mediated by antibodies against the LGI1 protein—would produce similar or divergent disruptions in latent strategy dynamics. Although both conditions target synaptic function within hippocampus-related circuits, they involve distinct molecular mechanisms: NMDA receptors mediate activity-dependent synaptic plasticity and memory updating (Malenka & Bear, 2004; Nakazawa et al., 2004), while LGI1 stabilizes excitatory synapses by regulating Kv1 channel localization and interacting with the postsynaptic adhesion molecule ADAM22 (Fukata et al., 2006; Lai et al., 2010). LGI1 autoantibodies reduce synaptic LGI1 levels and impair both hippocampal LTP and performance in hippocampal-dependent object-memory tasks (Petit-Pedrol et al., 2018; Kornau et al., 2020), indicating a distinct mode of synaptic disruption. As LGI1 encephalitis is the second most common form of autoimmune synaptopathy after anti-NMDAR (Graus et al., 2016), it offers a clinically relevant and mechanistically distinct model to test whether distinct molecular disruptions give rise to shared or unique patterns of memory-guided behavior. Similarly to the murine model of anti-NMDAR encephalitis, this paradigm represents the post-acute stage of the disorder, characterized by residual cognitive impairment in the absence of active inflammation.

LGI1-related synaptic disruption also altered the deployment of internal strategies, though in a subtler and more variable manner across animals. Animals exposed to anti-LGI1 antibodies showed increased expression of memory-guided states—particularly the 2h–memory strategy, accompanied by a modest increase in 24h–memory use and a reduction in lapse and avoidance behaviors. However, unlike the more polarized pattern observed in the anti-NMDAR model, these shifts were modest and accompanied by high inter-individual variability. This suggests that LGI1 dysfunction affects strategy allocation but does not impose a dominant or rigid behavioral regime. Rather than constraining the repertoire, the LGI1 model appeared to preserve a broader range of behaviors—suggesting a redistribution of strategy use rather than a reduction in strategic flexibility.

The temporal dynamics of strategy usage further highlight differences between the two models. In anti-LGI1 animals, the shift toward memory-guided behavior emerged early and remained relatively stable throughout treatment, suggesting a preserved, though altered, capacity to sustain internal memory-based policies. The brief increase in lapse expression near the end of the infusion period may reflect a transient destabilization in network function, potentially reflecting cumulative synaptic instability, as suggested by prior reports

(Petit-Pedrol et al., 2018). However, the rapid behavioral recovery after treatment ceased indicates that, as with anti-NMDAR exposure, the effects of LGI1 antibodies on strategy dynamics are reversible and likely reflecting a functional disturbance rather than long-term structural damage.

One of the most informative divergences between models emerged in the modulation of strategy use by spatial interference. In healthy animals, recall behavior flexibly adapts to spatial separation between previous rewards' locations—a proxy for resolving competition between overlapping memory traces (Missaire et al., 2017). This adaptive interference sensitivity was abolished in anti-NMDAR animals, consistent with prior evidence linking NMDA receptors to memory updating and conflict resolution (Nakazawa et al., 2004). LGI1-treated animals, by contrast, showed a preserved—but altered—sensitivity to spatial context: the modulation was non-monotonic, with strategy switching preserved, but unfolding in a less coordinated and more variable manner—suggesting partial disruption of contextual control. This pattern may reflect partial retention of adaptive control mechanisms despite LGI1-related synaptic instability—consistent with the role of LGI1 in maintaining synaptic strength rather than directly mediating activity-dependent plasticity (Fukata et al., 2006; Lai et al., 2010).

These findings support a broader interpretation: that distinct forms of autoimmune synaptopathy produce distinct disruptions in cognitive architecture. While both models affected strategy dynamics, the anti-NMDAR condition induced a stronger shift toward recent memory and away from adaptive flexibility. The LGI1 model, in contrast, led to a milder reorganization, with preserved engagement of memory-based strategies and altered—but not abolished—contextual modulation. These differences likely reflect the specific roles of NMDA receptors and LGI1 in synaptic plasticity and network stability. NMDA receptor loss disrupts LTP and trace updating, impairing competition resolution between memories. LGI1, by contrast, supports synaptic maintenance and excitability, and its disruption may degrade the regulation of strategy deployment without compromising the integrity of the stored representations themselves.

Importantly, neither model led to a collapse of recall behavior. Animals retained structured, internally organized strategies even under chronic synaptic perturbation. What changed was not the existence or accessibility of memory, but its regulation. This distinction underscores the value of latent-state modeling: by uncovering strategy-level dynamics invisible to standard performance metrics, this framework reveals how molecular dysfunction alters cognitive flexibility before accuracy breaks down.

In sum, the anti-LGI1 model reveals a distinct form of strategy reorganization under synaptic stress. Unlike anti-NMDAR antibodies, which bias behavior toward rigid recent memory use, LGI1 antibodies allow for broader engagement of memory strategies, but weaken their contextual modulation. These findings broaden our understanding of how distinct

autoimmune encephalitides impact the structure of memory-guided behavior, showing that distinct molecular disruptions produce distinguishable cognitive signatures. By linking synaptic pathology to latent strategy dynamics, we provide a framework for dissecting subtle cognitive effects in experimental models, setting the stage for clinical implication.

This pattern of preserved but variably expressed strategies resonates with the fluctuating recall profiles reported in LGI1 patients, providing a natural bridge to the clinical implications discussed next.

Clinical implications and translational outlook

From the results presented in this thesis using the anti-NMDAR murine model—intended to approximate the patient state after the treatment of severe symptoms but with persistent memory impairments—the mice displayed a consistent bias toward using more recent memories over older ones. This behavioral pattern was hypothesized as a reduced ability of the memory system to resolve interference among overlapping experiences rather than a generalized memory deficit.

A related phenomenon has been reported in humans, though at much shorter timescales: patients recovering from anti-NMDAR encephalitis show reduced serial dependence during spatial working-memory tasks, despite preserved accuracy (Stein et al., 2020). Serial dependence refers to the subtle bias of current perceptions toward recent information or short-term memories—a mechanism proposed to promote temporal continuity in perception and decision-making and has been linked, in computational models, to NMDAR-dependent short-term synaptic potentiation that maintains and integrates information across seconds (Barbosa et al., 2020;).

Although serial dependence operates over seconds, it may reflect a broader principle of NMDAR-dependent temporal integration extending across multiple timescales. In this view, the mechanisms that stabilize perception from moment to moment—described as continuity fields that integrate information across short timescales (Manassi & Whitney, 2023) may share computational and physiological principles with those that stabilize memory across longer intervals. Our findings may correspond to memory-stabilization processes acting on a slower timescale: in both perception and episodic recall, NMDAR dysfunction appears to reduce the temporal window over which past experiences influence current behavior. This time reduction has not yet been studied in human patients.

At the clinical level, converging evidence shows persistent cognitive sequelae after anti-NMDAR encephalitis, most prominently affecting memory and executive functions on long-term follow-up (Finke et al., 2012). Neuropsychological studies reveal enduring episodic and working-memory impairments consistent with hippocampal involvement (Finke et al., 2012; Finke et al., 2016; Heine et al., 2021), while neuroimaging demonstrates reduced hippocampal–DMN connectivity that predicts individual (subject-wise) memory performance (Finke et al., 2013). Together with the reduction of serial dependence observed at short timescales in perception and working memory (Stein et al., 2020), these findings point to a narrowed temporal influence of prior experience, potentially extending to long-term consolidation.

If future human studies confirm a structured pattern of interference observed in mice, where overlapping memories compete in a temporally biased rather than globally degraded way, it would indicate that anti-NMDAR encephalitis affects interference resolution more than

memory capacity per se. This mechanism would provide a clear rehabilitation target: re-establishing the temporal stability of memory traces. Ongoing work in collaboration with our labs is adapting multi-session associative-learning paradigms, which would help to probe such long-term interference in humans, but to date no results have yet been reported in patient populations.

Within this framework, spaced learning and spaced retrieval training represent promising approaches. These teaching methods distribute practice or recall attempts over increasing intervals, allowing memory traces to be reactivated after controlled delays. Extensive experimental evidence in healthy and clinical populations shows that spacing enhances long-term retention and reduces interference between overlapping associations (Cepeda et al., 2006; Kang, 2016; Maddox et al., 2011). In clinical settings such as mild cognitive impairment, dementia, or acquired brain injury, spaced-retrieval training, in which patients practice recalling target information after progressively longer delays, has repeatedly improved recall and transfer to everyday life (Haslam et al., 2011; Oren et al., 2014; Hopper et al., 2013). Adapting these paradigms to anti-NMDAR encephalitis could help re-expand the effective temporal window for consolidation and counteract the *recent* bias observed in our model. A practical implementation might combine short, high-frequency retrieval trials early in training/rehabilitation with progressively increasing inter-session intervals (e.g., 10–20 min → 2–3 h → 24–48 h → 7 days), using personally meaningful material such as medication routines, appointments, or autobiographical facts. Improvement in delayed recall and restore normal interference across sessions would indicate successful compensation for a narrowed integration window.

In addition, retrieval-practice principles, where memory is strengthened by the act of retrieval itself rather than by repeated study, could further enhance consolidation (Roediger & Karpicke, 2006; Pyc & Rawson, 2009). Combining retrieval practice with spacing ("distributed retrieval") produces additive gains in retention and generalization (Rawson et al., 2013).

Although these strategies have not yet been formally tested in autoimmune encephalitis, long-term outcome studies highlight enduring cognitive morbidity and explicitly encourage continued rehabilitation (proposed also by Finke et al., 2012; Heine et al., 2021; Brenner et al., 2024). Mechanism-guided interventions grounded in spacing and retrieval principles therefore represent a rational starting point for targeted cognitive therapy in anti-NMDAR encephalitis, aiming to restore continuity between successive experiences and strengthen long-term memory resilience after synaptic disruption.

In contrast, for patients with anti-LGI1 encephalitis, the specific behavioral dimensions examined in our model—particularly those related to interference dynamics and strategy organization—have not yet been systematically investigated. As a result, no direct parallelism can be established between our experimental findings and the existing clinical literature.

Human studies to date have primarily focused on broader cognitive outcomes, reporting persistent episodic-memory deficits and hippocampal network abnormalities (Finke et al., 2017; Heine et al., 2018; Qiao et al., 2020; Galioto et al., 2022). Future studies will therefore aim to identify parameters that can be meaningfully compared to murine models measures, facilitating a more direct translational link for the anti-LGI1 model.

CONCLUSIONS

General Conclusion

- Across all chapters, this thesis shows that memory recall is not a single process but a dynamic mixture of strategies shaped by temporal, spatial, and synaptic factors.
- In healthy animals, recall reflects the coexistence and competition of multiple memory traces, resolved through flexible switching between latent strategies.
- Synaptic dysfunction induced by anti-NMDAR or anti-LGI1 antibodies disrupts this internal architecture in distinct ways: NMDAR loss constrains flexibility and abolishes interference sensitivity, while LGI1 loss produces subtler redistributions of strategy use.
- These findings demonstrate that autoimmune synaptopathies reorganize the internal structure of recall in the 8-port maze before overt accuracy deficits appear, and that latent-state modeling provides a powerful tool to detect such hidden cognitive signatures.

Conclusions by Chapter

Chapter 1: Dissociating Learning and Memory in the 8-Port Maze

1. Learning performance during training did not predict recall strength as measured by the Memory Index (MI).

2. The MI is validated as a robust, learning-independent measure of memory recall, providing a reliable tool to assess long-term memory independently of session-by-session fluctuations in training performance.

3. Animals performing in the 8-port maze are able to recall both 2h and 24h memories within the same session, showing that multiple temporal traces coexist rather than overwrite one another, despite their similarities.

Chapter 2: Distance-Dependent Memory Interference

4. Recall strength in the 8-port maze depends on spatial distance between reward locations. Nearby memories reinforce one another, while distant ones interfere. These results provide behavioral evidence for competition and integration between memory traces.

Chapter 3: Latent States and Strategy Dynamics During Recall

5. Animals performing in the 8-port maze use four distinct strategies: goal-directed recall (2h memory), outdated recall (24h memory), memory-avoidance, and lapse/disengagement ,as uncovered by latent modeling.

6. Recall of animals performing in the 8-port maze was not governed by a single strategy but by a dynamic switching between multiple strategies within and across trials.

7. Strategy allocation varied substantially across animals, but the 2h memory-based strategy dominated animal behavior.

8. Memory interference was confirmed as redistribution of strategies, with reduced use of the 2h memory-based strategy and increased reliance on 24h memory when rewarded locations were farther apart.

Chapter 4: Strategy Disruption in the Anti-NMDAR Model

9. Anti-NMDAR antibodies increased reliance on the 2h memory-based strategy narrowing the repertoire of behavioral strategies.

10. Animals preserved recall ability but lost flexibility and sensitivity to spatial interference, showing a rigid bias toward recent memory and reduced strategy diversity.

11. The effects of NMDAR disruption were reversible once antibody infusion ended, confirming that the alteration of recall strategies was functional rather than structural

Chapter 5: Strategy Reorganization in the Anti-LGI1 Model

12. Anti-LGI1 antibodies altered strategy allocation, increasing memory-guided behavior (2h and 24h) while reducing lapses, but changes were subtler and more heterogeneous across animals than in the NMDAR model.
13. Spatial modulation was preserved but reorganized into a less structured, non-monotonic pattern.
14. As in the NMDAR model, effects were reversible, suggesting functional—not permanent—disruption.

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